

Webex Cloud-Connected UC

Technical Decision Maker Presentation

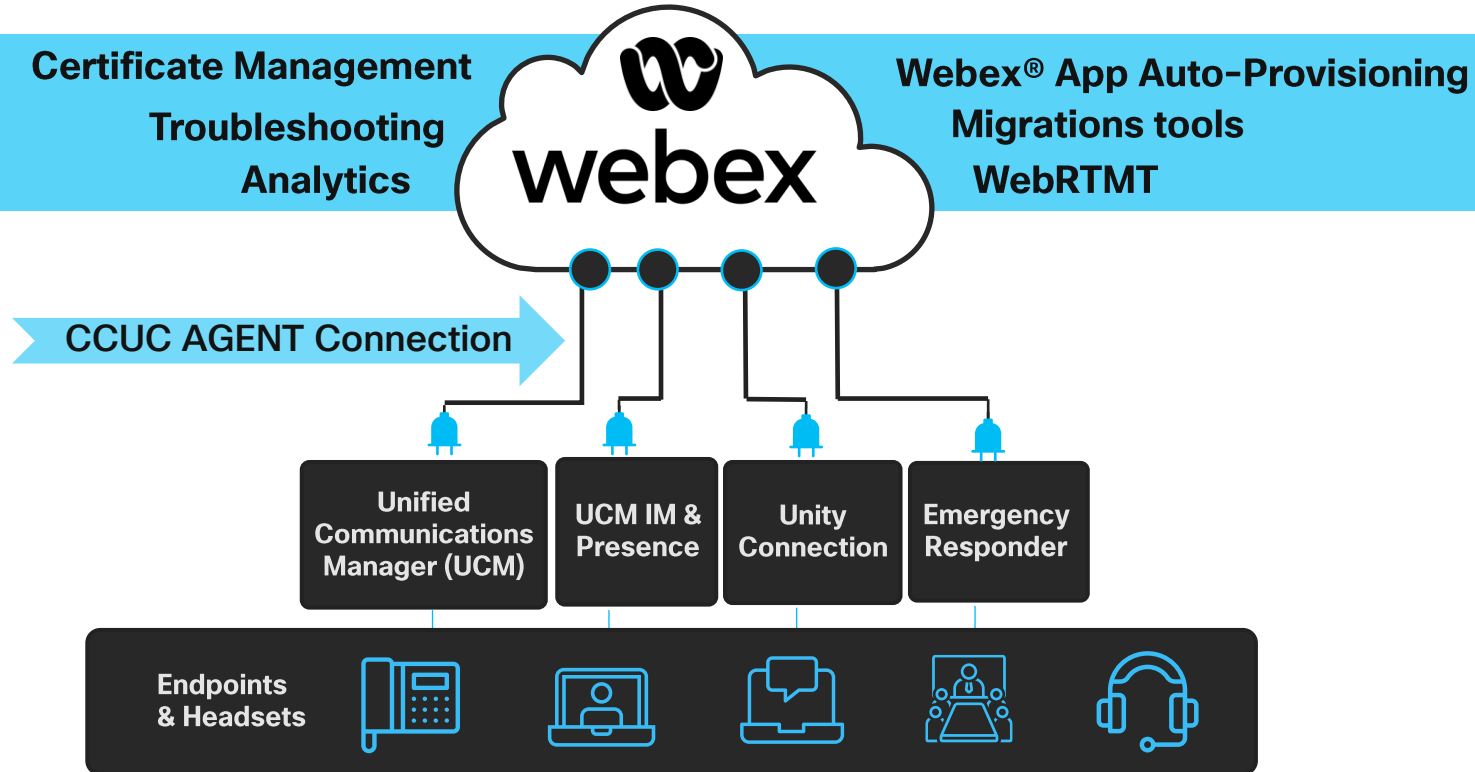
Webex Cloud-Connected UC PM and TME

July 2022



Webex Cloud-Connected UC (CCUC)

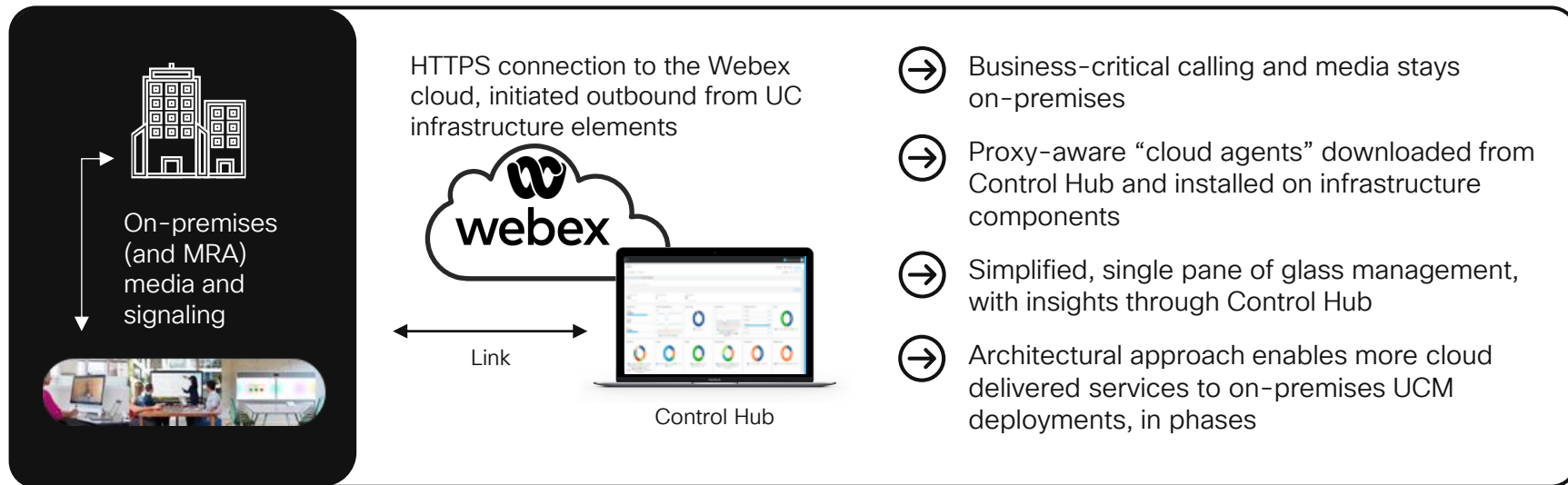
Feature Suite



Webex Cloud-Connected UC (CCUC)

High level overview

Provide analytical insights, enhanced management capabilities, and value-added services for Cisco® Unified Communications Manager (UCM) on-premises delivered through the Webex® cloud platform.



Webex Cloud-Connected UC in a nutshell

Simplify and augment configuration, provisioning, monitoring, and reporting features and workflows for Cisco's unified collaboration on-prem deployments

Why?

Cloud based service to collect telemetry, perform provisioning tasks, and provide value-add workflows to the Cisco® UCM infrastructure elements

Cloud agents installed on UC infrastructure nodes communicate with the cloud service

Deployed as another workflow in Control Hub for:

- Operations
- Analytics
- Web RTMT
- Troubleshooting

What?

Proxy aware CCUC agents installed (or activated) on UC infrastructure elements

CCUC is available by default in Control Hub org, only needs a click to activate

UC nodes are onboarded to Control Hub

CCUC Agents activate modules for specific features (e.g. telemetry) and communicate to the CCUC Cloud Service

CCUC Cloud Service manages its "agents" on each node (e.g. status checks, etc.)

CCUC cloud service surfaces admin functions and reports in customer's Control Hub "Services" and "Analytics"

How?

Content

1	Cloud-Connected Value Proposition	6	Troubleshooting
2	Onboarding	7	Webex App Auto-Provisioning
3	Analytics	8	Directory Service (LA)
4	Certificate Management	9	Security and Privacy
5	Web RTMT	10	Additional Resources

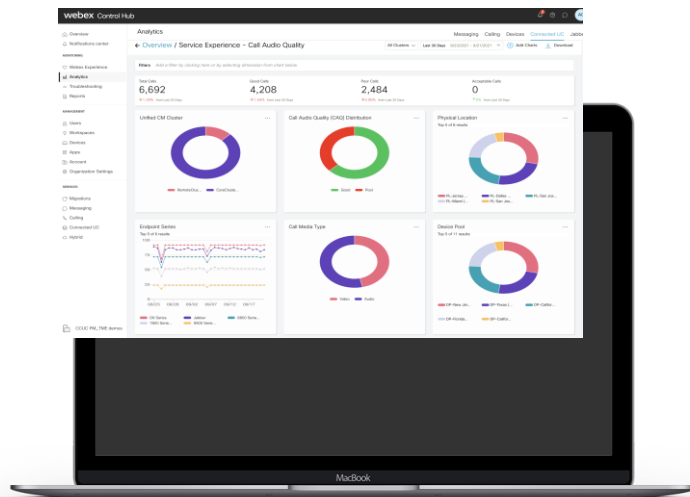
Cloud-Connected UC Value Proposition

Webex Cloud-Connected UC (CCUC)

A suite of Webex® cloud services that provide analytics, monitoring, troubleshooting, certificate management workflows to manage on-premises Cisco UC deployments

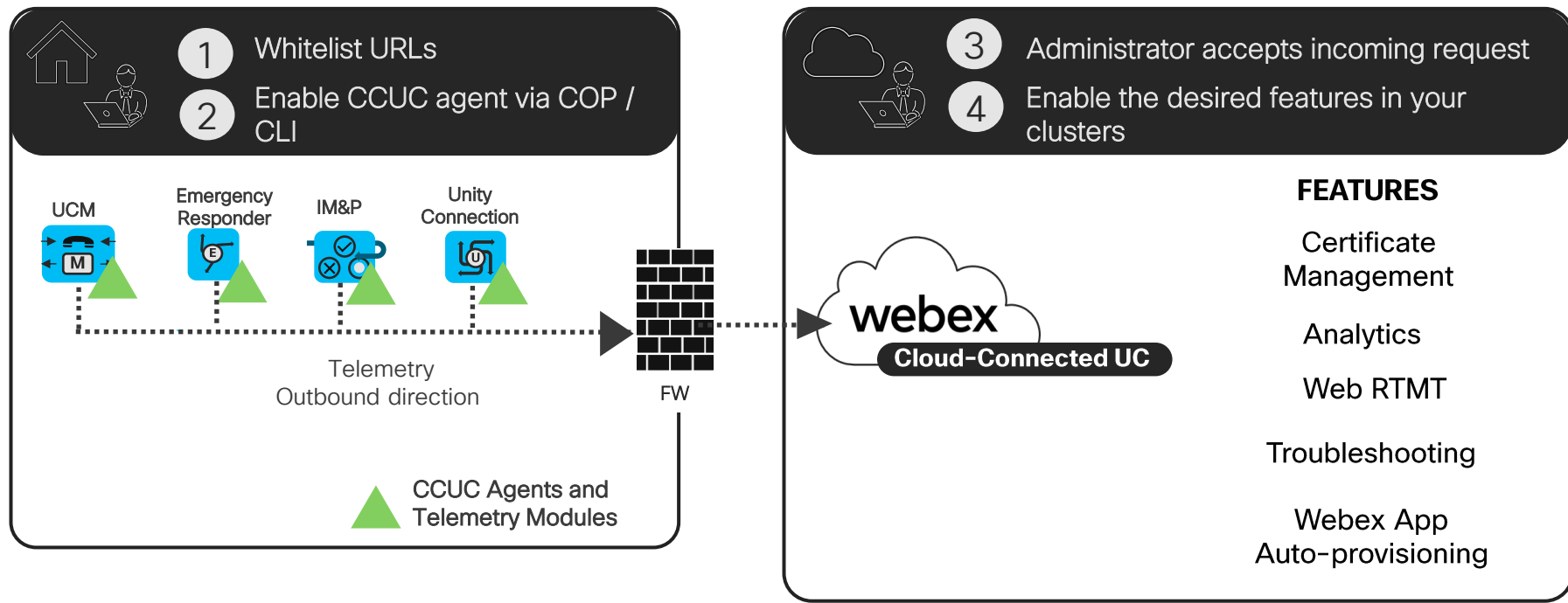
For customers who:

- Want the centralized administrative experience of the cloud, while keeping their UC services on-premises
- Desire a single pane of glass in Control Hub to manage mixed-mode deployments, such as Webex Calling, or Webex Meetings, with on-premises UCM
- Desire efficient, cloud-based and partner managed services for an on-premises UCM deployment



Webex Cloud-Connected UC

Solution overview



Webex CCUC value proposition – recap

Manage from the cloud

Feature velocity, added capabilities that are not feasible from current on-premises management platforms

Not a "transition-to-cloud" storyline, per se

Registration, signaling/call control, dial plans, conferencing resources, supplementary services, etc., all remain delivered from "Cisco UCM"

Future capabilities to aid in the cloud migration decision points

Minimal "cloud side" networking requirements

Outbound TLS connections over port 443

Specific URL's to allow

*.ucmgt.cisco.com

*.webex.com

Phased-in features deployed from the Webex® cloud

Once onboarded, CCUC agents and modules running on UC nodes are upgraded from the cloud

Modules would be added to bring about newer functionality

Onboarding

Onboarding Contents

01 CCUC Enablement

02 Pre-requisites

03 Onboarding flow with
Cop file

04 Onboarding flow CLI

05 Inventory Verification

06 Post install/assignment

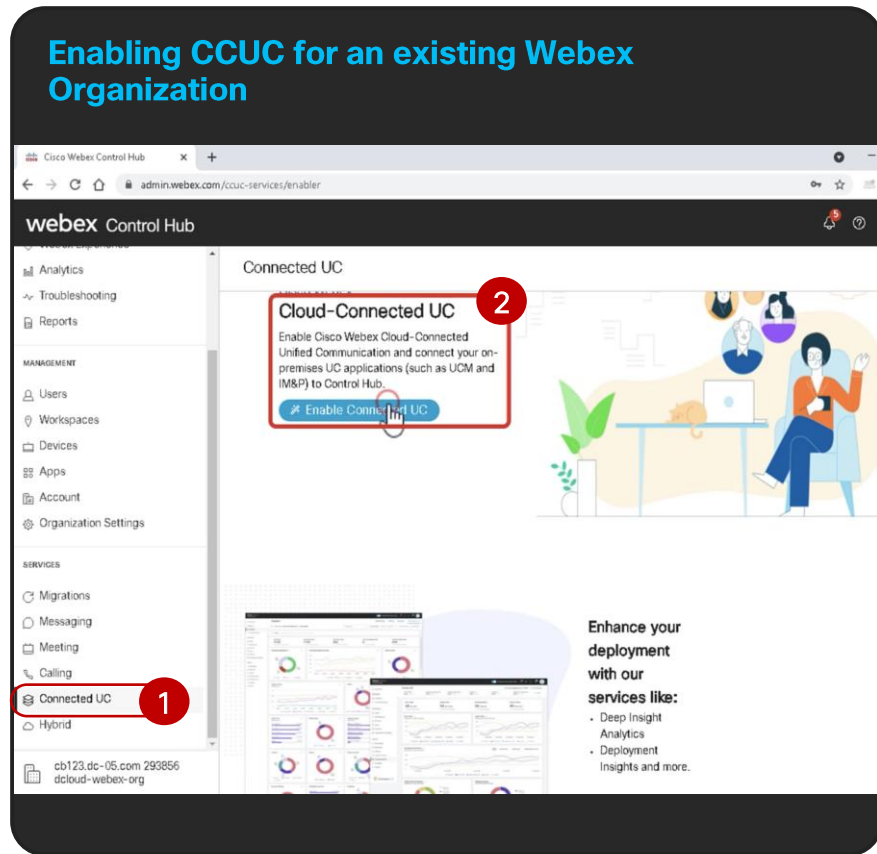
07 Im&p Onboarding

08 Server and Telemetry Module
Inventory

09 Recap of CCUC Onboarding

Two options for enabling Cloud-Connected UC

- No need to place an order. Customer with existing Control Hub/ organization ID can now enable CCUC directly under “Services > Connected UC > Enable Connected UC”
- Alternatively, new customers who do not have a Webex® Organization could place a \$0 CCW order



Data Collection Agreement

Service Management

Changes to these settings will take a short time to take effect.

Analytics	☒	Enabled
Certificate Management	☒	Enabled
Operational Metrics * #	☐	Disabled
Webex app Provisioning for Unified CM Calling *	☐	Disabled
Deployment Insights *	☐	Disabled

* Services require data collection of usage. Enabling these services will require explicit agreement to this collection.

By enabling this service, and using Troubleshooting Feature, additional data may be processed and stored outside your region as set forth in the Cisco Technical Assistance (TAC) Service Delivery [Privacy data sheet](#).

Data Collection Confirmation

One or more of the services that you selected require additional data collection. For privacy details of the information collected, see [Privacy data sheet](#).

This is a one time confirmation and applicable for the entire organization. The operation cannot be undone.

☐ Yes, I agree.

CancelSubmit

1

Enable the desired services in the cluster

Note: After cluster onboarding Analytics and Certificate Management are enabled by default

Agree Additional Data Collection

2

3

Submit changes

Control Hub roles for CCUC

CCUC Function	Control Hub Role		
	Full Admin	Read only	Support
Onboarding UC nodes: prepare agent install files, create cluster groups, verify nodes and clusters	✓	✗	✗
View onboarded UC infrastructure	✓	✓	✗
View Analytics Inventory	✓	✓	✗
Connected UC Analytics: up to 13 months of data with interactive charts, filters, ability to download reports as CSV	✓	✓	✓

Prerequisites

Cloud-Connected UC

UC infrastructure minimal versions

Cisco UCM, IM&P, Unity and CER: 11.5(x), 12.0(x), 12.5(x), 14

COP file to be installed except with release 12.5(1)SU4+ and 14

Network port requirements

TCP Port 443 outbound to the Webex®

Configure firewall to allow access to the following URLs:

- *.ucmgt.cisco.com
- *.webex.com

Network proxy support

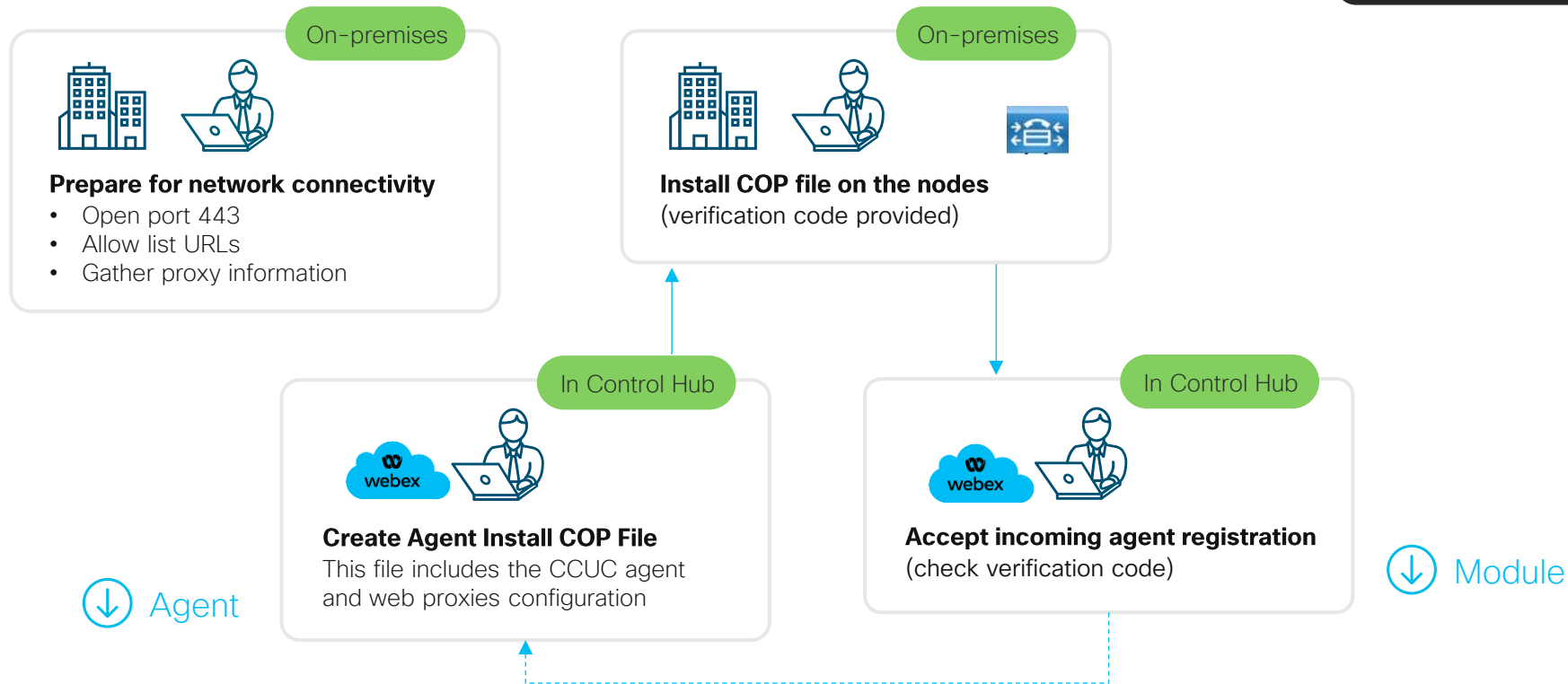
Unauthenticated and authenticated http proxies are supported

Onboarding flow is triggered by a "full admin" in Control Hub, who should preferably (but not necessarily) also be an "OS admin" for the UC nodes, so they can install COP file for CCUC agent

See [CCUC System Requirements](#) and [CCUC Network Requirements](#)

Onboarding flow with COP file

Onboarding flow with COP file



Onboarding Procedure with COP file

1. Create first agent file and cluster group
2. Verify and assign clusters
3. Set-up agents and cluster groups
4. View the analytics status of node inventory
5. Manage agents and cluster groups
6. Start and stop the Telemetry Service

See [Set up CCUC for on-premises devices](#)

First time wizard for onboarding COP File

The image displays three overlapping screenshots of the Cisco Webex Control Hub interface, illustrating the first-time wizard for onboarding a COP File.

1. Main Dashboard: The top screenshot shows the Cisco Webex Control Hub main dashboard. The left sidebar contains navigation links: Stratus Organiz..., Overview, Users, Workspaces, Services (highlighted), Devices, Analytics, Troubleshooting, and Settings. The main content area is titled 'Services' and shows 'Cloud-Connected UC' with a 'UC Management' card. The 'UC Management' card includes the text 'Create Install Agents and define Cluster Groups to begin connecting your UCM instances to cloud' and a 'Get Started' button.

2. Getting Started Page: The middle screenshot shows the 'Getting Started' page. It features a 'Welcome to Cisco Control Hub' message and instructions: 'Use this guide to create the basic components needed to connect and manage your on-premise appliances from the Cisco Control Hub Portal.' Below this, it states: 'Once complete you will be able to enable additional features to greater enhance the administrative experience. For additional information view these [Cisco Articles](#).' An illustration of a laptop and a coffee cup is shown at the bottom.

3. Getting Started Form: The right screenshot shows the 'Getting Started' form. It includes a progress bar at the top. The form contains the following sections:

- Filename:** A text input field with the placeholder 'Enter a filename'.
- File Description (Optional):** A text input field with the placeholder 'Optional File Description'.
- HTTP Proxies:** A text input field with the placeholder 'Enter HTTP proxies (i.e http(s)://IP Address or domain:port)'.

At the bottom right, there are 'Back' and 'Save' buttons.

Onboarding flow (1)

Control Hub > Services > Cloud-Connected UC > UC Management

The screenshot displays the Cisco Webex Control Hub interface. The left sidebar contains a navigation menu with the following items: Organization Health, Analytics, Troubleshooting, MANAGEMENT (Users, Workspaces, Devices, Apps, Account, Organization Settings), SERVICES (1), Updates, Messaging, Meeting, Calling, Connected UC (2), and Hybrid. The main content area is titled 'Connected UC' and contains a 'UC Management' section. This section includes a red-bordered box with the following items: Inventory, Agent Install Files (3), and Telemetry Module Inventory. The top right of the interface shows a notification bell with a red '3', a help icon, a chat icon, and a user profile icon labeled 'AC'.

Cisco Webex
Control Hub

Organization Health
Analytics
Troubleshooting

MANAGEMENT

- Users
- Workspaces
- Devices
- Apps
- Account
- Organization Settings

SERVICES 1

- Updates
- Messaging
- Meeting
- Calling
- Connected UC 2
- Hybrid

Connected UC

UC Management

- Inventory
- Agent Install Files 3
- Telemetry Module Inventory

3

AC

Onboarding flow (2)

Create Agent COP file and include proxy support if required

Create Agent Install File

Create an Agent Install file (.cop file). This file will be installed on each Unified Communications instance to connect the instance to UC management . Creation can take several minutes. When it's done, the file is available on the Agent Install File page to download and install.

Filename

ccuc-analytics-cucm

File Description (Optional)

ccuc-analytics-cucm

HTTP Proxies:

https://proxy1.sjc.cisco.com:80 x Enter HTTP proxies (i.e http(s)://IP Address:Port)

Optional: To use a proxy that supports authentication, use the Admin CLI on all Unified CM and IM&P server to configure it.

Cancel **Save**

UC Management

File Name ▾

Agent_Install_File

Alf-esl

test

Organization Settings

You can specify one or more HTTP proxies, one for each site.

If using an authenticated proxy, enter the proxy information later via CLI

Install Files

Add File

Supported Connections:

- No proxy
- Proxy no auth
- Proxy Basic auth
- Digest auth (From UC 14 version)

Created Date ▴	Action
12-01-2020 22:59:19	Download
04-15-2021 15:38:32	Download
05-28-2021 20:25:16	Download

Onboarding flow (3)

Download Agent COP file

webex Control Hub

UC Management

File Name **All 1**

AgentInstallFile

Agent File Download

Download the Agent file for installation on the corresponding version of the Unified CM, IM & Presence, Cisco Unity Connection, or Cisco Emergency Responder server.

Download Version	Actions
Version 11.5-11.5(SU6), Version 12.0.x, Version 12.5-12.5(SU1)	Download
Version 11.5(SU7)-11.5(SU9), Version 12.5(SU2)-12.5(SU3), Version 14	Download

Inventory **Install Files**

Created Date **Action**

03-11-2021 22:58:03 **Download**

Opening ciscocm-ucmgmt-agent-ccuc-analytic...

You have chosen to open:

...tics-cucm.1536827.v20120514-520.k3.cop.sgn
which is: DOCX file (2.1 MB)
from: ...production-installable.s3.amazonaws.com

What should Firefox do with this file?

☐ Open with **Choose...**

☒ Save File

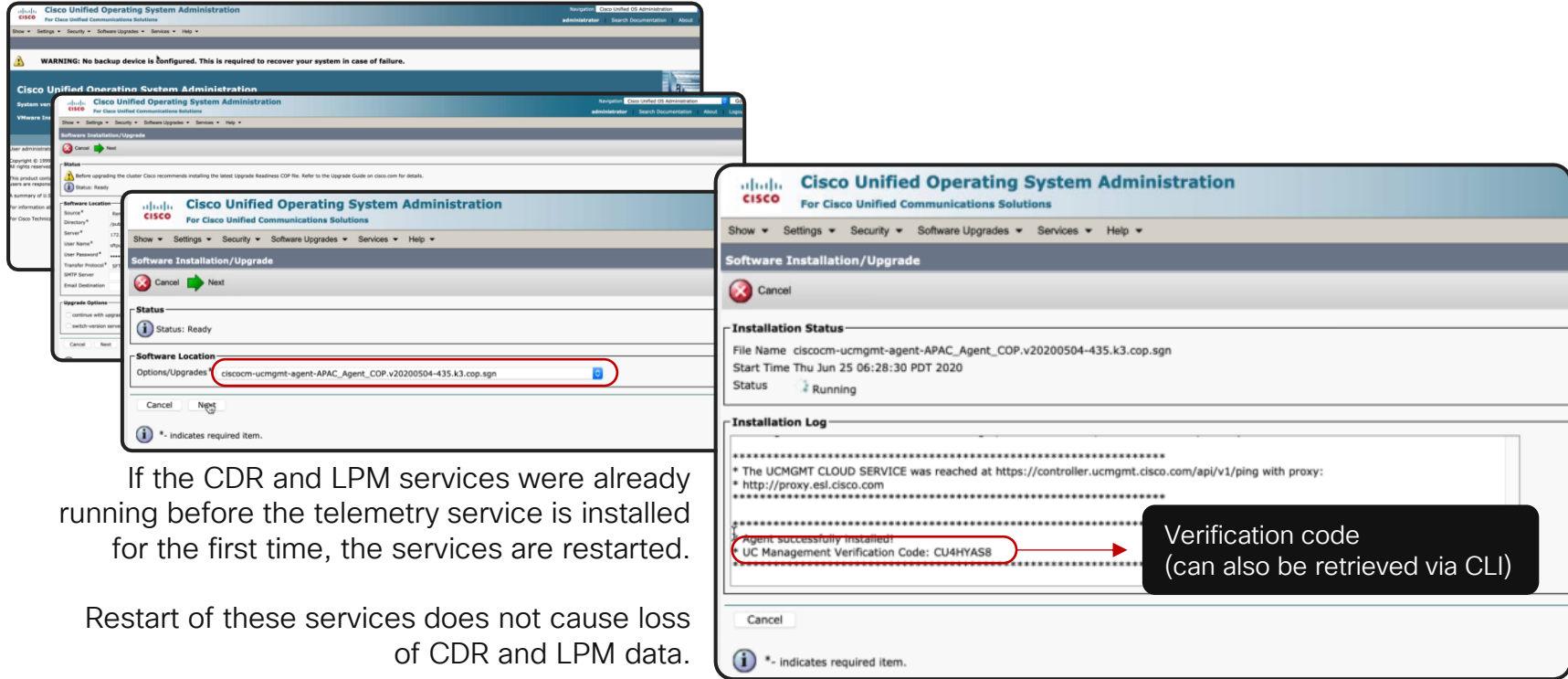
☐ Do this automatically for files like this from now on.

Cancel OK

Agent Install file is specific to the Control Hub org
The agent file can take up to a minute to generate.

Onboarding flow (4)

Install Agent COP file on the on-premises nodes



If the CDR and LPM services were already running before the telemetry service is installed for the first time, the services are restarted.

Restart of these services does not cause loss of CDR and LPM data.

Verification code (can also be retrieved via CLI)

Onboarding flow (5)

In Control Hub, accept new agent registration

Cisco Webex Control Hub

< UC Management Inventory Install Files

CUCC PM_TME ... All 5 Show Cluster Groups Add Cluster Group

Cluster Group	Description	Status	Clusters	Nodes	
Unassigned	-	Needs Assignment	1	1	Resolve
Cloud Connected UC Demo	First group	Offline	1	3	Details

Cisco Webex Control Hub

< Unverified Cluster Group

Cluster Name	Nodes	Product Type	
KoaCluster - koa-pub.cisco.com	1	CUCM	Verify

Verify and Assign Cluster

Cluster Group: PA set-up

Node Name	IP Address	Product	Type	Verification
koa-pub.cisco.c...	10.195.38.55	CUCM	PRIMARY	6/PYHTK1

webex | Collaboration Technical Networking

Onboarding flow (6)

Authenticated proxy configuration (if applicable)

authenticated proxies are deployed:

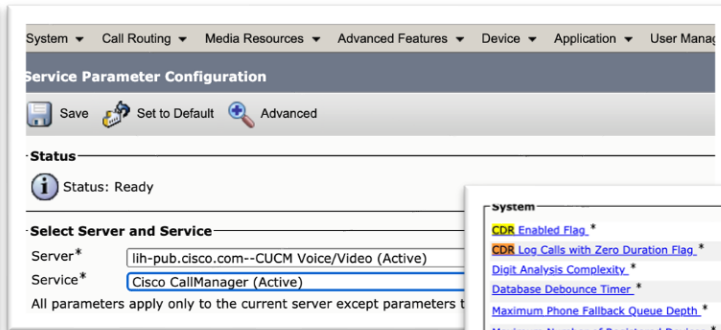
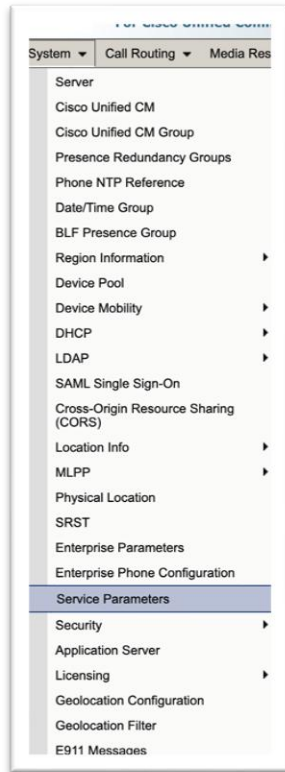
There is no need to configure the web proxy information when creating the Agent Install File

Once the Agent Install File is created and installed on the UC application server, use the CLI on the on-prem servers to add each authenticated web proxy information:

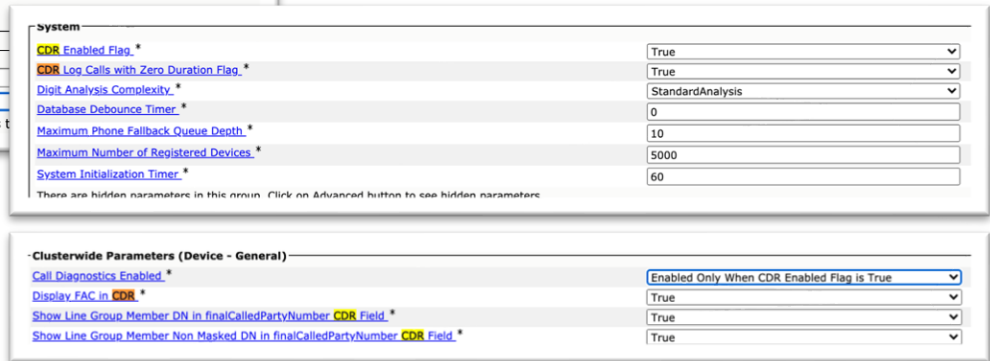
Proxy address and port

```
[admin:utils ucmgmt proxy add http://yourorgproxy.com:80 user1 ]  
[(Optional) Validating Proxy Password (won't display): ]  
[Re-enter Proxy Password (won't display): ]
```

UC server: enabling CDR/CMR

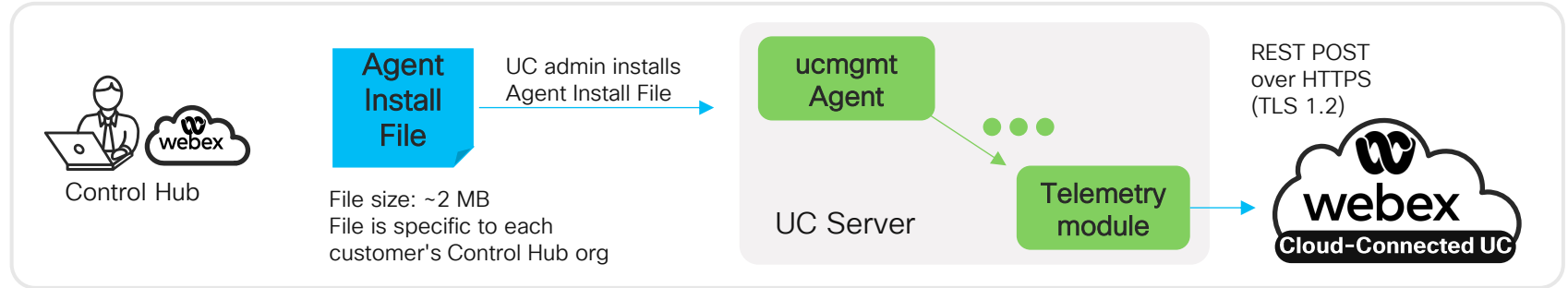


CDR/CMR must be enabled on the Cisco® UCM Publisher



<https://community.cisco.com/t5/collaboration-voice-and-video/understanding-cdr-call-detail-records/ta-p/3112476>

UC server "Telemetry Service"



CCUC "ucmgmt agent" installation (or activation) starts a "Telemetry Service" module on the UC Server.

The install also checks and verifies reachability to the "UC Management Cloud Service" (including through the proxy if configured on the "Agent Install File")

- "Telemetry Service" collects analytics data and sends it to the Webex® cloud
- From that point, this service is managed/upgraded from the cloud
- For future services: other "modules" may be spawned from the CCUC agent

UCM CLI – Basic CCUC agent sanity

Use UCM CLI commands to:

- Check ucmgmt agent status (incl. Control Hub verification code) as well as enable, disable, remove & restart agent
- Enable debug logging
- Validate or set Control Hub org
- Proxy: list, clear, add, force add

```
Command Line Interface is starting up, please wait ...
```

```
Welcome to the Platform Command Line Interface
```

```
VMware Installation:
```

```
2 vCPU: Intel(R) Xeon(R) CPU E5-2660 v3 @ 2.60GHz
```

```
Disk 1: 110GB, Partitions aligned
```

```
8192 Mbytes RAM
```

```
admin:utils ucmgmt
```

```
utils ucmgmt agent*
```

```
utils ucmgmt debug*
```

```
utils ucmgmt organization
```

```
utils ucmgmt proxy*
```

```
admin:utils ucmgmt agent status
```

```
Agent Config Information:
```

```
Verification Code: BWQIHPDP
```

```
Installed UCMGMT Agent version: 20200905-459
```

```
Agent's organization id: 22fd246b-9381-4f0b-ac0d-ee27ba85b8f2
```

```
Agent Runtime Information:
```

```
Agent process is running.
```

```
Agent process is enabled.
```

```
Agent watchdog is enabled.
```

```
Agent is verified in Webex Control Hub.
```

```
Recent agent logs:
```

```
2020-09-11T19:36:46Z cloud_agent (pid:2894) -> INFO Calling command handler: CHECK_IN_handler
```

```
admin:█
```

Performance considerations

Impact of cloud connectors on UC infrastructure nodes

CPU and memory (RAM) usage

The CCUC service runs with low priority. This helps in restricting the service to minimize CPU usage. The service runs with approximately 345 MB of RAM. The CPU usage is 12%.

Buffer size calculation details in [documentation](#)

Bandwidth usage

For a larger cluster, ensure that you have at least 2 Mbps of bandwidth to the cloud.

Disk Space Usage

Installed, the CCUC service takes about 155 MB of disk space for Voice Operating System (VOS) based products, such as Unified CM and IM & Presence. The space requirement for telemetry service is 3 MB for executables.

For log files, 10 MB in the /var/log/active folder path.

Call Detail Recording (CDR) and Call Management Recording (CMR)

Must be enabled in UCM (service parameter “CDR Enabled Flag” and “Call Diagnostics Enabled”)

To avoid delays or error in Webex Cloud-Connected UC analysis and operations, we recommend that you set CDR File Time Interval to 1.

Onboarding flow (No COP file)

Releases 12.5(1)SU4+ and 14

SSH Onboarding flow (No COP file)

```
admin:utils ucmgmt organization 22fd246b-9381-4f0b-ac0d-ee27ba85b8f2 1
admin:utils ucmgmt proxy add http://proxy.esl.cisco.com:80 2

Attempting to contact UCMGMT Cloud Controller... please wait (upto 30 seconds).
Successfully contacted controller.
{"serviceName": "Lookup Service", "serviceType": "REQUIRED", "serviceState": "online", "message": "Healthy",
"lastUpdated": "2021-07-19T22:05:10", "upstreamServices": []}

Adding http://proxy.esl.cisco.com:80 to proxy list.
```

- Configure org id
- Configure proxy if required*

```
admin:utils ucmgmt agent enable 3
Agent watchdog activated. Agent will start up shortly (~ 5 minutes).
admin:utils ucmgmt agent status
Agent Config Information:
Verification Code: STK+SKV1 4
Installed UCMGMT Agent version: 20210625-525
Agent's organization id: 22fd246b-9381-4f0b-ac0d-ee27ba85b8f2

Agent Runtime Information:
Agent process is running.
Agent process is enabled.
Agent watchdog is enabled.
Agent is not verified in Webex Control Hub.

Recent agent logs:
2021-07-19T22:13:37.999Z cloud_agent:process_handler_results (pid:16811) -> INFO Sending 'http_access --ti
meout 15 --proxy <(echo -n <SANITIZED> --header Content-type: application/json --header TrackingID: NA_b4565c
5e-40a9-4119-ad53-714cd48c8674 --header X-Cisco-Agent-Receipt: STK+SKV1:jPmpzKJGYTWK4h64rdYxEn4L93lhBgkE --me
thod "POST" --postData "<REDACTED>" --url "https://controller.na.ucmgmt.cisco.com/api/v1/unverified/agents/17
4445e3-76bf-40d6-8093-d350b04950b5/jobs/811c202c-870f-45a2-b045-02941ab5d288/result"'
```

- Enable ucmgmt agent
- Copy Verification Code

Note: Repeat in all nodes of the cluster

Inventory Verification

Inventory Verification – Verify Action

Admin has to click on "verify" to start the "verification" and "assignment" steps

"Verification" correlates the "UC Management Verification" code that was displayed after "Agent Install" file was successfully installed on the UC Server node

Control hub > Connected UC > Inventory > Unverified Clusters

< Unverified Cluster Group				
Cluster Name ▲	Nodes ▲	Product Type	Status	
LihCluster - lih-pub.cisco.com	2	CUCM,IM&P	● Needs Verification	Verify
StandAloneCluster - as-pub.cisco.com	1	CUCM	● Offline	Details

Inventory Verification – Assign Action

Verify and Assign Cluster

Cluster Name

LihCluster - lih-pub.cisco.com

Existing Cluster Group

PA

Change Cluster Group

Select An Option

Node Name	IP Address	Product	Type	Verification
lih-pub.cisco.com	10.195.38.50	CUCM	PRIMARY	STK+SKV1
lih-imp-pub.cisco.com	10.195.38.52	IM&P	PRIMARY	DH+OA.JC5

Cluster group: a logical group of UCM clusters for effective management; allows an admin to view the overall status and inventory of all the clusters under that cluster group.

"Assignment" maps UC Server nodes/clusters to "Cluster Groups"

Cluster group status

Post install/assignment

< UC Management Inventory Install Files

Search: All 6 Show Cluster Groups Add

Cluster Group	Description	Agent/Connector Status	Telemetry Module Status	Clusters	Nodes	
Cloud Connected UC ...	First group	Offline	Offline	1	3	Details
PA	PA lab	Online Info	Online Info	1	2	Details
SL Catalog Group	SL Catalog Group	Online	Online	3	13	

< Inventory

Cluster Group Name: PA ✎ Events History

Group Description: PA lab

☐ Cluster Name	Nodes	Product	Version	nodesListView.gridHeaderTitle.sta...	Telemetry Module Status	
☐ LihCluster - lih-pub.cisco.com	2	IM&PCUCM	12.5.1.14900-4	Online Info	Online	Details

Service Management

Enabling Services:



< tmedemo-cluster - cm-pub.tr

Cluster Group Name

Services

Node Name ▲ Node Version ...

cm-pub.tmedemo.com	14
cm-sub1.tmedemo.com	14
imp-sub1.tmedemo.com	14
cm-sub2.tmedemo.com	14
mr1-cm-sub3.tmedemo.cor	14
mr2-cm-sub4.tmedemo.cor	14
imp-pub.tmedemo.com	14

Core

Enabl

Disabl

10.50.55.25

IM&P

PUBLISHER

KE6YKMGY

Online

Service Management

Changes to these settings will take a short time to take effect.

Analytics

☒

Enabled

Directory Service

☒

Enabled

Certificate Management

☒

Enabled

Operational Metrics

☒

Enabled

Webex app Provisioning for Unified CM Calling

☒

Enabled

Deployment Insights

☐

Disabled

Cancel

Submit

Events History

App Provisioning for Unified CM Calling

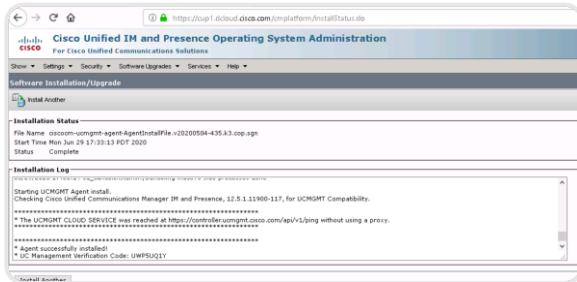
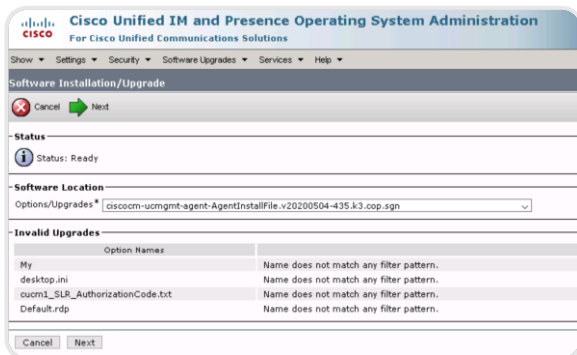
Events

Last Connection

Info	03-29-2022 23:53:31
Info	03-29-2022 23:52:57
Info	03-29-2022 23:53:15
Info	03-29-2022 23:53:25
	11-26-2021 19:12:15
Info	03-29-2022 23:53:43
Info	03-29-2022 23:52:51

IM&P onboarding

Same onboarding flow for
Cisco® UCM



IM&P Server will appear in
"Analytics Inventory"

< cucm1.dcloud.cisco.com

Cluster Group Cluster Group 2 ▾

Node Name	Node Version	IP Address	Service	Type	Verification	Status
cucm1.dcloud.cis...	12.5	198.18.133.3	CUCM	PRIMARY	X6ARYYYL	Offline
cup1.dcloud.cisc...	12.5	198.18.133.4	IM&P	SECONDARY	UWP5UQY1Y	Online
cucm2.dcloud.cis...	12.5	198.18.133.219	CUCM	SECONDARY	QUN8BGHZ	Offline

< Analytics Inventory

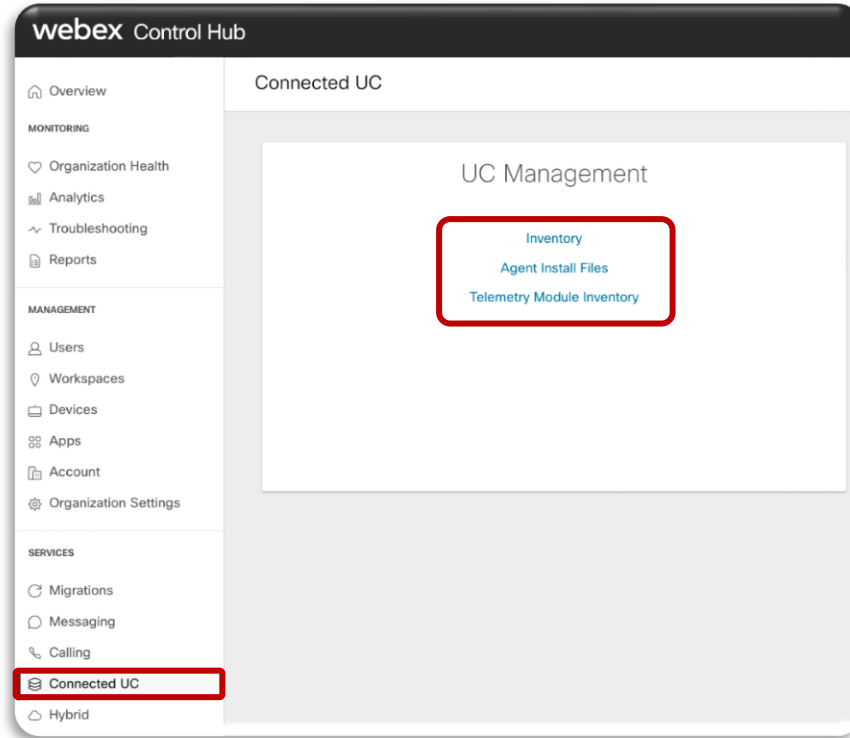
🔍 All 5

Node Name	Cluster Name	Cluster Group Name	Analytics Status	Last Connection
lih-pub.cisco.com	lih-pub.cisco.com	PA test set-up	Online	06-26-2020 16:45:27
cuc1.dcloud.cisco.com	cuc1.dcloud.cisco.com	Cluster Group 2	Unsupported	06-30-2020 00:10:08
cucm1.dcloud.cisco.com	cucm1.dcloud.cisco.com	Cluster Group 2	Offline	06-30-2020 00:33:11
cup1.dcloud.cisco.com	cucm1.dcloud.cisco.com	Cluster Group 2	Waiting for installation	06-30-2020 00:37:19
cucm2.dcloud.cisco.com	cucm1.dcloud.cisco.com	Cluster Group 2	Offline	06-30-2020 00:33:11

Server and Telemetry Module Inventory

Inventory

Control Hub > Services > Cloud-Connected UC > Inventory



Inventory status

UC Management						
Inventory Install Files						
All 4 Show Cluster Groups Add						
Cluster Group	Description	Agent/Connector Status	Telemetry Module Status	Clusters	Nodes	
Expressway	Expressway Devices	Online	Online	1	2	Details
PA	PA lab	No Agent running	Partially Online	2	6	Details
Cloud Connected UC Demo	First group	Offline	Offline	1	3	Details
SL Catalog Group	SL Catalog Group	Online	Online	3	13	Details

"Waiting for Installation" status: It takes around 2 hours for the installation of "telemetry service" module on UC Server to complete, then "Analytics Status" becomes "Online"

Telemetry Module Inventory

< Telemetry Module Inventory

All 7

Node Name ▲	Node Versio...	Cluster Name ▲	Cluster Group Name ▲	Produ..
cm-pub.tmedemo.com	14	tmedemo-cluster - cm-pub.tmedemo.con	Core Cluster	CUCM
cm-sub1.tmedemo.com	14	tmedemo-cluster - cm-pub.tmedemo.con	Core Cluster	CUCM
imp-sub1.tmedemo.com	14	tmedemo-cluster - cm-pub.tmedemo.con	Core Cluster	IM&P
cm-sub2.tmedemo.com	14	tmedemo-cluster - cm-pub.tmedemo.con	Core Cluster	CUCM
mr1-cm-sub3.tmedemo.com	14	tmedemo-cluster - cm-pub.tmedemo.con	Core Cluster	CUCM
mr2-cm-sub4.tmedemo.com	14	tmedemo-cluster - cm-pub.tmedemo.con	Core Cluster	CUCM
imp-pub.tmedemo.com	14	tmedemo-cluster - cm-pub.tmedemo.con	Core Cluster	IM&P

cm-pub.tmedemo.com

Node Details

Node Type

PUBLISHER

Node Version

14

Cluster Name

tmedemo-cluster - cm-pub.tmedemo.com

Cluster Group Name

Core Cluster

Agent Version

20220118-572

Agent Status

Online

Agent Last Connection

03-29-2022 23:56:37

Telemetry Module Details

Telemetry Module Version

1153

Telemetry Module Status

Online

Telemetry Module Last Connection

03-29-2022 23:56:27

Telemetry Module Action

Pause CCUC Services

"Waiting for Installation" status: It takes around 2 hours for the installation of "telemetry service" module on UC Server to complete, then "Analytics Status" becomes "Online"

Recap of CCUC onboarding

Onboarding UCM on-premises servers DOES NOT onboard users or devices into Control Hub, or any of its associated "cloud" services!

Endpoints (including Cisco® Jabber) and infrastructure nodes remain on-premises

CDR/CMR must be enabled on the Cisco® UCM Publisher

Noted in detail in "CCUC System Requirements"

CCUC enablement requires Data Collection Agreements to be accepted in Control Hub

"Agent File" COP file install does not require node restart. If the CDR and LPM services were already running before the telemetry service is installed for the first time, the services are restarted.

After first install, "Agent" and associated modules (e.g. telemetry service) upgrade and maintenance is orchestrated from the cloud

Onboarding flow for UC servers running 12.5SU4+ & 14 does not require COP file.

Analytics

Analytics Contents

01 Analytics Features

02 Basic Structure

03 Analytics Categories

Cloud-Connected UC analytics features

Asset usage and inventory

- Number of calls, talk time and usability

Quality of experience

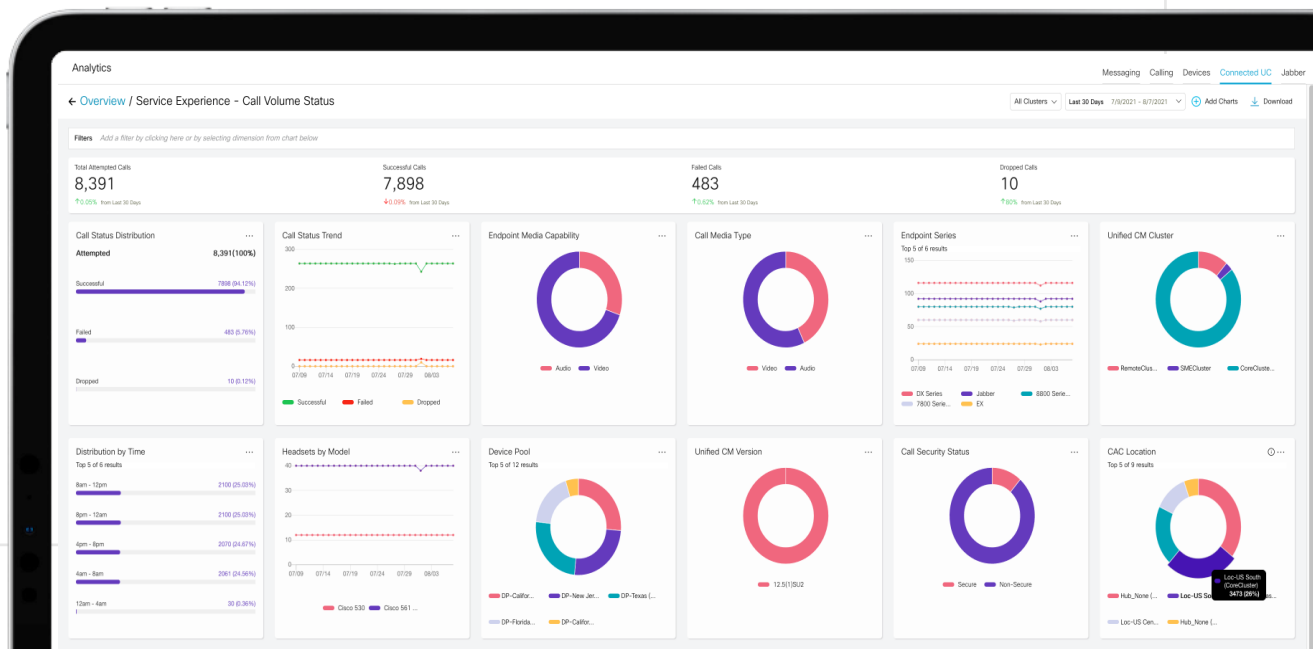
- Call success & failures
- Call quality metrics

Capacity analysis

- Trunk utilization

Traffic analysis

- Per call types



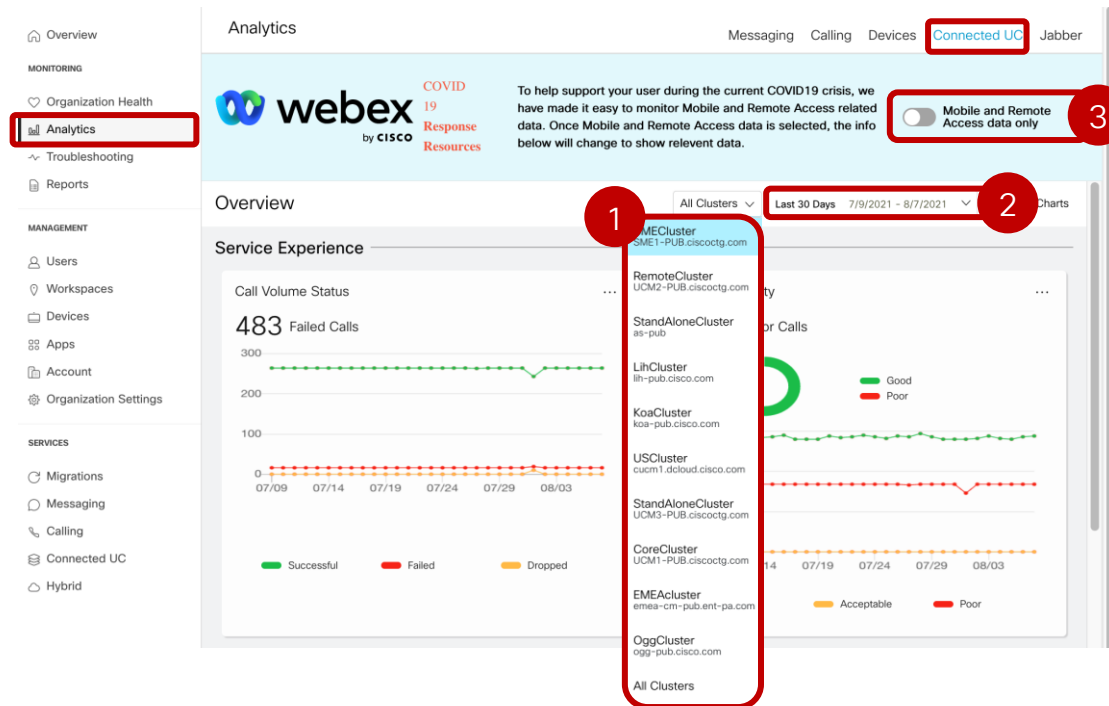
Analytics Basic Structure

Analytics: Basic structure

1 Cluster selector 2 Calendar selector 3 MRA filter toggle

Navigate to Monitoring > Analytics > Connected UC

Consistent "look and feel" and basic feature navigation to the rest of Control Hub Analytics



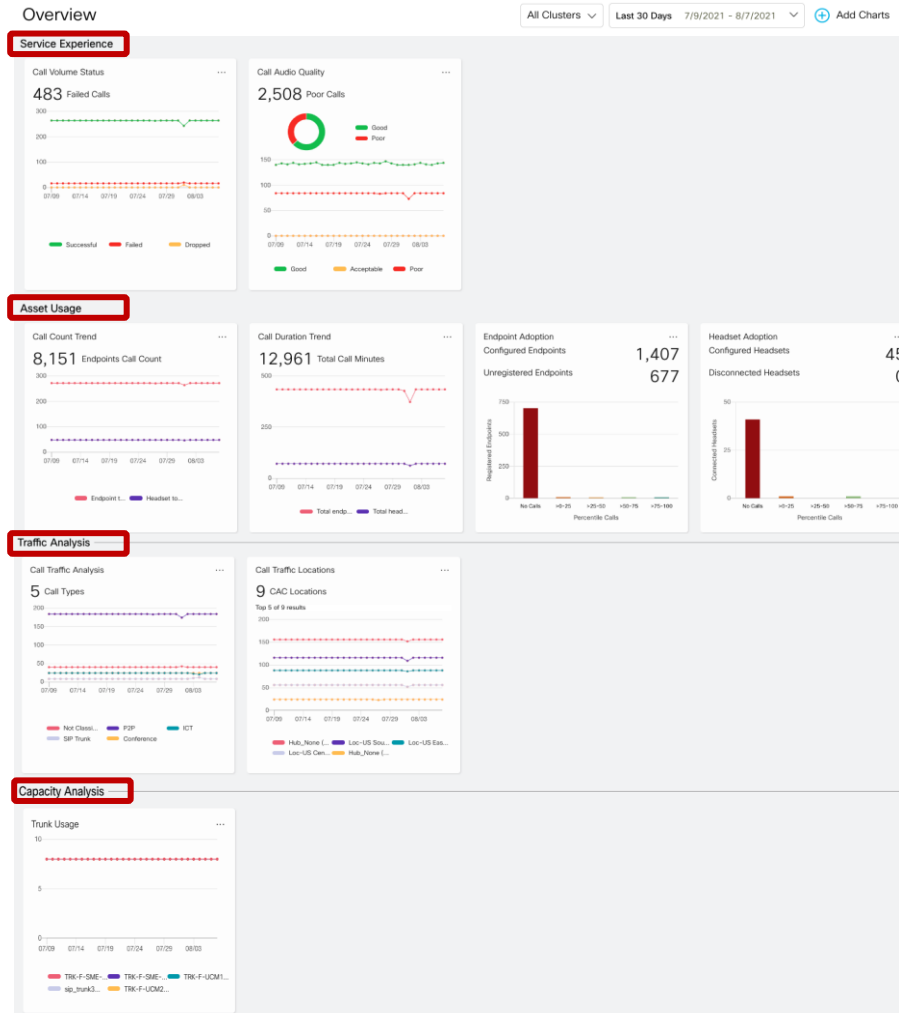
Analytics Basic Structure

4 Analytics categories

Navigate to Monitoring > Analytics > Connected UC > Overview

Overview page broken into four categories, with the option to click on each graph to drill down and display more specific data, based on these categories:

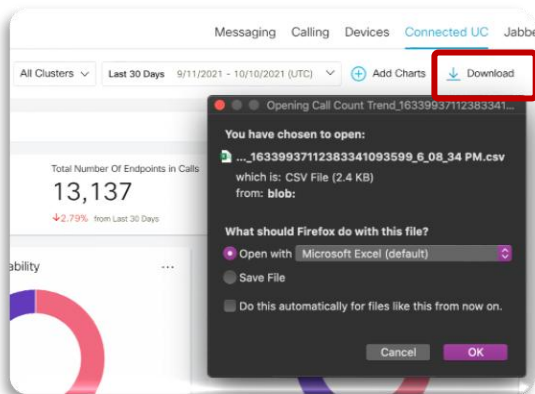
- Service Experience
- Asset Usage
- Traffic Analysis
- Capacity Analysis



Analytics Basic Structure

5 Filters

6 Download Data



← Overview / Asset Usage - Call Count Trend

Filters Add a filter by clicking here or by selecting dimension from chart below.

Mobile and Remote Access

User ID

URI

Phone Number

Calls Distribution by Asset

Endpoint Media Capability

Call Media Type

Headsets Call Count

1,416

↓0.49% from Last 30 Days

7779 (100.00%)

Analytics

← Overview /

User ID =

User Filters

User ID, URI, Phone Number

Q ansa

ansancho

ansancho2

Endpoints

7,77

↓3.56% from Last 30 Days

Headsets Call Count

,416

↓0.49% from Last 30 Days

Analytics Categories

Analytics Categories

Service Experience

Call Volume Status

Call Audio Quality

Traffic Analysis

Call Traffic Analysis

Call Traffic Locations

Asset Usage

KPI's for endpoints and headsets:

- Call Count Trend
- Call Duration Trend
- Endpoint Adoption
- Headset Adoption

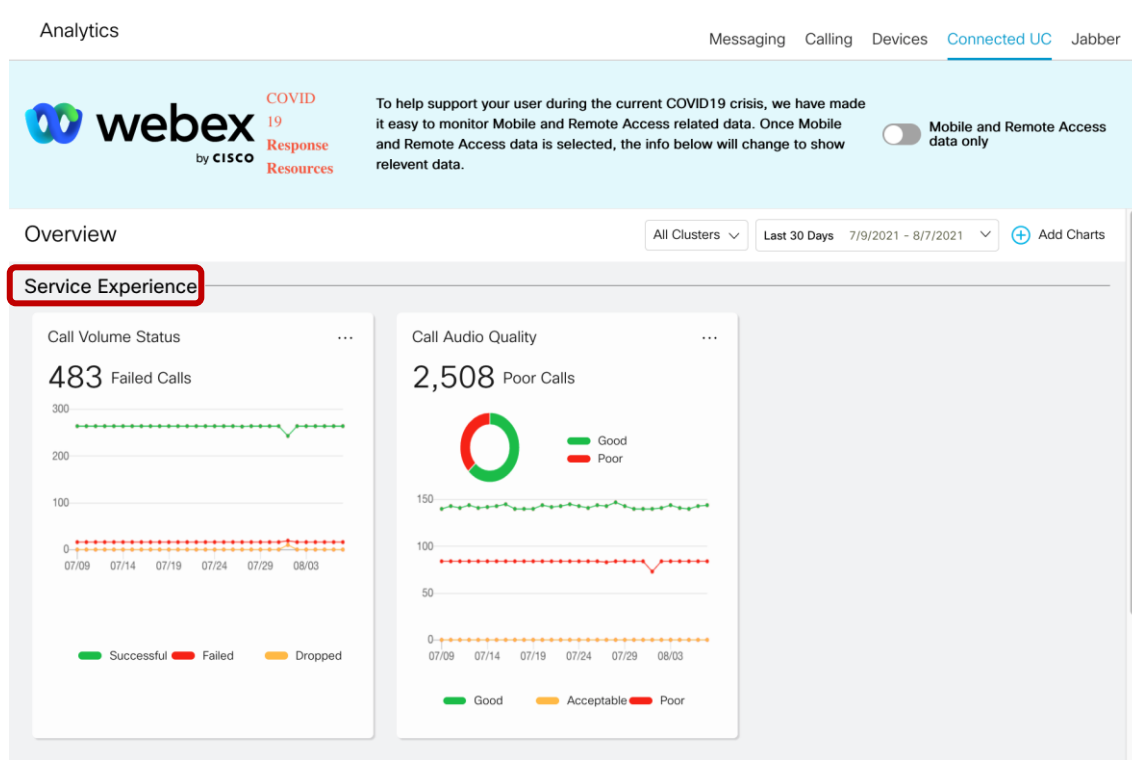
Capacity Analysis

Trunks Usage

CCUC analytics – service experience

Summary charts for

- Call Volume Status
- Call Audio Quality
- Drill down into each of the charts for more details



CCUC analytics – service experience

Call volume status

KPI's for calls: attempted, successful, failed, and dropped

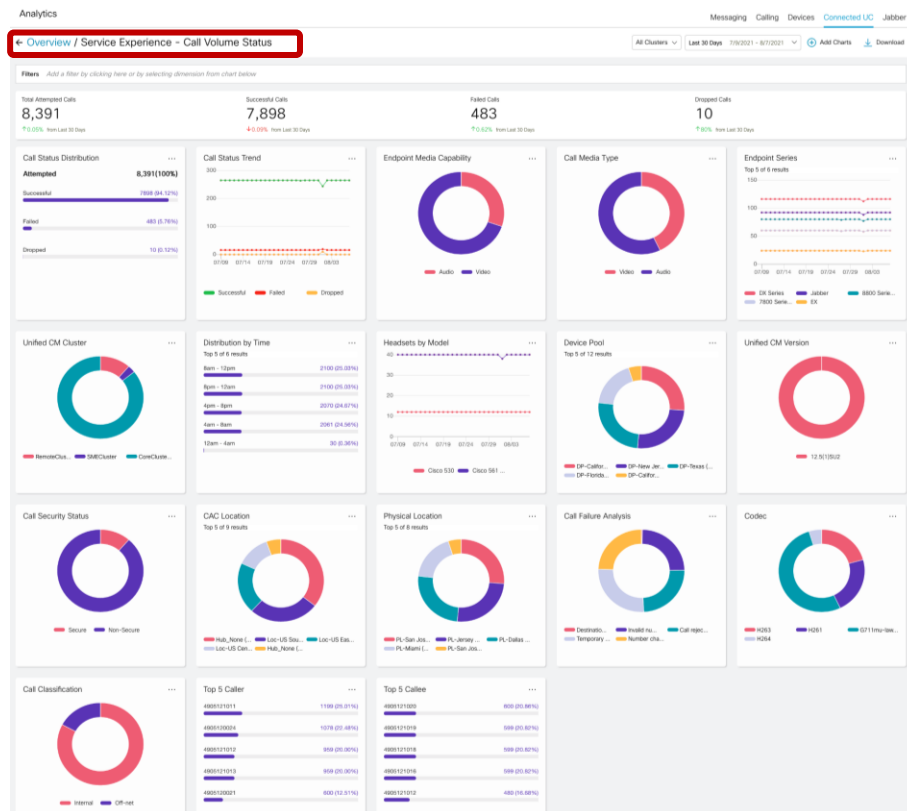
- Dropped call: any call > 0 min in duration and then failed

Interactive charts and graphs

- Call trends, endpoint media, type of endpoints and headsets, Cisco® UCM cluster, distribution of calls over time, device pools, CAC location bandwidth, etc.

"Filters" to guide admin on a particular aspect – all charts and graphs update accordingly

Ability to download CSV: snapshot of the entire data set, or of the filtered data view



CCUC analytics – service experience

Call audio quality

KPI's for total calls, then calls that were good/poor/acceptable

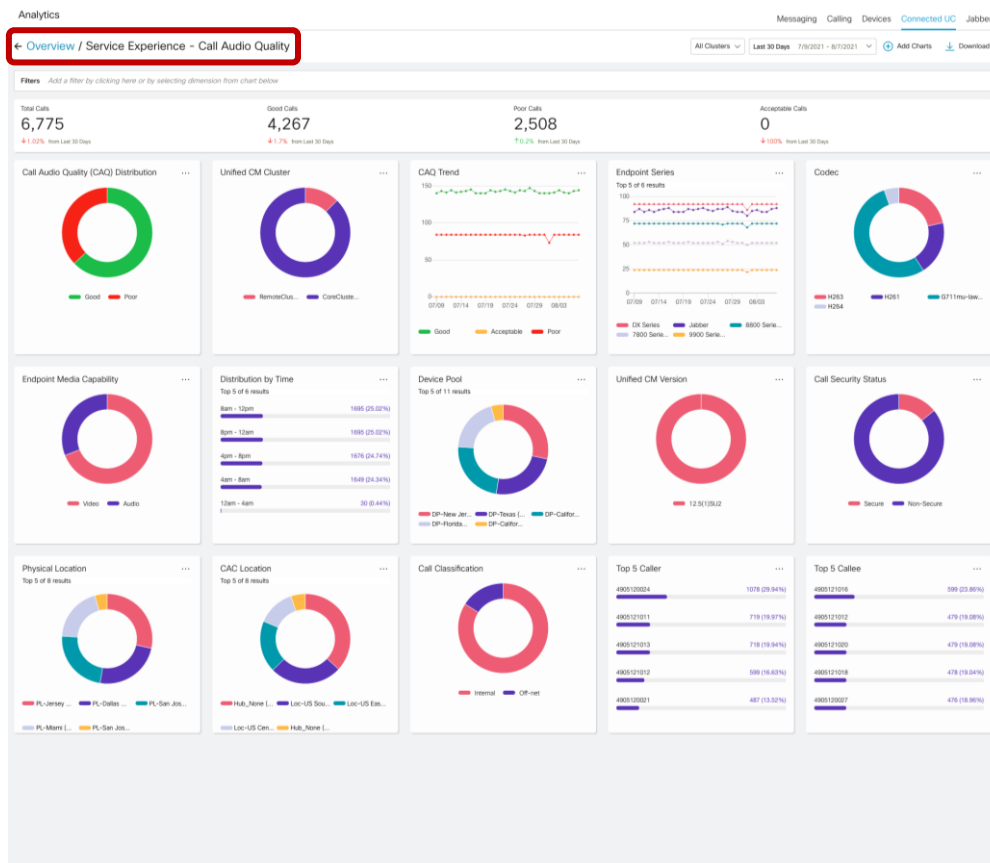
- Call Audio Grading is based on:
 - 1) Severely Concealed Seconds Ratio (SCS / Duration of Call)
 - 2) SCS information is available from CMR
 - 3) Thereby, only those calls which have a CMR will be graded for Call Audio Quality

Interactive charts and graphs

- Call audio quality trends, endpoint media, type of endpoints and headsets, Cisco® UCM cluster, distribution of calls over time, device pools, physical location, etc.

"Filters" to guide admin on a particular aspect – all charts and graphs update accordingly

Ability to download CSV: snapshot of the entire data set, or of the filtered data view



CCUC analytics – asset usage

**KPI's for endpoint and headsets
call counts, call minutes, and total
number in calls**

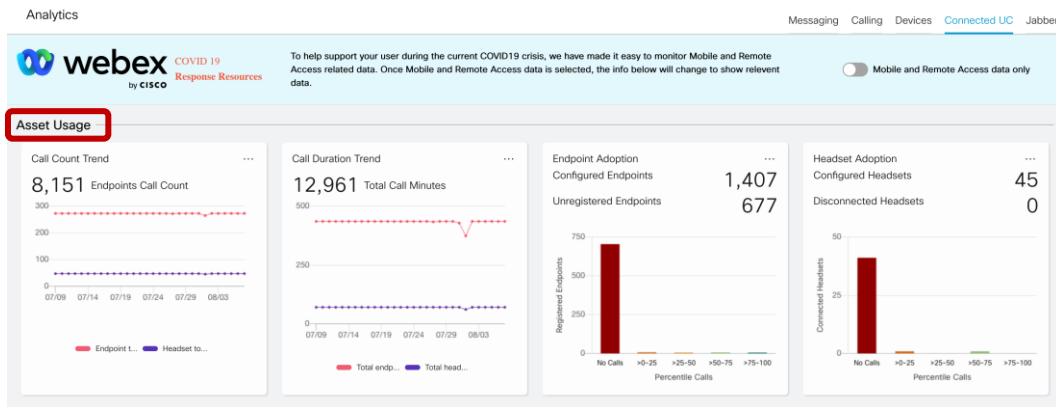
Summary charts for

- Call Count Trend
- Call Duration Trend
- Endpoint Adoption
- Headset Adoption
- Drill down into each of the charts for more details

**Summary view for "all clusters" or
focus on a particular one**

Date selector

Filter for "MRA data only"



CCUC analytics – asset usage

Call count trend

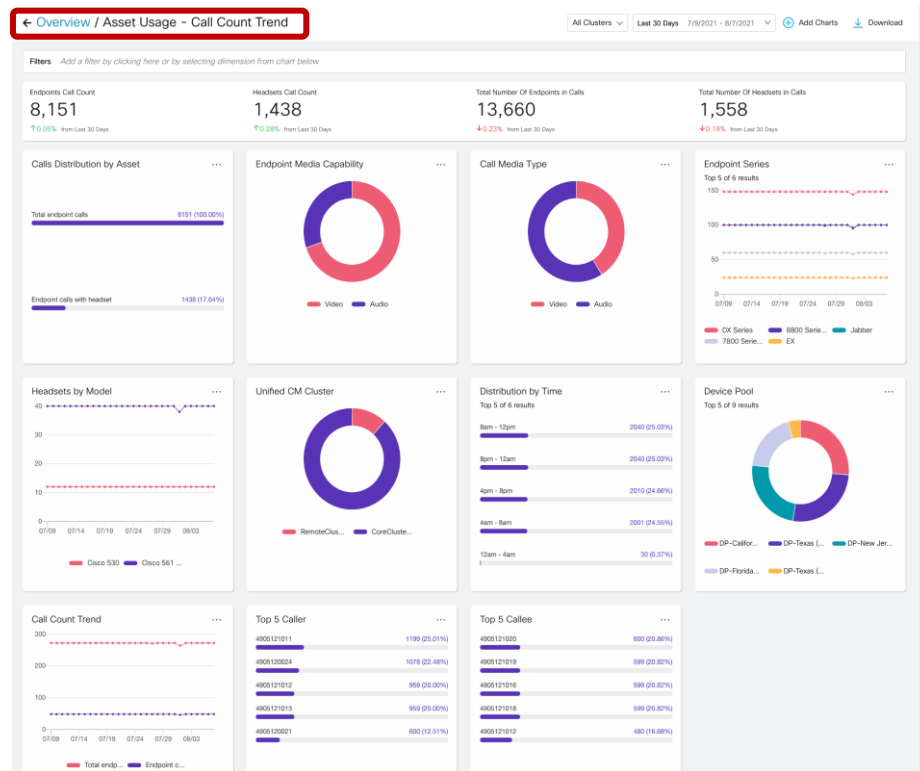
KPI's for endpoint and headset call count, number of endpoints and headsets in calls

Interactive charts and graphs

- Call distribution by asset, endpoint media, type of endpoints and headsets, Cisco® UCM cluster, distribution of calls over time, device pools, physical location, CAC location bandwidth, etc.

"Filters" to guide admin on a particular aspect – all charts and graphs update accordingly

Ability to download CSV: snapshot of the entire data set, or of the filtered data view



CCUC analytics – asset usage

Call duration trend

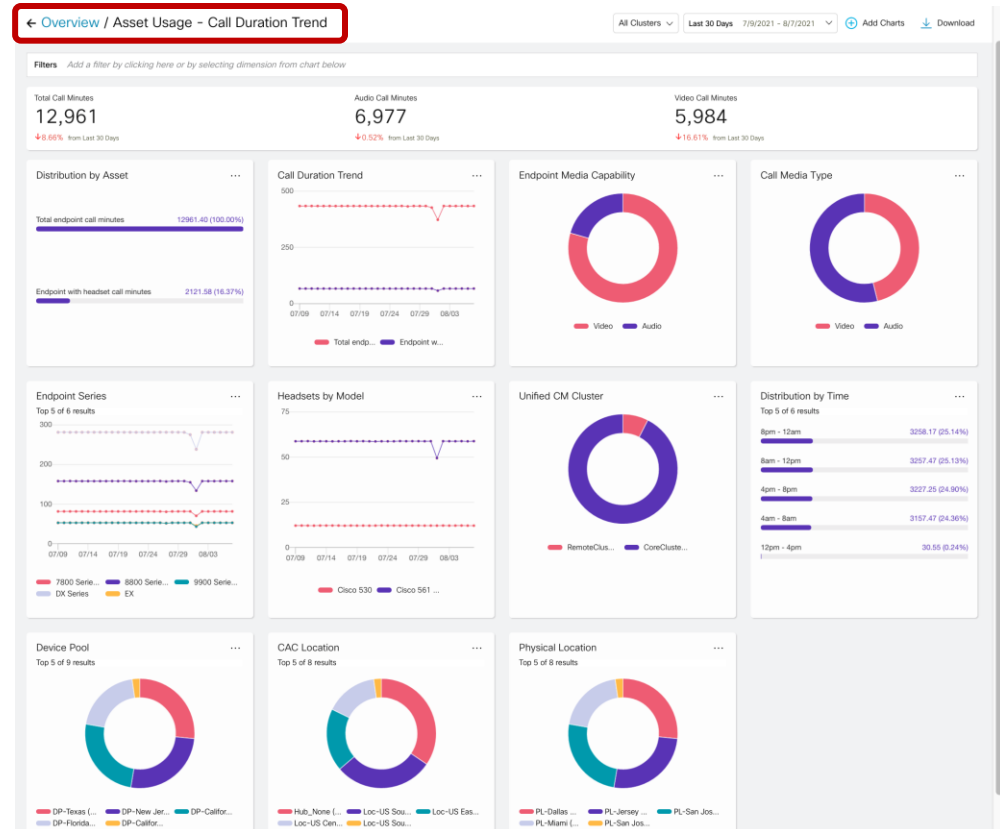
KPI's for total minutes, audio and video call minutes

Interactive charts and graphs

- Call distribution by asset (endpoint call minutes and endpoints that have headset call minutes), duration trend graph, endpoint media, type of endpoints and headsets, Cisco® UCM cluster, distribution of calls over time, device pools, physical location, CAC location bandwidth, etc.

"Filters" to guide admin on a particular aspect – all charts and graphs update accordingly

Ability to download CSV: snapshot of the entire data set, or of the filtered data view



CCUC analytics – asset usage

Endpoint Adoption

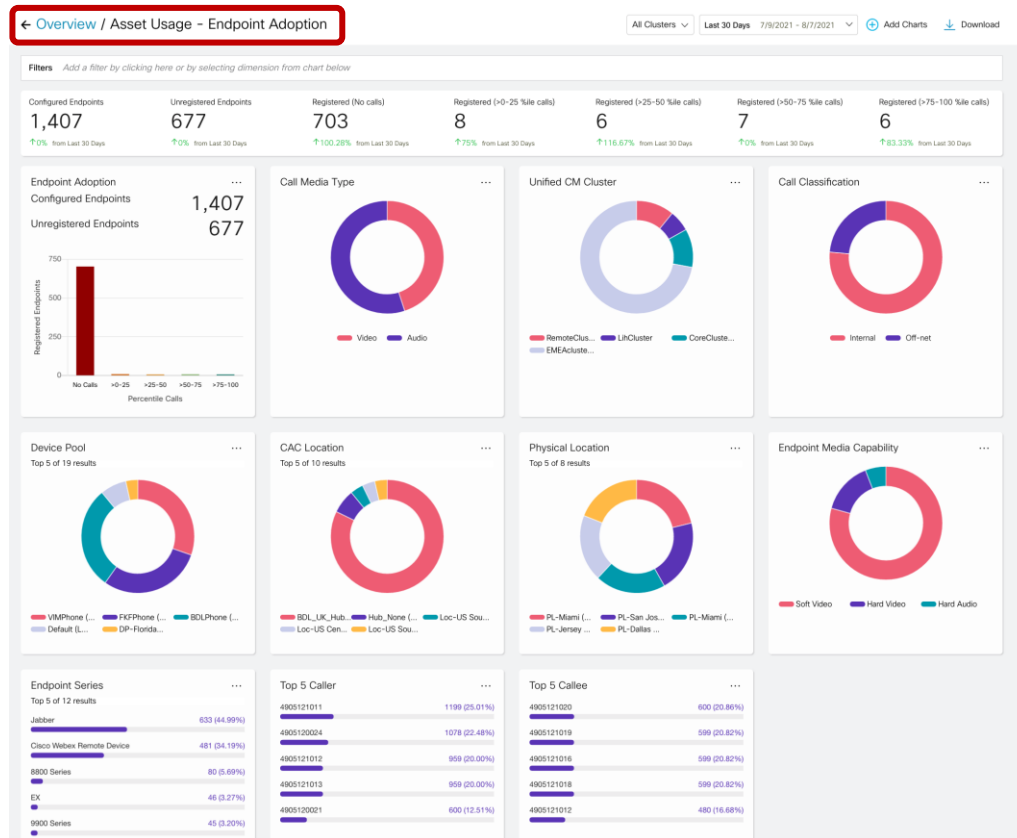
KPI's for total Configured endpoints, unregistered devices and percentile call distribution

Interactive charts and graphs

- Endpoint adoption charts based on Call Media Type, CM cluster, Call Classification, Device Pool, CAC location, Physical Location and Endpoint Media Capacity.

"Filters" to guide admin on a particular aspect - all charts and graphs update accordingly

Ability to download CSV: snapshot of the entire data set, or of the filtered data view



CCUC analytics – asset usage

Headset Adoption

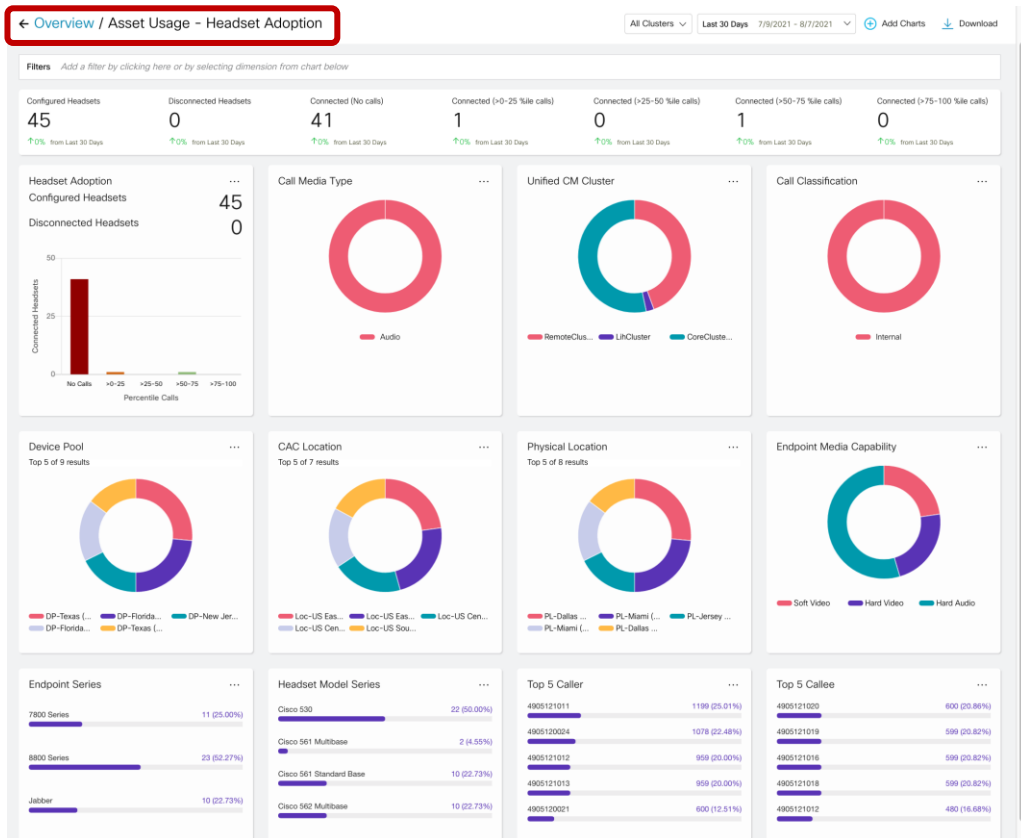
KPI's for total configured headsets, disconnected, connected headsets and percentile call distribution

Interactive charts and graphs

- Headset adoption by asset Call Media Type, Cisco® UCM cluster, Call classification, device pool, CAC Location, Physical Location, Endpoint Media Capacity, details based on Series, Model and 5 top callers and callees.

"Filters" to guide admin on a particular aspect – all charts and graphs update accordingly

Ability to download CSV: snapshot of the entire data set, or of the filtered data view



CCUC analytics – Traffic Analysis

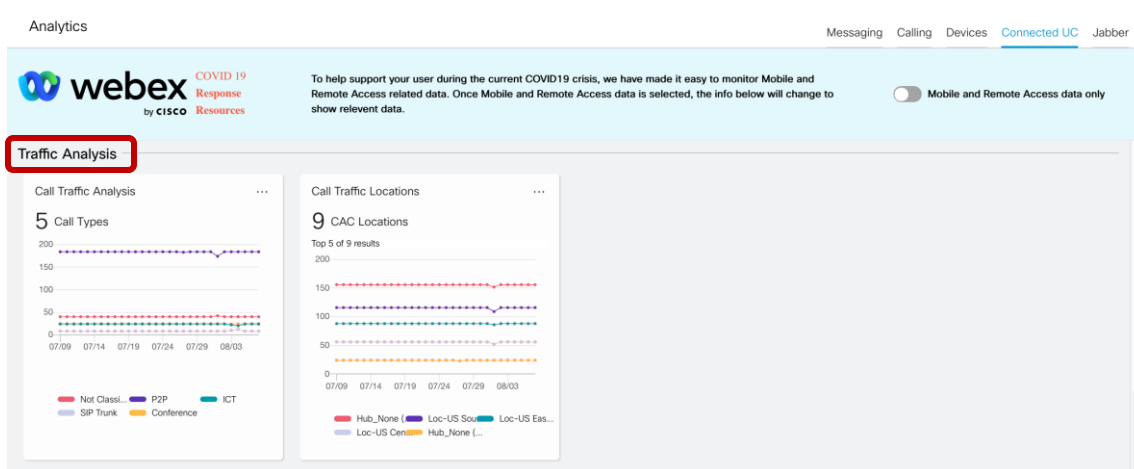
Summary charts for

- Call Traffic Analysis
- Call Traffic Locations
- Drill down into each of the charts for more details

Summary view for "all clusters" or focus on a particular one

Date selector

Filter for "MRA data only"



CCUC analytics – Traffic Analysis

Call Traffic Analysis

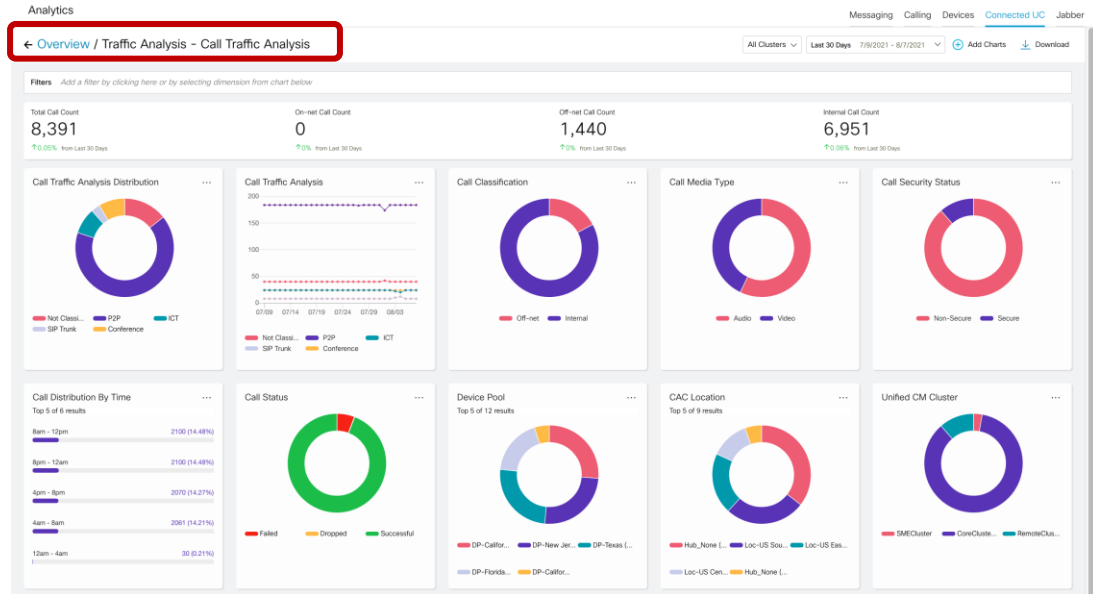
KPI's for total call count classified in on-net call count, off-net call count, internal call count

Interactive charts and graphs

- Call Traffic Analysis distribution by Cisco® UCM cluster, Call classification, Call Media Type, Call Security status, Time, Status, Device Pool, CAC Location.

"Filters" to guide admin on a particular aspect – all charts and graphs update accordingly

Ability to download CSV: snapshot of the entire data set, or of the filtered data view



CCUC analytics – Traffic Analysis

Call Traffic Locations

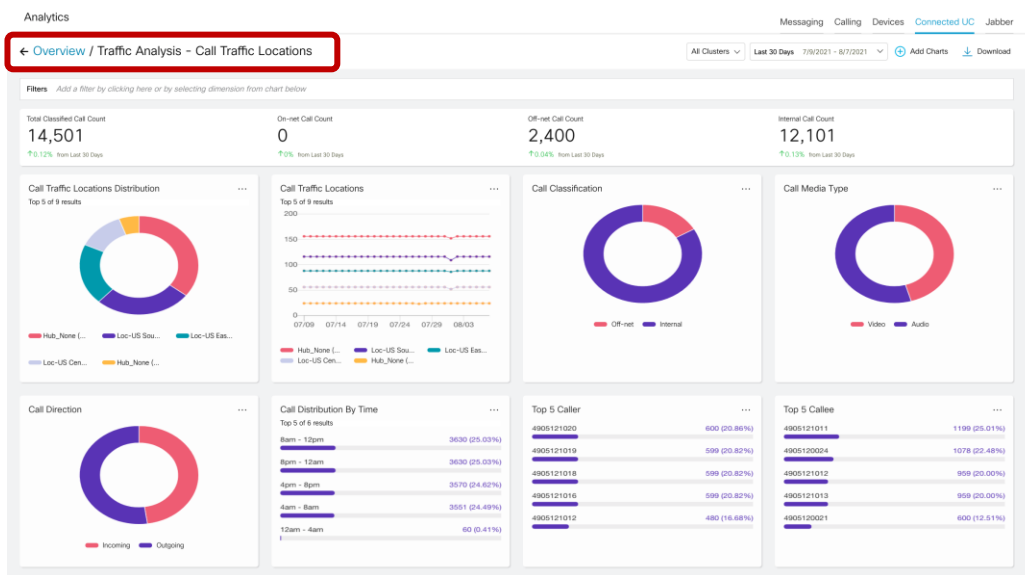
KPI's for total classified Call Count, On-net call count, off-net call count, Internal call count.

Interactive charts and graphs

- For Call Traffic Locations by Call classification, Call Media Type, Call Direction, Call Distribution by Time, Model and 5 top callers and callees.

"Filters" to guide admin on a particular aspect – all charts and graphs update accordingly

Ability to download CSV: snapshot of the entire data set, or of the filtered data view



CCUC analytics – Capacity Analysis

**KPI's for endpoint and headsets
call counts, call minutes, and total
number in calls**

Summary charts for

- Trunk Usage
- Drill down into the chart for more details

**Summary view for "all clusters" or
focus on a particular one**

Date selector

Filter for "MRA data only"



CCUC analytics – Capacity Analysis

Trunk Usage

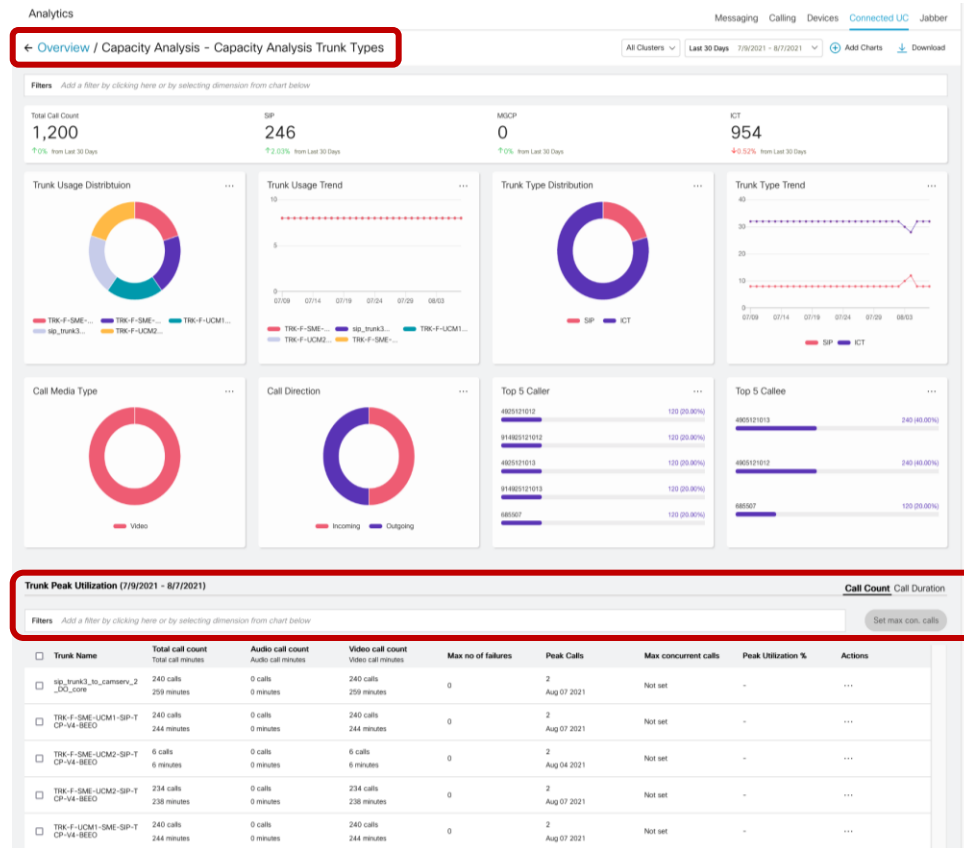
KPI's for total call Count, SIP, MGCP and ICT type calls

Interactive charts and graphs

- Trunk Usage Distribution, Trunk Usage Trend, Type Distribution, Trunk Type Trend, Call Media Type, Call Direction, Top 5 callers and Top 5 Calleees.

Trunk Peak Utilization filter for "Call Count" and "Call Duration"

Ability to download CSV: snapshot of the entire data set, or of the filtered data view



Certificate Management

Certificate Management Contents

01 Overview

General Summary

02 Prerequisites

Necessary steps to get started with Certificate Management

03 Certificate Management in CCUC

04 Components

Alert Dashboard, Cluster view, Profiles, Settings

Overview

Certificate Management



Certificate Profiles

Profiles to setup
certificate settings per
cluster



Alerts and notifications

Dashboard to view
expired/expiring
soon certificates
and set email
notifications



Management Operations

Renew, generate
CSR, upload and
more...



Admin productivity

Single pane of glass
for all clusters in
your environment.

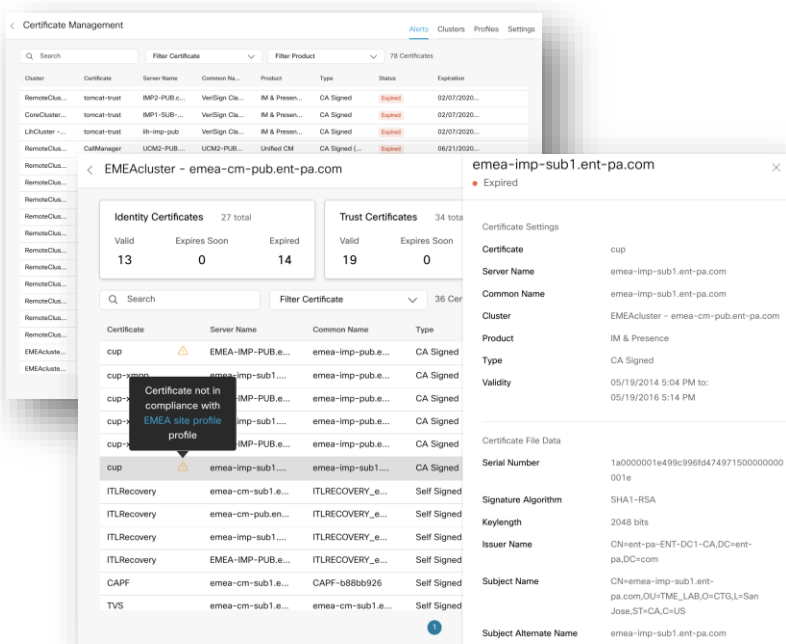


Insights

For certificates out
of compliance

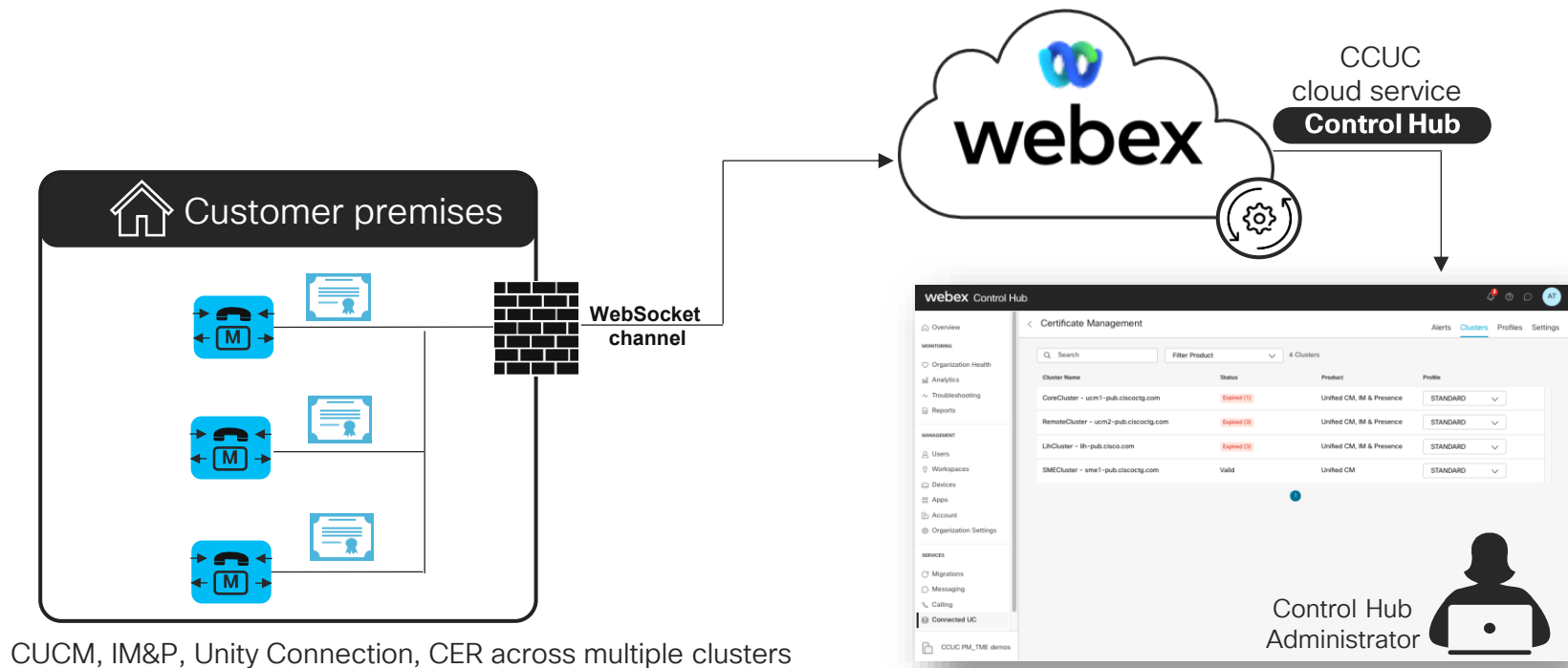
Overview

Certificate Management



- Single place to view, manage and distribute certificates of CUCM, IM&P, Unity and CER across multiple clusters and trust stores
- Centralized certificate actions – CSR generation, upload, renew, download, copy, replace, remove, etc.
- Quick dashboard to view expired or expiring soon certificates
- Certificate insights – out of compliance warnings with profiles
- Multi-cluster listing with certificate summary for each cluster and detailed view of identity certificates and trust stores.
- Notification settings for certificate expiration
- Jobs tab to view certificate actions in progress and history of actions taken in the past
- Define certificate profile with various settings and assign certificate profile to cluster

CCUC and Certificate Management



Pre-requisites

Pre-requisites

Platform

- Supported versions:
11.5, 12.5 and 14 for CUCM, IM&P, CUC and CER products.
- Onboard cluster(s) to Webex® Cloud-Connected-UC.

[How to onboard clusters HERE](#)

Control Hub

- Certificate Management service activated.
- Note: After cluster onboarding, Certificate Management is activated by default

Connectivity

- For a larger cluster, ensure that you have at least 2 MBPS of bandwidth.
- Cloud-Connected UC cloud tcp port 443 must be reachable from customer environment.

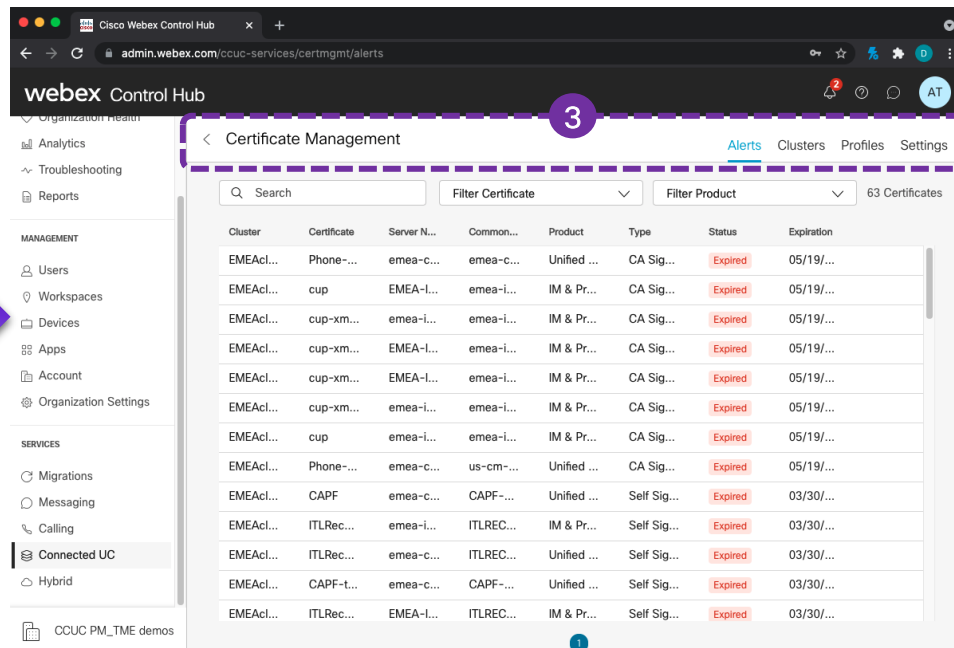
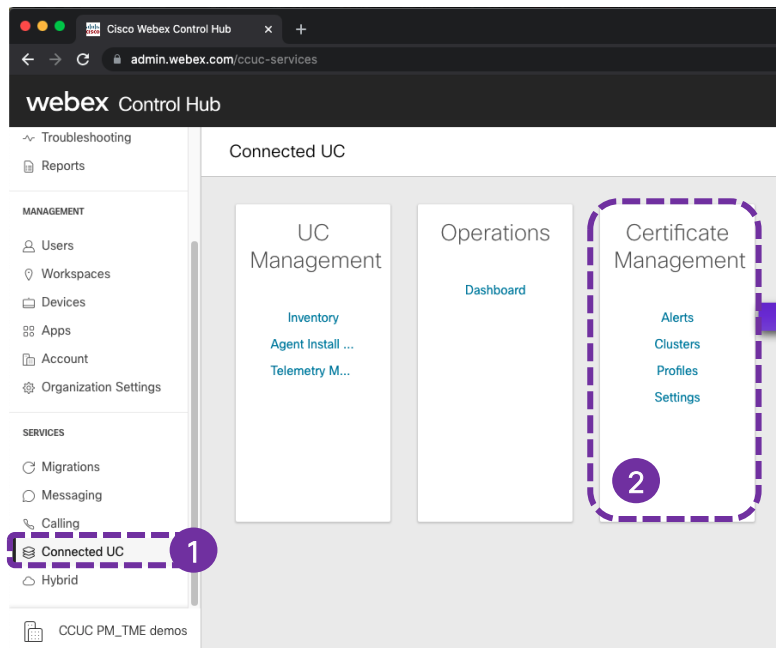
*.ucmgmt.cisco.com

*.webex.com

Certificate Management in CCUC

Certificate Management in CCUC

Navigate to Control Hub > Connected UC > Certificate Management>



Alerts Dashboard

Alerts Dashboard

- Quick dashboard to view expired and expiring soon certificates
- Allows administrators to take suitable actions to keep the certificates up to date.

The screenshot shows the Cisco Webex Control Hub interface. The left sidebar contains navigation options: Organization Health, Analytics, Troubleshooting, Reports, MANAGEMENT (Users, Workspaces, Devices, Apps, Account, Organization Settings), and SERVICES (Migrations, Messaging, Calling, Connected UC, Hybrid). The main content area is titled 'Certificate Management' and features a search bar, filter options, and a table of certificates. A red box highlights the 'Alerts' tab in the top right corner of the main content area.

Alerts Clusters Profiles Settings

Search Filter Certificate Filter Product 63 Certificates

Cluster	Certificate	Server N...	Common...	Product	Type	Status	Expiration
EMEAcl...	Phone-...	emea-c...	emea-c...	Unified ...	CA Sig...	Expired	05/19/...
EMEAcl...	cup	EMEA-I...	emea-i...	IM & Pr...	CA Sig...	Expired	05/19/...
EMEAcl...	cup-xm...	emea-i...	emea-i...	IM & Pr...	CA Sig...	Expired	05/19/...
EMEAcl...	cup-xm...	EMEA-I...	emea-i...	IM & Pr...	CA Sig...	Expired	05/19/...
EMEAcl...	cup-xm...	EMEA-I...	emea-i...	IM & Pr...	CA Sig...	Expired	05/19/...
EMEAcl...	cup-xm...	emea-i...	emea-i...	IM & Pr...	CA Sig...	Expired	05/19/...
EMEAcl...	cup	emea-i...	emea-i...	IM & Pr...	CA Sig...	Expired	05/19/...
EMEAcl...	Phone-...	emea-c...	us-cm-...	Unified ...	CA Sig...	Expired	05/19/...
EMEAcl...	CAPF	emea-c...	CAPF-...	Unified ...	Self Sig...	Expiring Soon	03/30/...
EMEAcl...	ITLRec...	emea-i...	ITLREC...	IM & Pr...	Self Sig...	Expiring Soon	03/30/...
EMEAcl...	ITLRec...	emea-c...	ITLREC...	Unified ...	Self Sig...	Expiring Soon	03/30/...
EMEAcl...	CAPF-t...	emea-c...	CAPF-...	Unified ...	Self Sig...	Expiring Soon	03/30/...
EMEAcl...	ITLRec...	EMEA-I...	ITLREC...	IM & Pr...	Self Sig...	Expiring Soon	03/30/...

CCUC PM_TME demos

Alerts Dashboard

- Search and filtering by certificate type, product or string contained in the certificate decoded chain.
- Management operations:
 - Replace
 - Remove
 - Download
 - Regenerate.

Note: Available actions depend on the type of certificate.

The screenshot displays the 'Alerts' tab of the 'Certificate Management' interface. A purple dashed box highlights the search and filter section, which includes a search bar and two dropdown menus: 'Filter Certificate' and 'Filter Product'. The 'Filter Certificate' dropdown is shown open, listing options: All, Phone-SAST-trust, cup, cup-xmpp, cup-xmpp-s2s, CAPF, ITLRecovery, and CAPF-trust. The 'Filter Product' dropdown is also shown open, listing: All, Unified CM, and IM & Presence. Below these, a table lists certificates with columns for Cluster, Certificate, Server Name, Common Name, Product, Type, Status, and Expiration. Several certificates are marked as 'Expired'. A context menu is open over one of the expired certificates, showing actions: Replace Certificate, Remove Certificate, Download .der, and Download .pem. A blue circle with the number '1' is located at the bottom right of the table.

Cluster	Certificate	Server Name	Common Name	Product	Type	Status	Expiration
EMEAcl...	Phone-...	emea-c...	emea-c...	Unified ...	CA Sig...	Expired	05/19/...
EMEAcl...	cup	EMEA-l...	emea-l...	IM & Pr...	CA Sig...	Expired	05/19/...
EMEAcl...	cup-xm...	emea-l...	emea-l...	IM & Pr...	CA Sig...	Expired	05/19/...
EMEAcl...	cup-xm...	EMEA-l...	emea-l...	IM & Pr...	CA Sig...	Expired	05/19/...
EMEAcl...	cup-xm...	EMEA-l...	emea-l...	IM & Pr...	CA Sig...	Expired	05/19/...
EMEAcl...	cup-xm...	emea-l...	emea-l...	IM & Pr...	CA Sig...	Expired	05/19/...
EMEAcl...	cup	emea-l...	emea-l...	IM & Pr...	CA Sig...	Expired	05/19/...
EMEAcl...	Phone-...	emea-c...	us-cm-...	Unified ...	CA Sig...	Expired	05/19/...
CAPF	emea-c...	CAPF-...	Unified ...	Self Sig...	Expired	03/30/...	
ITLRec...	emea-l...	ITLREC...	IM & Pr...	Self Sig...	Expired	03/30/...	
ITLRec...	emea-c...	ITLREC...	Unified ...	Self Sig...	Expired	03/30/...	
CAPF-t...	emea-c...	CAPF-...	Unified ...	Self Sig...	Expired	03/30/...	
ITLRec...	EMEA-l...	ITLREC...	IM & Pr...	Self Sig...	Expired	03/30/...	

Alerts Dashboard

The interface is divided into two main sections. The left section, titled 'Certificate Management', contains a search bar and a table of certificates. The right section, titled 'emea-cm-pub.ent-pa.com', shows details for a specific certificate, including its subject name, fingerprint, and usage.

Cluster	Certificate	Server Name	Common Na...	Product
EMEAclust...	ITLRecovery	emea-cm-...	ITLRECOVE...	Unified CM
EMEAclust...	CAPF-trust	emea-cm-...	CAPF-170...	Unified CM
EMEAclust...	ITLRecovery	EMEA-IMP...	ITLRECOVE...	IM & Prese...
EMEAclust...	CalManag...	emea-cm-...	CAPF-170...	Unified CM
EMEAclust...	TVS	emea-cm-...	emea-cm-...	Unified CM
EMEAclust...	CalManag...	emea-cm-...	CAPF-b88...	Unified CM
CoreCluste...	tomcat-trust	IMP1-PUB...	VeriSign Cl...	IM & Prese...

Certificate Details (emea-cm-pub.ent-pa.com):

- Status: Expired
- Subject Name: CN=emea-cm-pub.ent-pa.com,OU=TME_LAB,O=CTG,L=San Jose,ST=CA,C=US
- Subject Alternate Name: CN=emea-cm-pub.ent-pa.com,OU=TME_LAB,O=CTG,L=San Jose,ST=CA,C=US
- Certificate Fingerprint (SHA1): AA:12:CF:75:36:DC:87:26:00:EC:CC:F9:7D:8E:A2:4A:65:B9:E6:65 (SHA256): 5F:0E:18:DD:01:BC:47:5E:EF:12:B5:8D:4D:33:FB:1A:FE:44:82:54:67:E0:33:CA:64:B D:BE:4B:B3:F3:57:0F
- Extended Key Usage: critical : false; usage : ServerAuth,ClientAuth,IPSECEndSystem
- Key Usage: critical : false; usage : DigitalSignature,KeyEncipherment,DataEncipherment,KeyAgreement,CertSign
- CRL Distribution Points: (empty)
- PEM Format Certificate: (button)
- BEGIN CERTIFICATE----- MIDvTCCaQWgAwIBAgIQXpW6B2w...

Actions: Renew Certificate, Download .der, Download .pem

1 Certificate View Inspection Available

2 Available Actions per certificate type

Detailed messages to guide administrators

Renew Certificate

The settings defined in the profile for this certificate are not supported by the software version running on a cluster. This operation will be performed with certificate attributes that matches the cluster capability.

The certificate will be renewed for the nodes listed below with the following attributes.

Warning: Perform this action only during a maintenance window as it requires service(s) restarts on node(s). Cisco Trust Verification Service need to be restarted manually on node(s) after the successful completion of this action.

Node(s) where certificate will be renewed: emea-cm-pub.ent-pa.com

Certificate Attributes

Key Type	RSA
Key Length	2048
Hash Algorithm	SHA256
Expiration	07/20/2026

3

Renew Certificate

Renew Certificate has been added to the Jobs list. View progress under the [Jobs](#) tab.

Ok

4

All actions can be monitored in the Jobs Tab afterwards.

Profiles

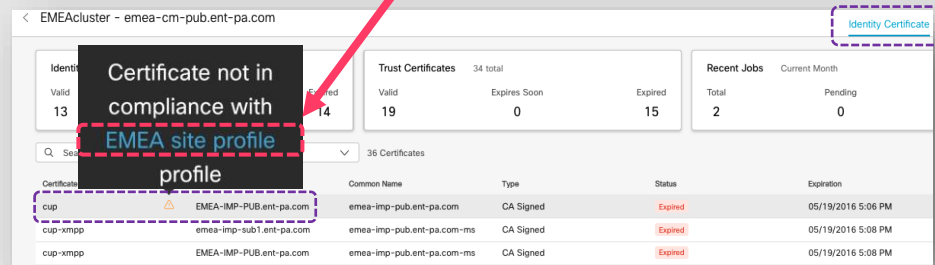
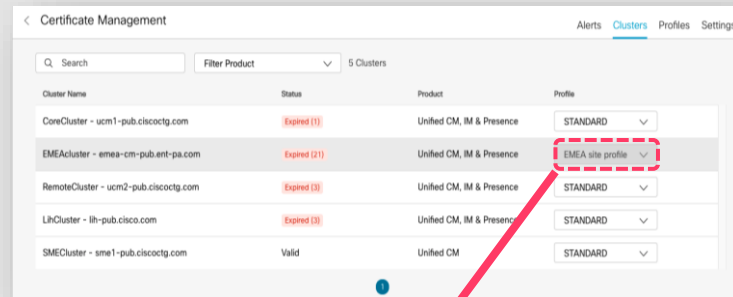
Profiles - Overview

Certificate Management comes with a Standard Profile which is assigned by default to all onboarded clusters

Profiles enforce certificate settings based on platform type.

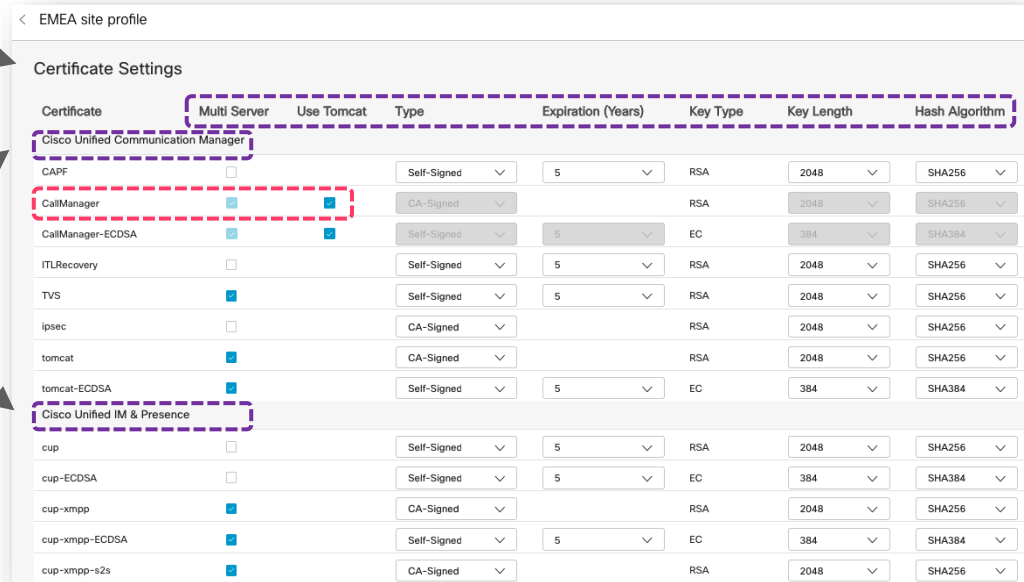
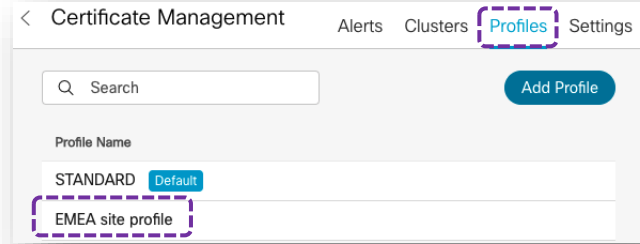
Administrator can customize the settings based on deployment needs and assign to desired clusters.

Certificate Management Service monitors **certificates** in the onboarded servers to check if they **comply with the settings** defined in the associated profile. If it's out of compliance, it will be flagged in the identity certificates list of the cluster



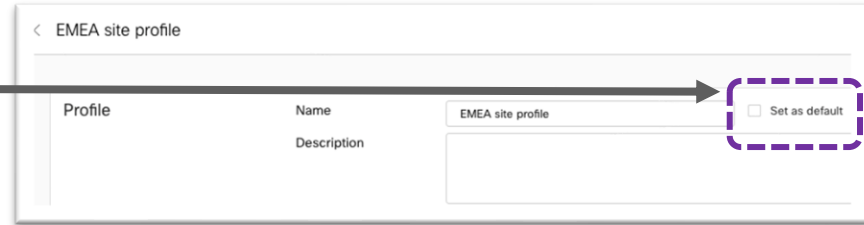
Profiles

- In Profiles tab you can add or edit existing profiles
- Configurable Certificate Settings: Multi-server/Multi-san, CA-signed vs self-signed, validity period, RSA vs ECDSA, key length Hash Algorithm
- Certificate settings based on platform, e.g. Call Manager and IM & P
- Note: CallManager certificate has the option to re-use Tomcat Certificate.

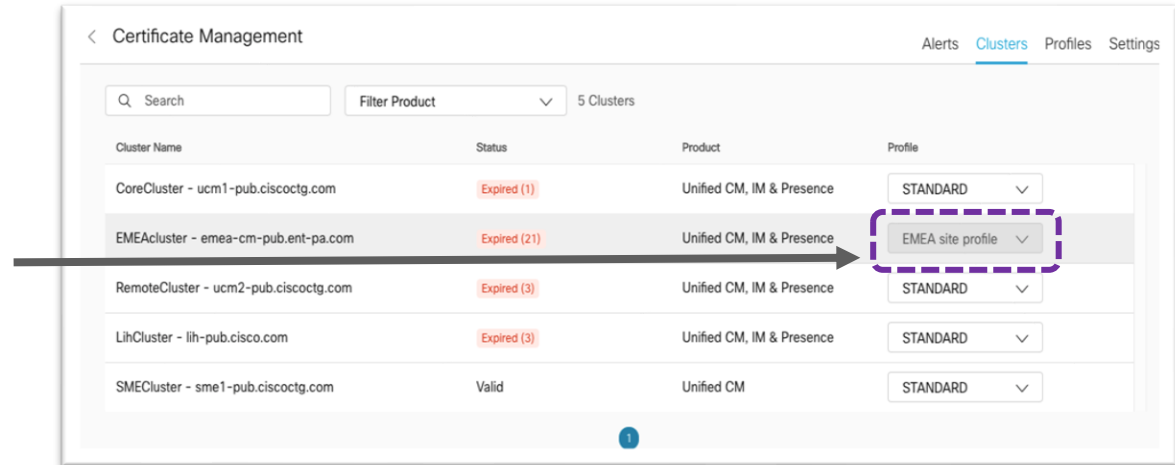


Profiles

- You can designate a profile as the default profile that will be applied to all on-boarded clusters.



- Clusters view provides an option to change the profile



Profiles

“View Profile” available from Cluster view

Actions Menu Available in this view as well

- Warning when a certificate is not in compliance with the assigned profile.

Note: Use separate profiles for each version of the cluster.

The screenshot displays the 'EMEAcluster - emea-cm-pub.ent-pa.com' interface. At the top, there are summary boxes for 'Identity Certificates' (27 total: 13 Valid, 14 Expired), 'Trust Certificates' (34 total: 19 Valid, 0 Expires Soon, 15 Expired), and 'Recent Jobs' (Current Month: 2 Total, 0 Pending, 2 Complete). Below these is a search bar and a 'View Profile' button. The main section is a table of certificates. Two rows are highlighted with a dashed purple border and a warning icon: 'cup' (EMEA-IMP-PUB.ent-pa.com) and 'cup' (emea-imp-sub1.ent-pa.com). A tooltip for the first row states: 'Certificate not in compliance with EMEA site profile'. A context menu is open over the 'View Profile' button, showing options: 'Generate CSR', 'Download .der', and 'Download .pem'.

Certificate	Common Name	Type	Status	Expiration	
cup	EMEA-IMP-PUB.ent-pa.com	emea-imp-pub.ent-pa.com	CA Signed	Expired	05/19/2016 5:06 PM
cup-xmpp	emea-imp-sub1.ent-pa.com	emea-imp-pub.ent-pa.com-ms	CA Signed	Expired	05/19/2016 5:08 PM
cup-xmpp	EMEA-IMP-PUB.ent-pa.com	emea-imp-pub.ent-pa.com-ms	CA Signed	Expired	05/19/2016 5:08 PM
cup-xmpp-s2s	emea-imp-sub1.ent-pa.com	emea-imp-pub.ent-pa.com-ms	CA Signed	Expired	05/19/2016 5:11 PM
cup-xmpp-s2s	EMEA-IMP-PUB.ent-pa.com	emea-imp-pub.ent-pa.com-ms	CA Signed	Expired	05/19/2016 5:11 PM
cup	emea-imp-sub1.ent-pa.com	emea-imp-sub1.ent-pa.com	CA Signed	Expired	05/19/2016 5:14 PM
ITLRecovery	emea-cm-sub1.ent-pa.com	ITLRECOVERY_emea-cm-pub....	Self Signed	Expired	03/30/2019 12:58 AM
ITLRecovery	emea-cm-pub.ent-pa.com	ITLRECOVERY_emea-cm-pub....	Self Signed	Expired	03/30/2019 12:58 AM
ITLRecovery	emea-imp-sub1.ent-pa.com	ITLRECOVERY_emea-cm-pub....	Self Signed	Expired	03/30/2019 12:58 AM
ITLRecovery	EMEA-IMP-PUB.ent-pa.com	ITLRECOVERY_emea-cm-pub....	Self Signed	Expired	03/30/2019 12:58 AM
CAPF	emea-cm-sub1.ent-pa.com	CAPF-b88bb926	Self Signed	Expired	03/30/2019 3:34 AM
TVS	emea-cm-sub1.ent-pa.com	emea-cm-sub1.ent-pa.com	Self Signed	Expired	03/30/2019 3:34 AM

Insights



Enables administrators with data to take proactive decisions and offers a centralized interface to take actions.



Certificate Management service identify and notifies administrator with a list of certificates that are about to expire



Automatic detection of out of compliance certificates saving time and effort of the administrator.

Clusters View

Clusters view

Clusters view summarizes certificate information per on-boarded cluster

Overall status of certificates (Expired/Expiring Soon/Valid) is shown under status.

Clicking on Cluster name will display certificates in that specific cluster.

< Certificate Management Alerts **Clusters** Profiles Settings

Q Search Filter Product 5 Clusters

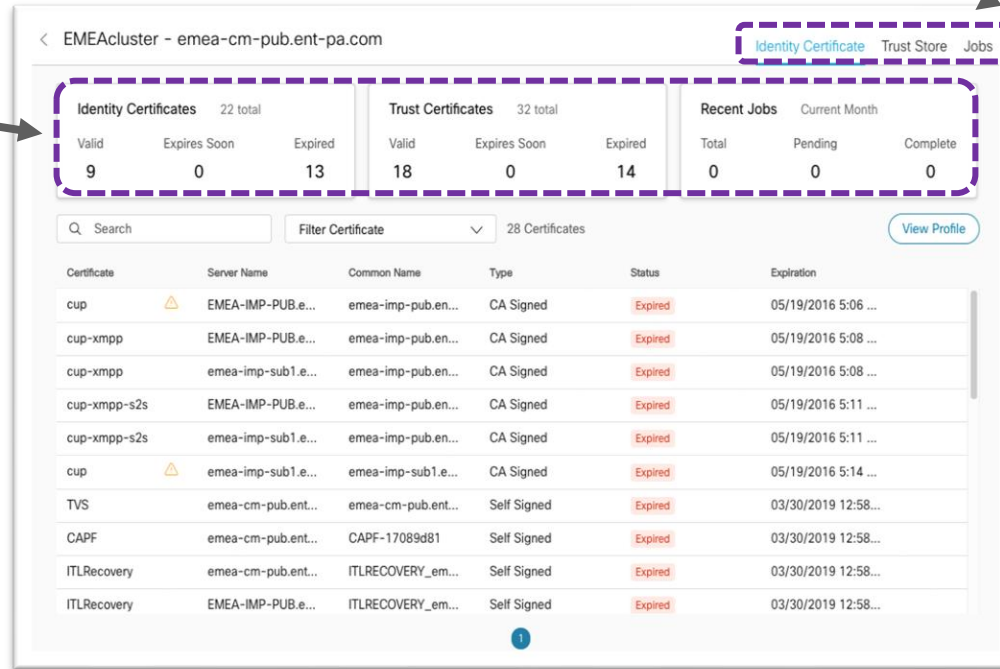
Cluster Name	Status	Product	Profile
CoreCluster - ucm1-pub.ciscoctg.com	Expired (1)	Unified CM, IM & Pr...	STANDARD
EMEAcluster - emea-cm-pub.ent-pa.com	Expired (21)	Unified CM, IM & Pr...	STANDARD
RemoteCluster - ucm2-pub.ciscoctg.com	Expired (3)	Unified CM, IM & Pr...	STANDARD
LihCluster - lih-pub.cisco.com	Expired (3)	Unified CM, IM & Pr...	STANDARD
SMECluster - sme1-pub.ciscoctg.com	Valid	Unified CM	STANDARD

1

Searching and filtering per product is available.

Individual Cluster View – Summary view

Summary of existing certificates is shown.



Three tabs available:

- *Identity Certificate* to display all server certificate belonging to the cluster shown.
- *Trust Store* to display all certificates in the trust stores of the cluster.
- *Jobs* to display and monitor the status of certificate management actions.

Individual Cluster View – Identity Certificate

Click on Identity Certificate Tab to view all server certificates in this cluster

Search and filter options available

Identity Certificate list with details of certificates

The screenshot shows the 'Identity Certificate' tab for the cluster 'emea-cm-pub.ent-pa.com'. It features a summary section with counts for Identity Certificates (27 total: 15 Valid, 0 Expires Soon, 12 Expired), Trust Certificates (35 total: 21 Valid, 0 Expires Soon, 14 Expired), and Recent Jobs (7 Total: 0 Pending, 7 Complete). Below this is a search and filter section with a search bar and a 'Filter Certificate' dropdown. The main area is a table of certificates with columns: Certificate, Server Name, Common Name, Type, Status, and Expiration. The table lists various certificates including 'ipsec', 'CallManager-ECDSA', and 'cup-xmpp-s2s-ECDSA'. A 'View Profile' button is located in the top right corner of the certificate list area.

Certificate	Server Name	Common Name	Type	Status	Expiration
ipsec	emea-cm-pub.ent-pa.c...	emea-cm-pub.ent-pa.c...	Self Signed	Valid	06/07/2026 6:16 PM
ipsec	emea-cm-sub1.ent-pa.c...	emea-cm-sub1.ent-pa.c...	Self Signed	Valid	06/07/2026 6:16 PM
ipsec	EMEA-IMP-PUB.ent-pa.c...	EMEA-IMP-PUB.ent-pa.c...	Self Signed	Valid	06/07/2026 6:16 PM
ipsec	emea-imp-sub1.ent-pa.c...	emea-imp-sub1.ent-pa.c...	Self Signed	Valid	06/07/2026 6:16 PM
CallManager-ECDSA	emea-cm-pub.ent-pa.c...	emea-cm-pub-EC.ent-pa.c...	Self Signed	Valid	06/08/2026 1:44 PM
CallManager-ECDSA	emea-cm-sub1.ent-pa.c...	emea-cm-sub1-EC.ent-pa.c...	Self Signed	Valid	06/08/2026 1:44 PM
cup-xmpp-ECDSA	EMEA-IMP-PUB.ent-pa.c...	EMEA-IMP-PUB-EC.ent-pa.c...	Self Signed	Valid	06/08/2026 1:44 PM
cup-xmpp-s2s-ECDSA	EMEA-IMP-PUB.ent-pa.c...	EMEA-IMP-PUB-EC.ent-pa.c...	Self Signed	Valid	06/08/2026 1:44 PM
cup-xmpp-ECDSA	emea-imp-sub1.ent-pa.c...	emea-imp-sub1-EC.ent-pa.c...	Self Signed	Valid	06/08/2026 6:35 PM
cup-xmpp-s2s-ECDSA	emea-imp-sub1.ent-pa.c...	emea-imp-sub1-EC.ent-pa.c...	Self Signed	Valid	06/08/2026 6:36 PM
TVS	emea-cm-pub.ent-pa.c...	emea-cm-pub.ent-pa.c...	Self Signed	Valid	07/20/2026 10:36 PM
CAPF	emea-cm-pub.ent-pa.c...	CAPF-04b20cb4	Self Signed	Valid	07/22/2026 1:44 PM
cup	EMEA-IMP-PUB.ent-pa.c...	EMEA-IMP-PUB.ent-pa.c...	Self Signed	Valid	08/01/2026 9:05 PM

Visibility of the profile assigned to the cluster

Certificate actions

Certificate detailed view also available in the Identity Certificate Tab

The screenshot shows a detailed view of a certificate for the cluster 'emea-cm-pub.ent-pa.com'. It indicates the certificate is 'Valid'. The 'Certificate Settings' section lists the following details:

- Certificate: TVS
- Server Name: emea-cm-pub.ent-pa.com
- Common Name: emea-cm-pub.ent-pa.com
- Cluster: EMEAcluster - emea-cm-pub.ent-pa.com
- Product: Unified CM
- Type: Self Signed
- Validity: 07/21/2021 10:36 PM to: 07/20/2026 10:36 PM

Individual Cluster View – Trust Store

Click on Trust Store Tab to view all certificates in various trust stores in this cluster.

Search and filtering by trust store

EMEACluster - emea-cm-pub.ent-pa.com

Identity Certificates 27 total Trust Certificates 35 total Recent Jobs Current Month

Valid	Expires Soon	Expired	Valid	Expires Soon	Expired	Total	Pending	Complete
15	0	12	21	0	14	7	0	7

Search Filter Certificate 107 Certificates

Common Name	Server Name	Serial Number	Issued by	Status	Expiration
Cisco Root CA 2048	emea-cm-sub1.ent...	5f87b282b54dc8d...	CN=Cisco Root CA ...	Valid	05/14/2029 3:25 PM
ACT2 SUDI CA	emea-cm-sub1.ent...	61096e7d0000000...	CN=Cisco Root CA ...	Valid	05/14/2029 3:25 PM
ACT2 SUDI CA	emea-cm-sub1.ent...	61096e7d0000000...	CN=Cisco Root CA ...	Valid	05/14/2029 3:25 PM
Cisco Manufacturin...	emea-cm-sub1.ent...	6a6967b30000000...	CN=Cisco Root CA ...	Valid	05/14/2029 3:25 PM
Cisco Manufacturin...	emea-cm-sub1.ent...	6a6967b30000000...	CN=Cisco Root CA ...	Valid	05/14/2029 3:25 PM
Cisco Manufacturin...	emea-cm-sub1.ent...	6a6967b30000000...	CN=Cisco Root CA ...	Valid	05/14/2029 3:25 PM
Cisco Manufacturin...	emea-cm-sub1.ent...	6a6967b30000000...	CN=Cisco Root CA ...	Valid	05/14/2029 3:25 PM
ACT2 SUDI CA	emea-cm-sub1.ent...	61096e7d0000000...	CN=Cisco Root CA ...	Valid	05/14/2029 3:25 PM
Cisco Root CA 2048	emea-cm-pub.ent...	5f87b282b54dc8d...	CN=Cisco Root CA ...	Valid	05/14/2029 3:25 PM
Cisco Root CA 2048	emea-cm-pub.ent...	5f87b282b54dc8d...	CN=Cisco Root CA ...	Valid	05/14/2029 3:25 PM
ACT2 SUDI CA	emea-cm-pub.ent...	61096e7d0000000...	CN=Cisco Root CA ...	Valid	05/14/2029 3:25 PM
Cisco Root CA M2	emea-cm-sub1.ent...	1	CN=Cisco Root CA ...	Valid	11/12/2037 7:00 AM
Cisco Root CA M2	emea-cm-sub1.ent...	1	CN=Cisco Root CA ...	Valid	11/12/2037 7:00 AM
Cisco Manufacturin...	emea-cm-sub1.ent...	2	CN=Cisco Root CA ...	Valid	11/12/2037 7:00 AM

Filter Certificate: ☐ All ☐ Phone-SAST-trust ☐ CallManager-trust ☐ CAPF-trust ☐ tomcat-trust ☐ cup-trust ☐ ipsec-trust

Upload to Trust

Upload Certificate

Cisco DRF Local and Cisco DRF Master need to be restarted manually on node(s) after the successful completion of this action.

File Name	Common Name	Serial Number
ipsec.pem	emea-cm-pub.ent-pa...	57981937924061180...

Choose File

Associate with Clusters and Trust

Cluster	Trust Store
☐ cucm1.dcloud.cisco.com	0
☐ SMECluster - sme-1-pub.ciscoctg.com	0
☒ LihCluster - lih-pub.cisco.com	1
☐ tomcat-trust	CUCM
☐ ipsec-trust	CUCM
☐ CallManager-trust	CUCM
☐ CAPF-trust	CUCM
☐ TVS-trust	CUCM
☐ Phone-trust	CUCM
☐ Phone-VPN-trust	CUCM
☐ Phone-SAST-trust	CUCM
☐ Phone-CTL-trust	CUCM
☐ Phone-CTL-ASA-trust	CUCM
☐ Phone-Edge-trust	CUCM
☐ tomcat-trust	IMS&P
☒ ipsec-trust	IMS&P
☐ cup-trust	IMS&P

Cancel Upload

Click on "Upload to trust" and administrator will view the options to upload a certificate to one or many clusters and trust stores.

Individual Cluster View – Jobs

Jobs tab displays tasks that were performed, node info, certificate type, products and status whether is success or failure.

< EMEAcluster - emea-cm-pub.ent-pa.com

Identity Certificate Trust Store **Jobs**

Identity Certificates 22 total

Valid	Expires Soon	Expired
10	0	12

Trust Certificates 32 total

Valid	Expires Soon	Expired
18	0	14

Recent Jobs Current Month

Total	Pending	Complete
1	0	1

Current Month ▾ 1 Job

Job Type	Node	Time Stamp	Certificate	Product	Status
Renew Self Signed	emea-cm-pub.ent-...	07/21/2021 10:36 ...	TVS	Unified CM	Success

Settings

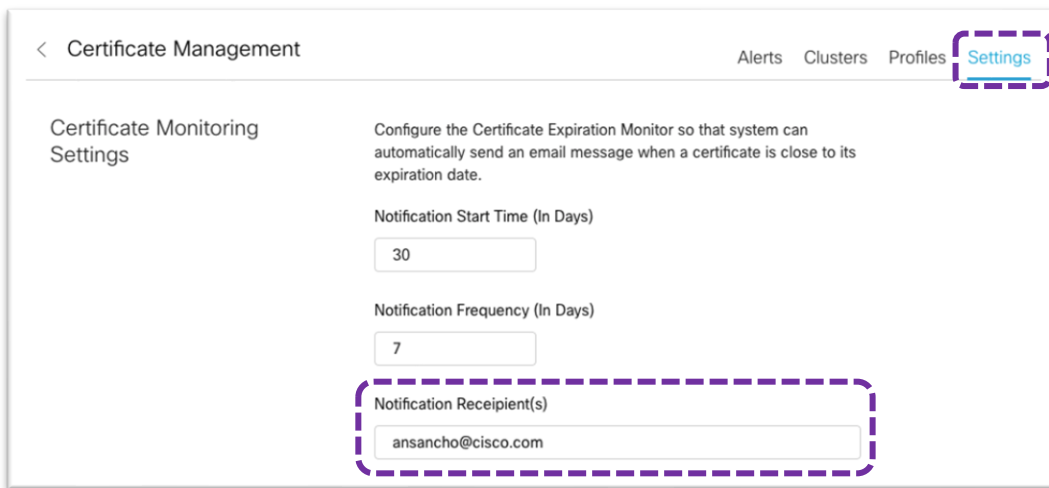
Setup Notification Receipients

Automatic notifications available in certificate management with the capability to define:

Notification Start Time

Notification Frequency

Notification Recipients



The screenshot shows the 'Certificate Management' settings page. The 'Settings' link in the top right navigation bar is highlighted with a dashed purple box. The 'Notification Recipient(s)' field, containing the email 'ansancho@cisco.com', is also highlighted with a dashed purple box. The page title is '< Certificate Management'. The section is 'Certificate Monitoring Settings'. The description states: 'Configure the Certificate Expiration Monitor so that system can automatically send an email message when a certificate is close to its expiration date.' The 'Notification Start Time (In Days)' is set to 30. The 'Notification Frequency (In Days)' is set to 7.

< Certificate Management

Alerts Clusters Profiles **Settings**

Certificate Monitoring Settings

Configure the Certificate Expiration Monitor so that system can automatically send an email message when a certificate is close to its expiration date.

Notification Start Time (In Days)

30

Notification Frequency (In Days)

7

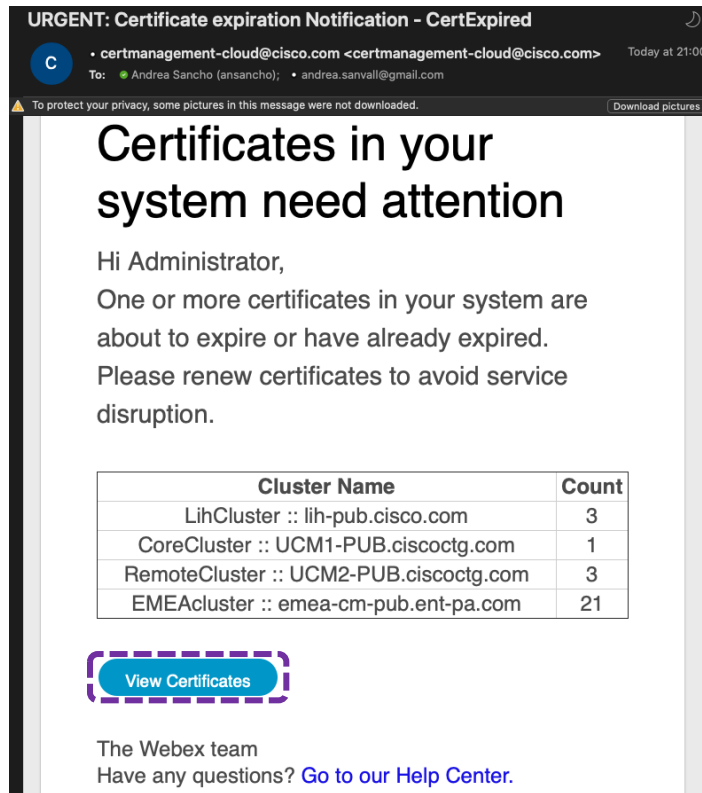
Notification Recipient(s)

ansancho@cisco.com

Email Notification

A report is generated per cluster and administrator is notified via email that action is required.

Administrator can visit Alert Dashboard to review the the certificates.



URGENT: Certificate expiration Notification - CertExpired

• certmanagement-cloud@cisco.com <certmanagement-cloud@cisco.com> Today at 21:00

To: • Andrea Sancho (ansancho); • andrea.sanval@gmail.com

To protect your privacy, some pictures in this message were not downloaded. [Download pictures](#)

Certificates in your system need attention

Hi Administrator,

One or more certificates in your system are about to expire or have already expired. Please renew certificates to avoid service disruption.

Cluster Name	Count
LihCluster :: lih-pub.cisco.com	3
CoreCluster :: UCM1-PUB.ciscoctg.com	1
RemoteCluster :: UCM2-PUB.ciscoctg.com	3
EMEAcluster :: emea-cm-pub.ent-pa.com	21

[View Certificates](#)

The Webex team
Have any questions? [Go to our Help Center.](#)

Certificate Regeneration Example

Example: Regenerate IM&P certificate expired and out of compliance

IM&P certificate expired in 2016.

Out of compliance due to the Signature Algorithm

EMEAcluster - emea-cm-pub.ent-pa.com

[Identity Certificate](#) [Trust Store](#) [Jobs](#)

Identity Certificates			Trust Certificates			Recent Jobs		
Valid	Expires Soon	Expired	Valid	Expires Soon	Expired	Total	Pending	Complete
13	0	14	19	0	15	5	0	5

Search Filter Certificate 36 Certificates

Certificate	Server Name	Common Name	Type	Status	Expiration
cup	EMEA-IMP-PUB.ent-pa...	emea-imp-pub.ent-pa...	CA Signed	Expired	05/19/2016 5:06 PM

[View Profile](#)

< EMEA site profile

Cisco Unified IM & Presence

cup ☐

Self-Signed 5 RSA 2048 SHA256

EMEA Profile defines that IM&P certificate is to be Self-signed, expire in 5 years with RSA SHA 256, 2058 bit key length

Example: Regenerate IM&P certificate expired and out of compliance

< EMEAcluster - emea-cm-pub.ent-pa.com

Identity Certificate						Trust Store	Jobs
Identity Certificates 27 total			Trust Certificates 34 total			Recent Jobs Current Month	
Valid	Expires Soon	Expired	Valid	Expires Soon	Expired	Total	Pending Complete
13	0	14	19	0	15	4	0 4
Q Search			Filter Certificate			36 Certificates	
						View Profile	
Certificate	Server Name	Common Name	Type	Status	Expiration		
cup	EMEA-IMP-PUB.ent-pa....	emea-imp-pub.ent-pa....	CA Signed	Expired	05/19/2016 5:06 PM		
cup-xmpp	emea-imp-sub1.ent-pa....	emea-imp-pub.ent-pa....	CA Signed	Expired	05/19/2016 5:08		
cup-xmpp	EMEA-IMP-PUB.ent-pa....	emea-imp-pub.ent-pa....	CA Signed	Expired	05/19/2016 5:08		
cup-xmpp-s2s	emea-imp-sub1.ent-pa....	emea-imp-pub.ent-pa....	CA Signed	Expired	05/19/2016 5:11		
cup-xmpp-s2s	EMEA-IMP-PUB.ent-pa....	emea-imp-pub.ent-pa....	CA Signed	Expired	05/19/2016 5:11		

Click on the three dots to display the available action and select “Renew Certificate”

CCUC pops up a window with information of the Profile assigned and applicable warnings including the services that need a mandatory restart. The view of the certificate attributes is included.

When ready click on “Renew”

Renew Certificate

The certificate will be renewed for the nodes listed below with the following attributes. The attributes are managed via "EMEA site profile" profile.

Warning: Perform this action only during a maintenance window as it requires service(s) restarts on node(s).
Cisco Presence Engine and Cisco SIP Proxy need to be restarted manually on node(s) after the successful completion of this action.

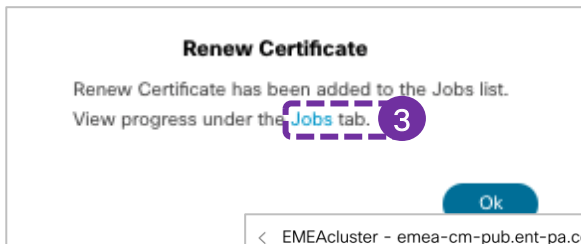
Node(s) where certificate will be renewed
emea-imp-pub.ent-pa.com

Certificate Attributes

Key Type	RSA
Key Length	2048
Hash Algorithm	SHA256
Expiration	08/01/2026

Cancel Renew

Example: Regenerate IM&P certificate expired and out of compliance



After clicking “Renew”, the certificate task is added to the job list. Click on Jobs tab to see the status.

Jobs
List →

< EMEAcluster - emea-cm-pub.ent-pa.com

Identity CertificateTrust StoreJobs

Identity Certificates27 total

Valid

Expires Soon

Expired

13

0

14

Trust Certificates34 total

Valid

Expires Soon

Expired

19

0

15

Recent JobsCurrent Month

Total

Pending

Complete

5

0

5

Current Month5 Jobs

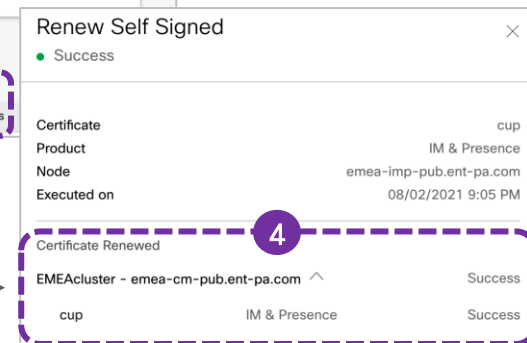
Job Type	Node	Time Stamp	Certificate	Product	Status
Renew Self Signed	emea-imp-pub.ent-pa.com	08/02/2021 9:05 PM	cup	IM & Presence	Success

Renew Self Signed

● Success

Certificate

In the Jobs details page you can confirm the status of the job and expand for details →



Example: Regenerate IM&P certificate expired and out of compliance

< EMEAcluster - emea-cm-pub.ent-pa.com

Identity Certificate Trust Store Jobs

Search Filter Certificate 36 Certificates View Profile

Certificate	Server Name	Common Name	Type	Status	Expiration
cup-xmpp-s2s-ECDSA	EMEA-IMP-PUB.ent-pa.com	EMEA-IMP-PUB-EC.ent-pa.com	Self Signed	Valid	06/08/2026 1:39 PM
cup-xmpp-ECDSA	emea-imp-sub1.ent-pa.com	emea-imp-sub1-EC.ent-pa.com	Self Signed	Valid	06/08/2026 6:35 PM
cup-xmpp-s2s-ECDSA	emea-imp-sub1.ent-pa.com	emea-imp-sub1-EC.ent-pa.com	Self Signed	Valid	06/08/2026 6:36 PM
TVS	emea-cm-pub.ent-pa.com	emea-cm-pub.ent-pa.com	Self Signed	Valid	07/20/2026 10:36 PM
CAPF	emea-cm-pub.ent-pa.com	CAPF-04b20cb4	Self Signed	Valid	07/22/2026 1:44 PM
cup	EMEA-IMP-PUB.ent-pa.com	EMEA-IMP-PUB.ent-pa.com	Self Signed	Valid	08/01/2026 9:05 PM

5

VERIFICATION: In the Identity Certificate List show the new regenerated IM&P certificate as valid and complying with the settings defined in the profile.

Certificate List (1 - 31 of 31)								Rows per Page 50
Find Certificate List where Certificate begins with Find Clear Filter								
Certificate	Common Name	Type	Key Type	Distribution	Issued By	Expiration	Description	
cup	EMEA-IMP-PUB.ent-pa.com	Self-signed	RSA	EMEA-IMP-PUB.ent-pa.com	EMEA-IMP-PUB.ent-pa.com	08/01/2026	Self-signed certificate generated by system	

You can verify now that the new certificate is in the trust store IMP-PUB and proceed with the required services restart.

Redistribute a certificate across multiple trust stores/clusters Example

Example: Redistribute a Certificate across multiple trust stores and clusters.

In the Identity Certificate Tab you select the desired certificate to distribute across multiple clusters and trust stores, in this case a Tomcat Multi Server Certificate from EMEAcluster.

< EMEAcluster - emea-cm-pub.ent-pa.com

Identity Certificates 27 total

Valid	Expires Soon	Expired
14	0	13

Trust Certificates 35 total

Valid	Expires Soon	Expired
20	0	15

Recent Jobs Current Month

Total	Pending	Complete
5	0	5

Search 1 Items Selected 4 Certificates View Profile

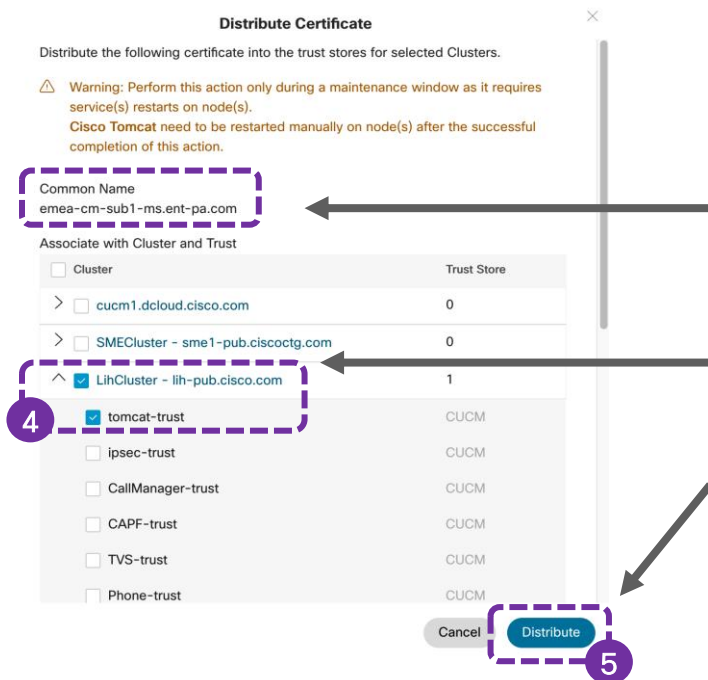
Certificate	Server Name	Common Name	Type	Status	Expiration	
tomcat	emea-cm-pub.ent-pa.com	emea-cm-sub1-ms.ent-pa...	CA Signed (Multi Server)	Valid	05/17/2026 1:27 PM	⋮
tomcat	emea-imp-sub1.ent-pa.com	emea-cm-sub1-ms.ent-pa...	CA Signed (Multi Server)	Valid	05/17/2026 1:27	Generate CSR
tomcat	EMEA-IMP-PUB.ent-pa.com	emea-cm-sub1-ms.ent-pa...	CA Signed (Multi Server)	Valid	05/17/2026 1:27	Download .der
tomcat	emea-cm-sub1.ent-pa.com	emea-cm-sub1-ms.ent-pa...	CA Signed (Multi Server)	Valid	05/17/2026 1:27	Download .pem

Distribute Certificate

Click on the three dots to display the available action and select "Distribute Certificate"

Use the filters as needed

Example: Redistribute a Certificate across multiple trust stores and clusters.



After you click in “Distribute certificate” a new window pops up to select the recipient clusters out of all the onboarded clusters, specific trust stores can be selected as well.

In this example the administrator selects LihCluster and tomcat-trust as recipients of the distributed certificate coming from EMEA cluster.

When ready click on “Distribute”

Example: Redistribute a Certificate across multiple trust stores and clusters.

Distribute Certificate

Distribute Certificate has been added to the Jobs list. View progress under the **Jobs** tab.

6

After clicking “Distribute”, the certificate task is added to the job list. Click on Jobs tab to see the status.

OK

< EMEAcluster - emea-cm-pub.ent-pa.com

Identity Certificate Trust Store

Jobs

Identity Certificates 27 total

Valid	Expires Soon	Expired
14	0	13

Trust Certificates 34 total

Valid	Expires Soon	Expired
20	0	14

Recent Jobs Current Month

Total	Pending	Complete
6	0	

Current Month 6 Jobs

Job List

Job Type	Node	Time Stamp	Certificate	Product
Distribute Certificate	emea-cm-pub.ent-pa...	08/02/2021 10:08 PM	tomcat	Unified CM

Status
Success

Distribute Certificate

● Success

Certificate tomcat
Product Unified CM
Node emea-cm-pub.ent-pa.com
Executed on 08/02/2021 10:08 PM

In the Jobs details you can confirm the status of the job and expand for details.

Certificate Distributed
LihCluster - lih-pub.cisco.com
tomcat Unified CM Success

Web RTMT

Contents

01 Overview

General Summary

02 Requirements

Pre-requisites to enable Web RTMT

03 Web RTMT in CCUC

04 Features

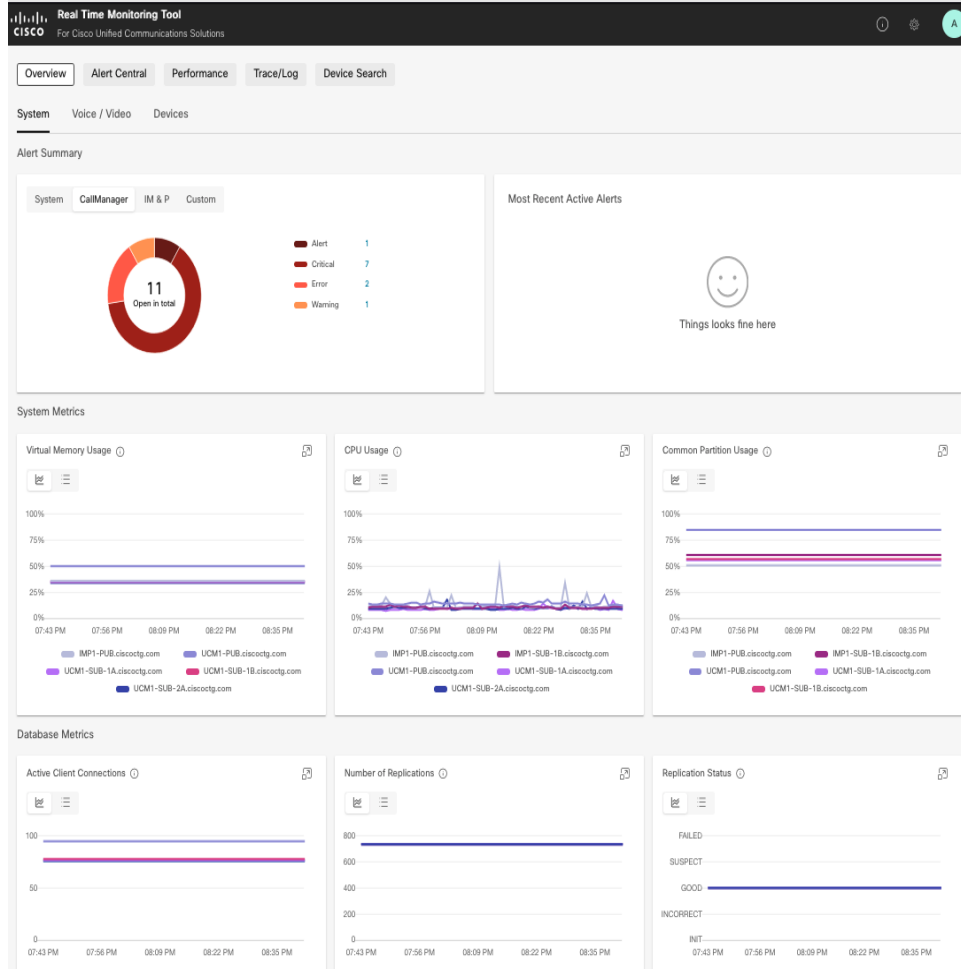
Overview - Alert Central - Performance - Trace and log -
Device Search

01 Overview

Web RTMT Overview

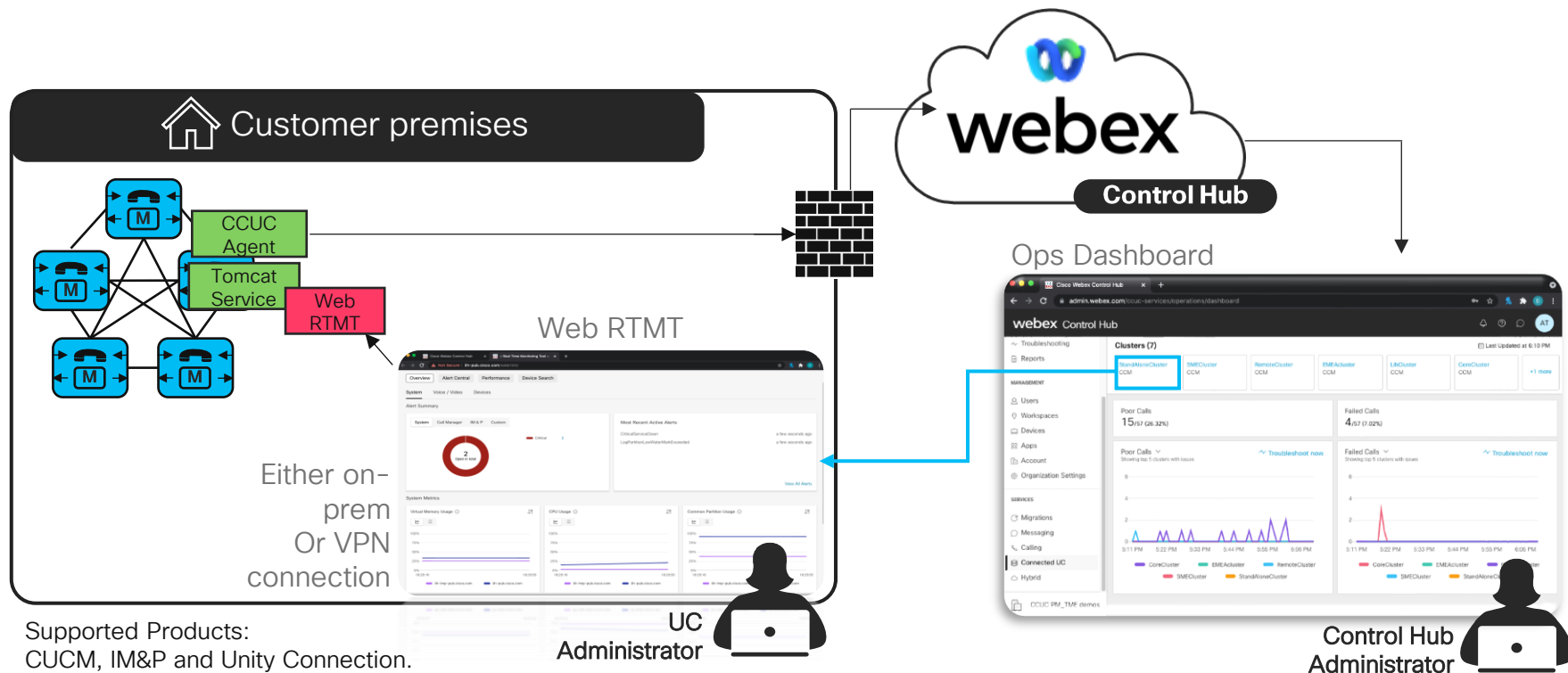
New tool that simplifies and enhances the administration and troubleshooting experience.

- Unified Web-based interface to monitor the real-time behavior of your system components, configure alerts to take suitable action in CUCM, IM&P and Unity Connection.
- Available via Operations Dashboard to troubleshoot and collect logs per cluster or node basis.
- No dependency on Java and OS/Browser platform.
- The data displayed remains on-prem.
- Cloud upgradeable via Cloud-Connected UC.



Web RTMT

High Level Overview



Features Overview



System Overview

Displays Alert Summary, Systems and Database objects, Voice/video and device metrics to monitor the health of the UC environment.



Alert Central

Displays Active Alerts, Alerts history and configuration of alerts settings.



Device Search

Allows administrators to search, filter and create/export custom reports.



Performance Monitoring Counters

Offers administrators to create multiple dashboards and add counters to monitor system components and services.



Trace and Log

Allows administrators to collect logs on demand or schedule log collection.

02 Requirements

Requirements

Platform

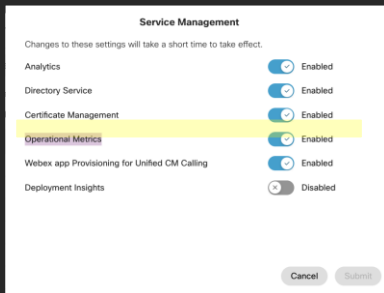
Supported versions:

- 11.5 SU9, 12.5 and 14 for CUCM, IM&P and CUC products
- Onboard cluster(s) to Webex® Cloud-Connected-UC.

[How to onboard clusters HERE](#)

Control Hub

- Administrator must enable Operational Metrics Service.
- Administrator must cross launch the Web RTMT from Operations Dashboards at least once to activate the service.



Connectivity

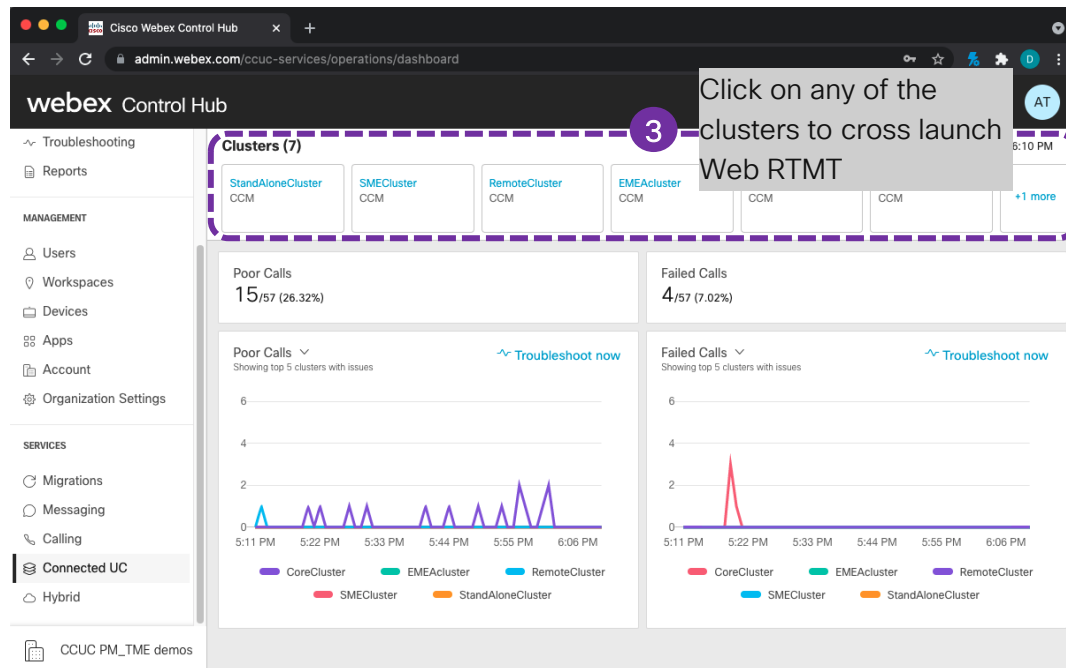
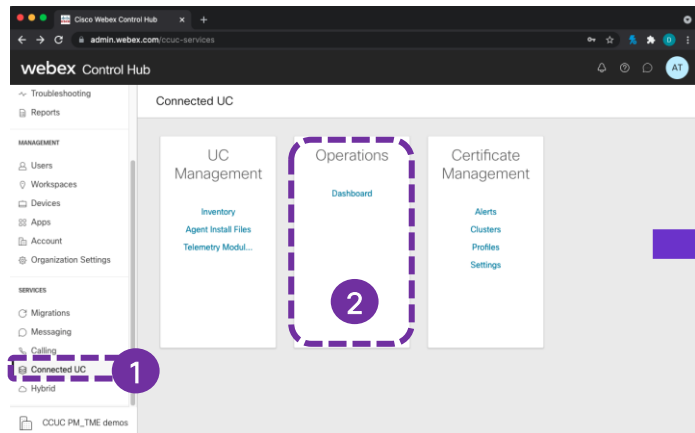
- Administrator will cross launch the Web RTMT from Operations Dashboard and authenticate with On-premises UC credentials.
- Local connection or VPN connectivity required to access Web RTMT.

03

Web RTMT in CCUC

Web RTMT in CCUC

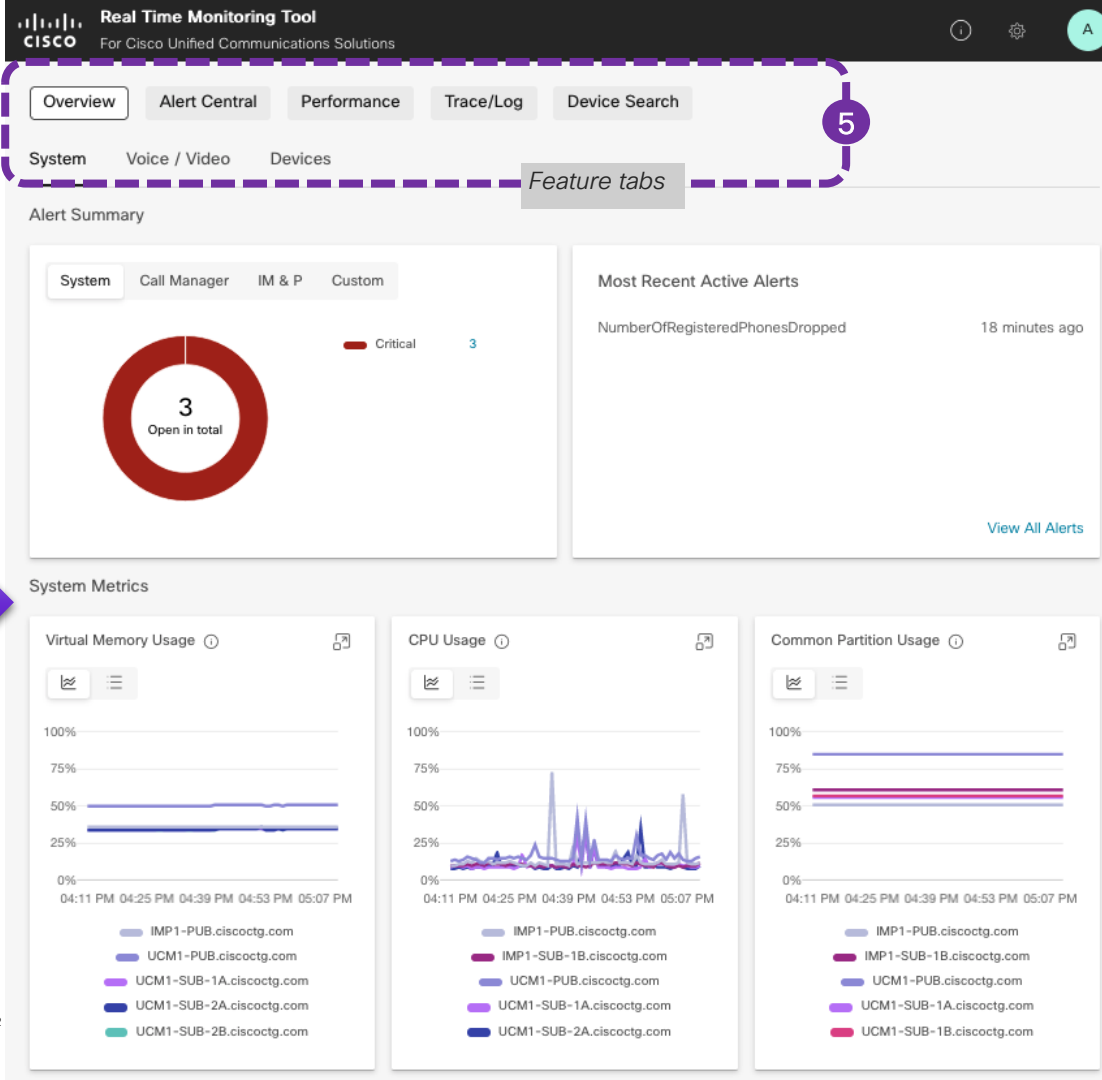
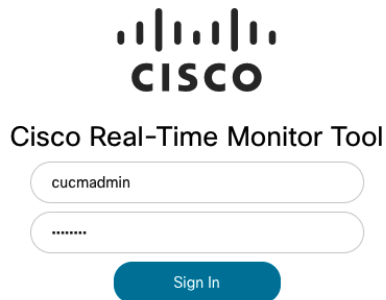
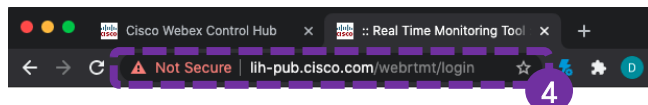
Navigate to Connected UC > Operations > Dashboard



Web RTMT in CCUC

*A new tab opens to authenticate an UC administrator

*VPN access or local connectivity required.



04 Features – Overview

Overview > System

Provides a dashboard with a Summary of alerts across several categories and a quick snapshot of the Most Recent Active Alerts.

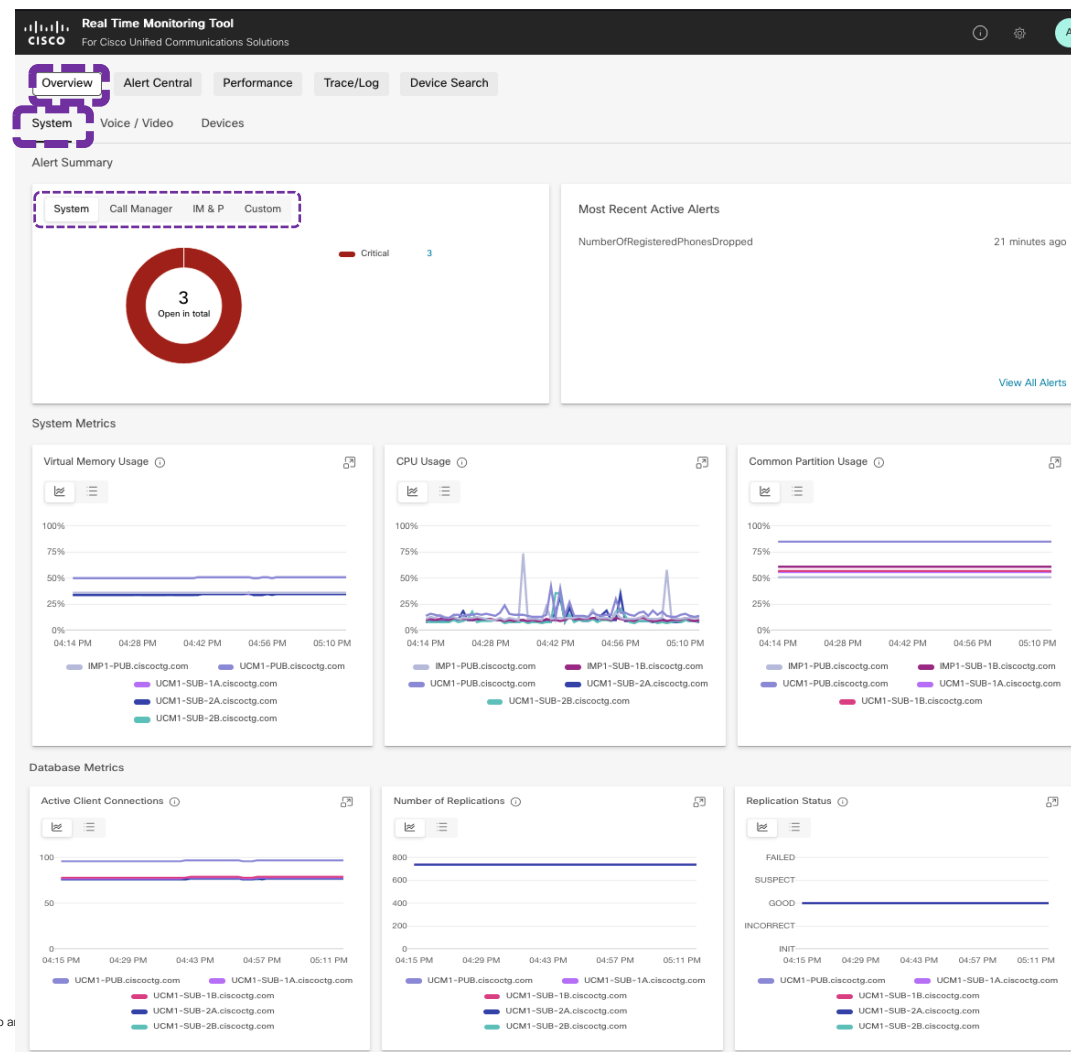
System tab displays System Metrics and Database Metrics for the below counters:

System Metrics:

- Virtual Memory Usage
- CPU Usage
- Common Partition Usage

Database Metrics

- Active Client Connections
- Number of Replications
- Replication Status



Overview > System

Web RTMT offers the option to switch charts between Chart View and Table View. Expanded view available as well by clicking in the desired icon on the right top corner of the chart.

Chart View

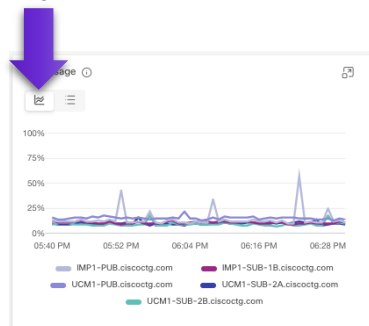
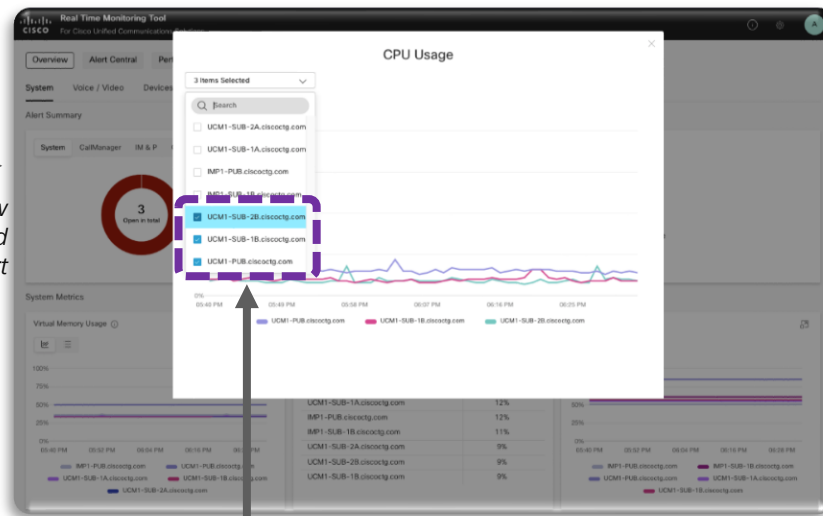


Table View

A table showing the same data as the chart view, but in a tabular format. The table has two columns: 'Servers' and 'Usage'. The data is as follows:

Servers	Usage
UCM1-PUB.ciscoctg.com	15%
IMP1-PUB.ciscoctg.com	12%
IMP1-SUB-1B.ciscoctg.com	11%
UCM1-SUB-2B.ciscoctg.com	10%
UCM1-SUB-2A.ciscoctg.com	9%
UCM1-SUB-1A.ciscoctg.com	9%
UCM1-SUB-1B.ciscoctg.com	9%

Click to view expanded chart



Expanded View:

- Select the desired nodes to display.
- Chart will update accordingly to the selection of nodes

Overview > Voice/Video

Voice/Video tab offers summary of voice and video calls statistics.

Available categories:

Call Activity

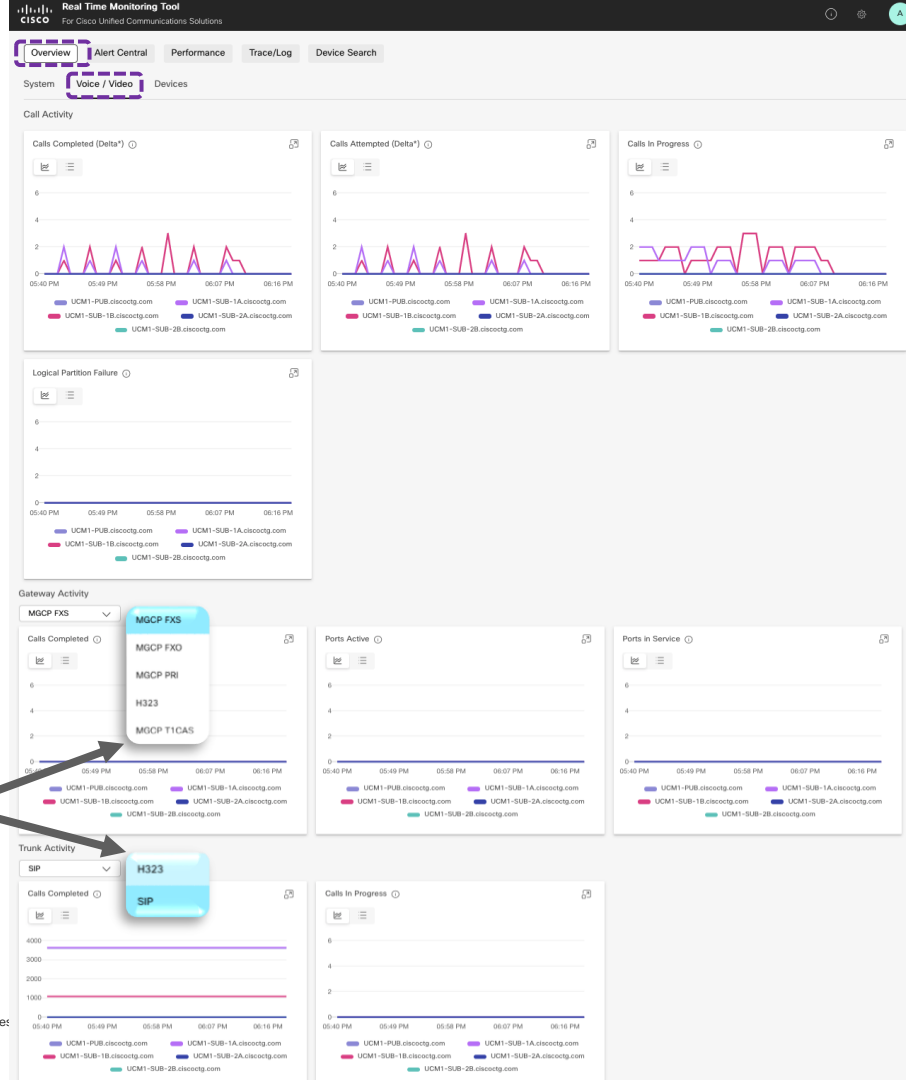
- Call Completed
- Calls Attempted
- Calls in Progress
- Logical Partition Failure

Gateway Activity

- Call Completed
- Ports active
- Ports in service

Trunk Activity

- Calls completed
- Calls in Progress



Overview > Devices

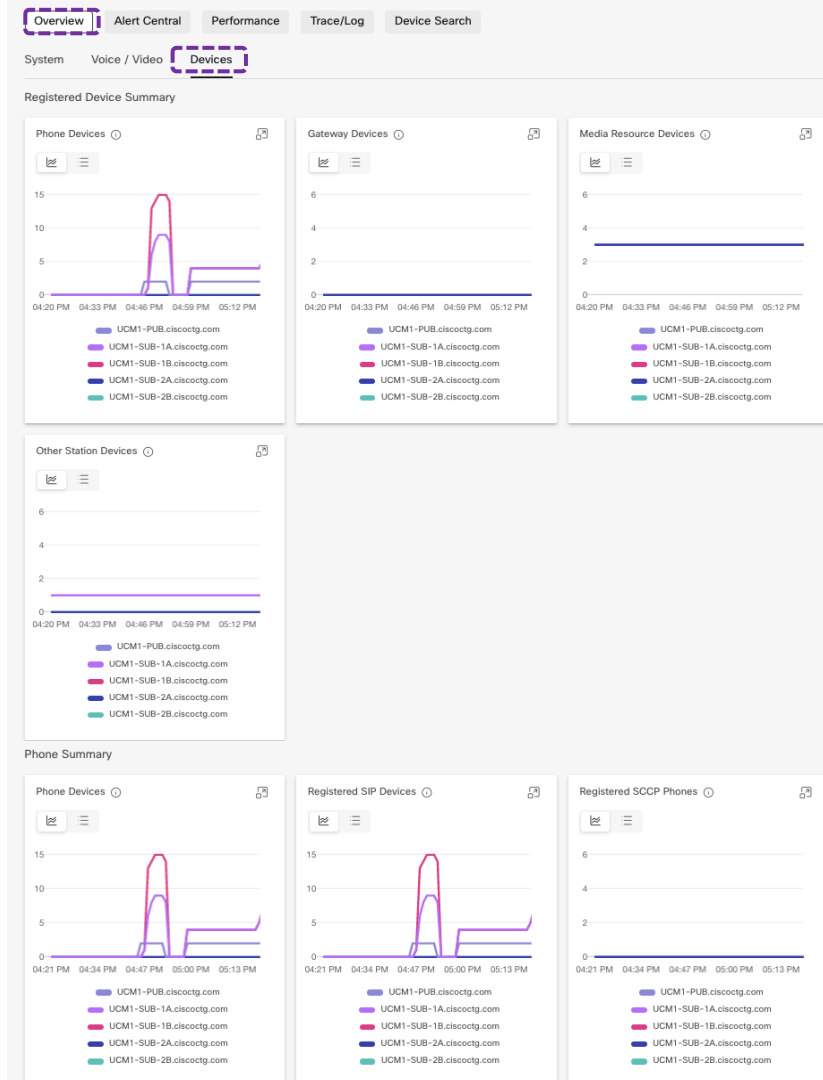
Devices Tab offers statistics for devices, sorted in the following categories:

Registered Device Summary

- Phone Devices
- Gateway Devices
- Media Resource Devices
- Other Station Devices

Phone Summary

- Phone Devices
- Registered SIP Devices
- Registered SCCP Phones
- Partially Registered Phones
- Failed Registration Attempts



04

Features – Alert Central

Alert Central > Active

Single view of active alerts across all nodes in your cluster to take suitable action

The screenshot displays the Cisco Real Time Monitoring Tool interface, specifically the Alert Central section. The top navigation bar includes tabs for Overview, Alert Central (selected), Performance, Trace/Log, and Device Search. Below this, the sub-navigation bar shows Active (selected), History, Settings, and Email Group. The main content area shows a list of 15 alerts, with 3 selected. A 'Clear' button is visible. The table lists alerts with columns for Last Alert Raised, Alert Name, Severity, Group, In Safe Range, and System Cleared. Three alerts are selected: 'NumberOfRegisteredPhonesDropped' (Critical), 'MediaListExhausted' (Warning), and 'SmartLicenseRenewAuthFailed' (Error). A detailed view of the 'SmartLicenseRenewAuthFailed' alert is shown below the table, including a timestamp and a description of the event. An 'Export' button is located in the top right corner of the alert list.

Clear Selected Alarms

Export report of Active Alerts

Alarm details

Last Alert Raised	Alert Name	Severity	Group	In Safe Range	System Cleared
Aug 30, 2021, 8:57:06 PM	NumberOfRegisteredPhonesDropped	Critical	CallManager	No	N/A
Aug 30, 2021, 5:33:36 PM	MediaListExhausted	Warning	CallManager	Yes	Aug 30, 2021, 6:30:36 PM
Aug 6, 2021, 4:47:25 AM	SmartLicenseRenewAuthFailed	Error	CallManager	No	N/A
Jun 28, 2021, 4:49:40 PM	SyslogSeverityMatchFound	Critical	System	No	N/A
Jun 26, 2021, 11:35:09 PM	SDLLinkOutOfService	Critical	CallManager	Yes	Jun 26, 2021, 11:35:39 PM
Jun 9, 2021, 7:47:01 AM	EspConfigAgentNetworkRestored	Warning	IM & P	No	N/A
Jun 9, 2021, 7:28:45 AM	ServerDown	Critical	System	Yes	Jun 9, 2021, 7:43:31 AM

At Mon Jun 28 17:49:40 EDT 2021 on node UCM1-SUB-1A.ciscoctg.com, the following CoreDumpFileFound events generated: TotalCoresFound : 2 CoreDetails : The following lists up to 6 cores dumped by corresponding applications. Core1 : Unknown (core.24788.11.sudo.1624916950) Core2 : Cisco Extended Functions (core.24790.11.csf.1624916950) AppID : Cisco Log Partition Monitoring Tool ClusterID : NodeID : UCM1-SUB-1A.ciscoctg.com TimeStamp : Mon Jun 28 17:49:14 EDT 2021

Alert Central > History

History Tab displays past alerts from the last 30 minutes in the systems.
Export alerts history to pdf or csv formats.

Real Time Monitoring Tool
For Cisco Unified Communications Solutions

Overview **Alert Central** Performance Trace/Log Device Search

Active **History** Settings Email Group

Total 2 Alerts **Export**

Time	Alert Name	Description	Severity	Group	Node	Role	Email
Aug 30, 2021, ...	MediaListExha...	Number of Me...	Warning	CallManager			N/A
Aug 30, 2021, ...	NumberOfRegi...	Number of regi...	Critical	CallManager			N/A

Alert Central > Settings

Settings tab allows administrators to select Alerts and set the desired configuration for each one in an easy to access UI

Each alarm can be associated with an email group and if required the email group can be configured on the fly.

Enable, Disable, Reset and Delete settings for each alarm in the same page.

Real Time Monitoring Tool
For Cisco Unified Communications Solutions

Overview Alert Central Performance Trace/Log Device Search

Active History Settings Email Group

Total 211 Alerts 1 Alert(s) Selected Enable Disable Reset Delete

Alert Name	Email Group	Status	Group
☒ HardwareFailure	Default	Enabled	System
☐ ICSCertificateCAConflict	Default	Enabled	IM & P
☐ ICSCertificateCASignedTrustCertFound	Default	Enabled	IM & P
☐ ICSCertificateFingerPrintMismatch	Default	Enabled	IM & P
☐ ICSCertificateValidationFailure	Default	Enabled	IM & P
☐ ILSDuplicateURI	Default	Enabled	CallManager
☐ ILSHubClusterUnreachable	Default	Enabled	CallManager
☐ ILSPeerLimitApproachingWarning	Default	Enabled	CallManager
☐ ILSPeerLimitExceeded	Default	Enabled	CallManager
☐ ILSPwdAuthenticationFailed	Default	Enabled	CallManager
☐ ILSTLSAuthenticationFailed	Default	Enabled	CallManager
☐ IMEDistributedCacheInactive	Default	Enabled	CallManager
☐ IMEOverQuota	Default	Enabled	CallManager
☐ IMEQualityAlert	Default	Enabled	CallManager

HardwareFailure

General

☒ Enabled Alert

Severity

Critical

Enable or Disable this alert

☒ UCM1-SUB-1B.ciscoctg.com

☒ UCM1-SUB-2A.ciscoctg.com

☒ IMP1-SUB-1B.ciscoctg.com

☒ UCM1-SUB-1A.ciscoctg.com

☒ UCM1-PUB.ciscoctg.com

☒ UCM1-SUB-2B.ciscoctg.com

☒ IMP1-PUB.ciscoctg.com

Frequency

Trigger alert on every poll

Trigger alert on every poll

Trigger alert up to 3 alert within 30 minutes

Schedule

☒ Trigger alert when it occurs. (Non-Stop Monitoring)

☐ Trigger alert everyday (Scheduled Monitoring) between

Start Time End Time

Email Notification

☒ Enable Email

Email Group

Default

New Group

Default

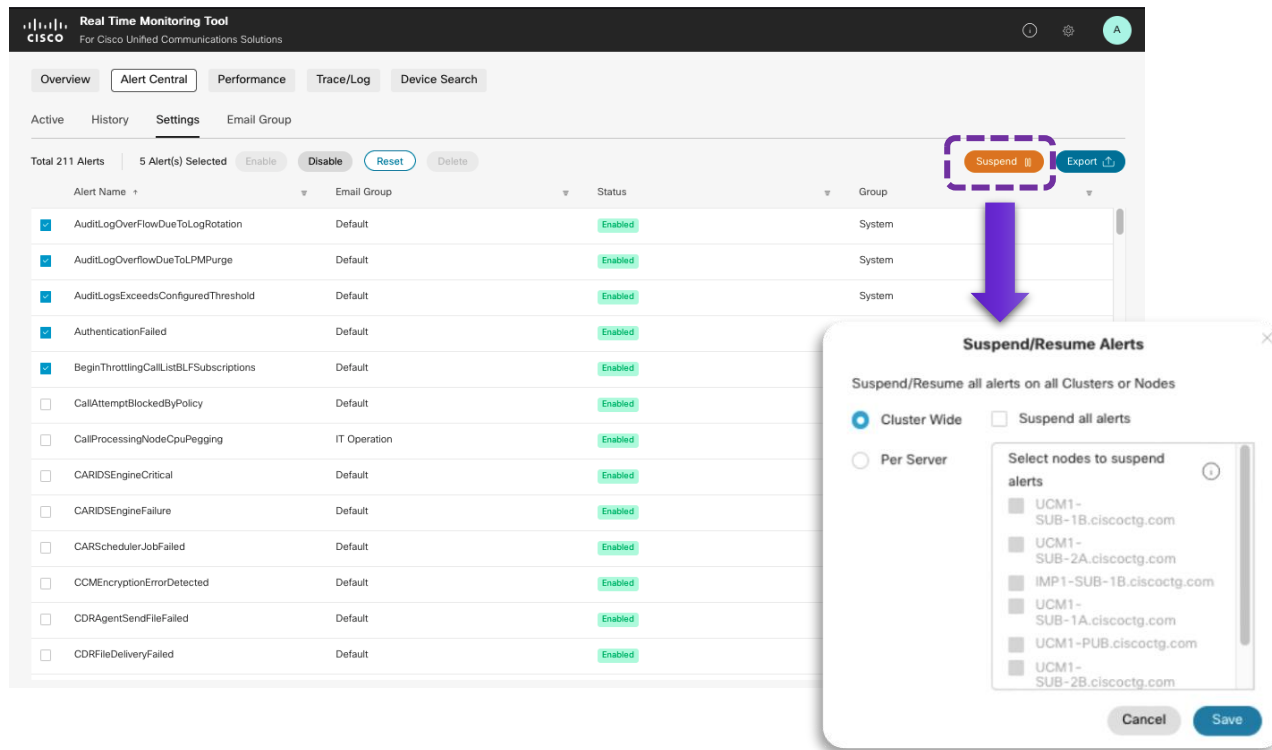
IT Operation

UC Team

Reset Save

Alert Central > Settings

Suspend button allows administrators to quickly suspend/resume alerts in the desired node, or cluster wide



Real Time Monitoring Tool
For Cisco Unified Communications Solutions

Overview | **Alert Central** | Performance | Trace/Log | Device Search

Active | History | **Settings** | Email Group

Total 211 Alerts | 5 Alert(s) Selected | Enable | Disable | **Reset** | Delete

Alert Name	Email Group	Status	Group
☒ AuditLogOverflowDueToLogRotation	Default	Enabled	System
☒ AuditLogOverflowDueToLPM Purge	Default	Enabled	System
☒ AuditLogsExceedsConfiguredThreshold	Default	Enabled	System
☒ AuthenticationFailed	Default	Enabled	
☒ BeginThrottlingCallListBLFSubscriptions	Default	Enabled	
☐ CallAttemptBlockedByPolicy	Default	Enabled	
☐ CallProcessingNodeCpuPegging	IT Operation	Enabled	
☐ CARIDEngineCritical	Default	Enabled	
☐ CARIDEngineFailure	Default	Enabled	
☐ CARSchedulerJobFailed	Default	Enabled	
☐ CCMEncryptionErrorDetected	Default	Enabled	
☐ CDRAgentSendFileFailed	Default	Enabled	
☐ CDRFileDeliveryFailed	Default	Enabled	

Suspend/Resume Alerts

Suspend/Resume all alerts on all Clusters or Nodes

☒ Cluster Wide ☐ Suspend all alerts

☐ Per Server

Select nodes to suspend alerts

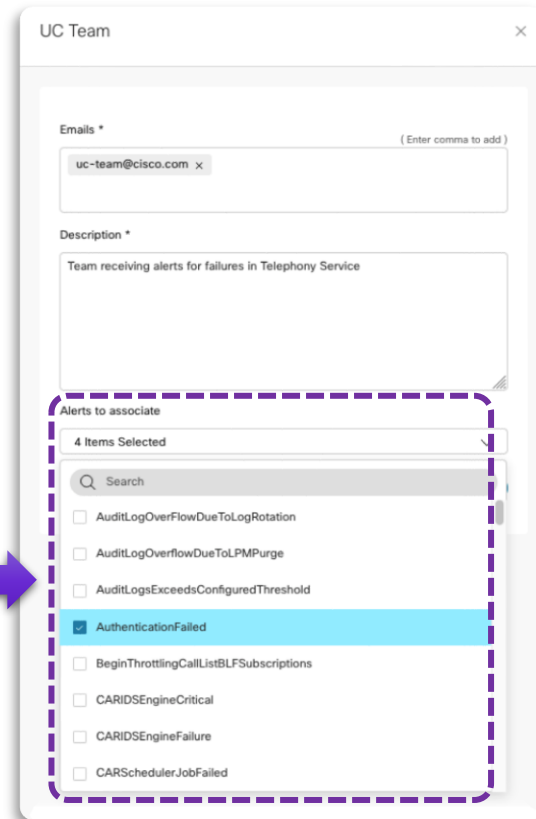
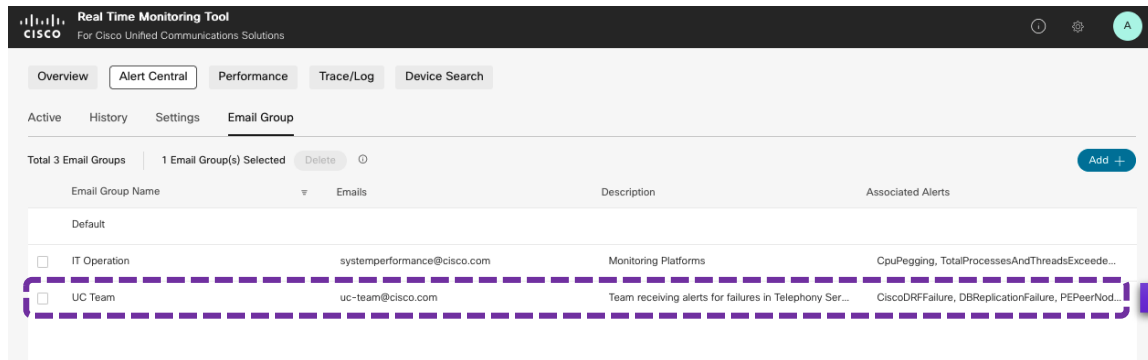
- ☐ UCM1 - SUB-1B.ciscoctg.com
- ☐ UCM1 - SUB-2A.ciscoctg.com
- ☐ IMP1-SUB-1B.ciscoctg.com
- ☐ UCM1 - SUB-1A.ciscoctg.com
- ☐ UCM1-PUB.ciscoctg.com
- ☐ UCM1 - SUB-2B.ciscoctg.com

Cancel Save

Alert Central > Email Group

Email group tab allows administrators to set up email groups and associate specific alerts per groups.

Now the alerts can be delivered to the specific group/team to take action.



04

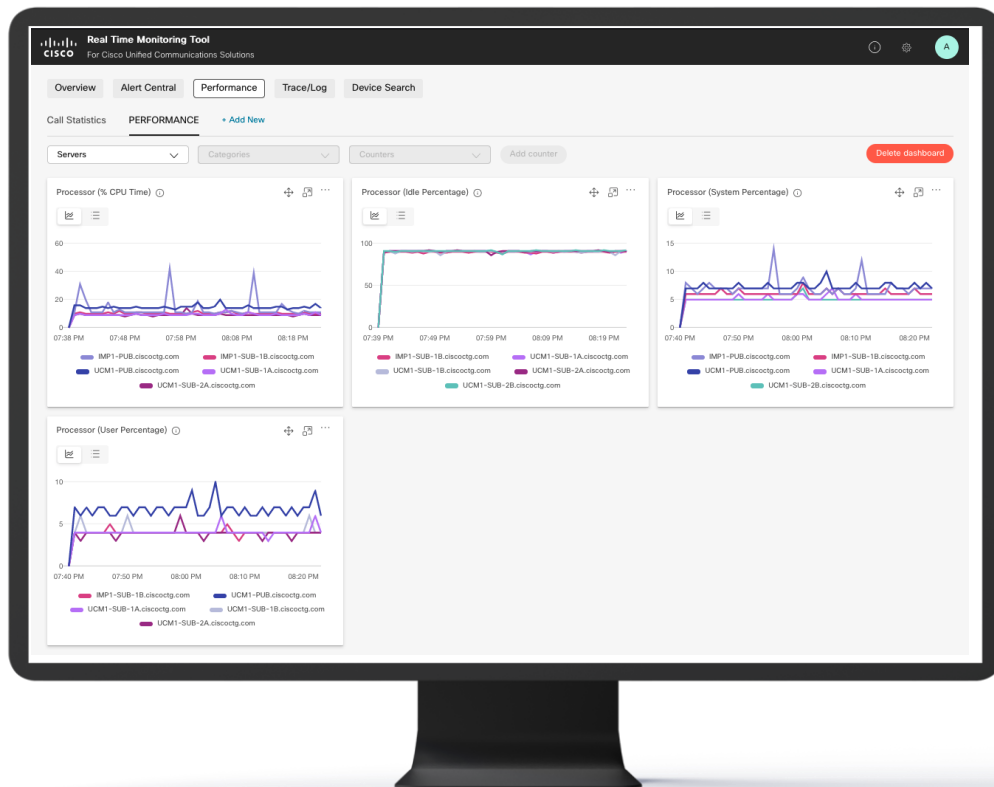
Features – Performance

Performance

Performance tab offers the capability to create multiple dashboards to view critical statistics, monitor and troubleshoot your environment.

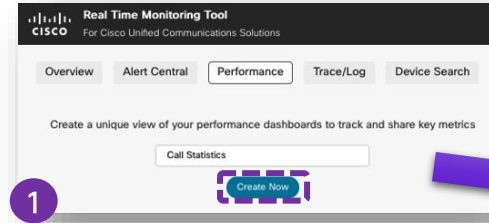
Easy and intuitive interface to add up to 16 counters based on categories with the flexibility to add a cluster or specific nodes only.

Grid view is flexible, administrator can rearrange counters, change chart views and edit counters.

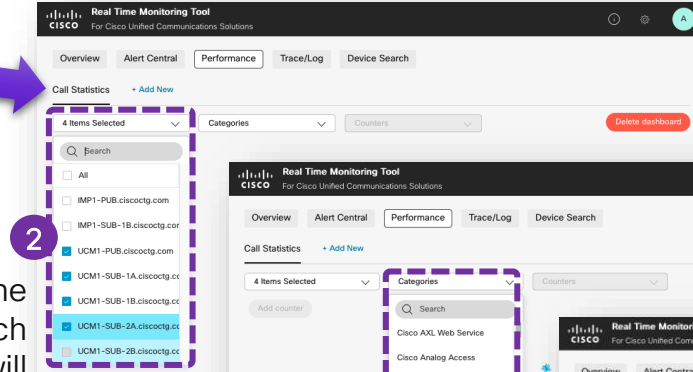


Performance Dashboard creation

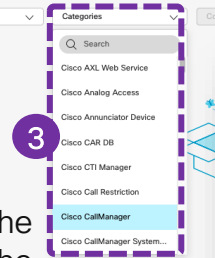
Configure a name for your dashboard and click on “Create Now”.



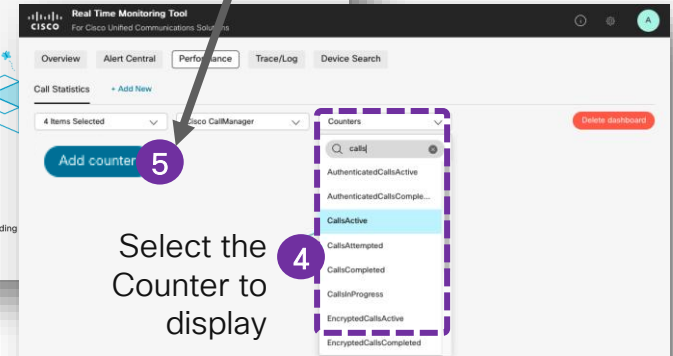
Select the node(s) on which the counters will be monitored



Choose the category of the Counters



Select the Counter to display

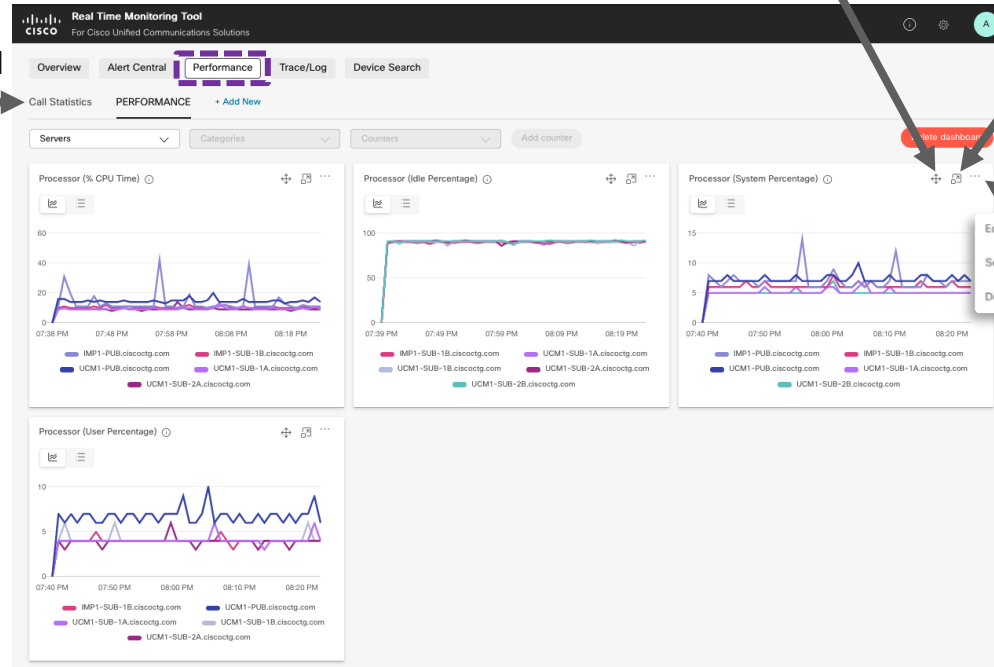


Add up to 16 counters, as you need, and rearrange the charts at your convenience.

Performance Dashboard creation

Now the Dashboard is created and new dashboards for different purposes can be created and displayed in multiple screens.

Dashboard created
+ Add New



Re-arrange
charts

Show
Extended View

Edit Counter
Set Alert Properties
Delete Counter

Edit Counters, set
alerts or delete it

04

Features – Trace and Log

Trace and Log > Service Logs

Intuitive interface to collect system, service and audit logs on demand or Schedule log collection

Service Logs:

- Logs for various services
- Crash dumps

Each type of log collection can be scheduled and logs are stored in SFTP server at scheduled time

The screenshot shows the 'Real Time Monitoring Tool' interface for Cisco Unified Communications Solutions. The 'Trace/Log' tab is selected. Under the 'Service Logs' sub-tab, the 'Collect Files' option is chosen. The interface is divided into several sections:

- Select System Services / Applications:** A dropdown menu.
- Select UCM Services / Applications:** A dropdown menu.
- Select IM & Presence Services / Applications:** A dropdown menu.
- Action:** Radio buttons for 'Download Logs' and 'Schedule Collection' (selected).
- Schedule Options:**
 - Schedule Time:** Includes a 'Select Reference Server Time Zone' dropdown (set to 'lth-imp-pub.cisco.com :: (GMT-8:0)Pacific Daylight Time-America/Los_Angeles') and date/time pickers for 'From' (31 Aug 2021, 12:00 PM) and 'To' (31 Aug 2021, 12:15 PM).
 - Schedule Frequency:** A dropdown menu set to 'Weekly'.
 - Collect Files generated in the last:** A dropdown menu set to '1' and a unit dropdown set to 'Hours'.
 - Search Options:** Includes a 'Search String' field, a 'Case Sensitive' checkbox, and a text input field.
 - Action Options:** Includes checkboxes for 'Zip Files' (checked), 'Delete Collected Logs from Server' (checked), and 'Generate Syslog' (unchecked).
- Trace Download Configuration:**
 - Radio buttons for 'Localhost' and 'SFTP/FTP Server' (selected).
 - Protocol:** A dropdown menu set to 'SFTP'.
 - Host IP Address:** A text input field containing '192.168.1.1'.
 - User Name:** A text input field containing 'admin'.
 - Password:** A text input field with masked characters.
 - Port:** A text input field containing '22'.
 - Download Directory Path:** A text input field containing '/'.
 - Buttons for 'Test Connection' and 'Reset'.

A 'Schedule' button is located at the bottom right of the interface.

Trace and Log > System Logs >

System Logs tab displays the option to collect:

- Install logs
- Audit logs

Schedule the download and specify a log collection time

Select the partition and decide to delete or not the collected logs from the server

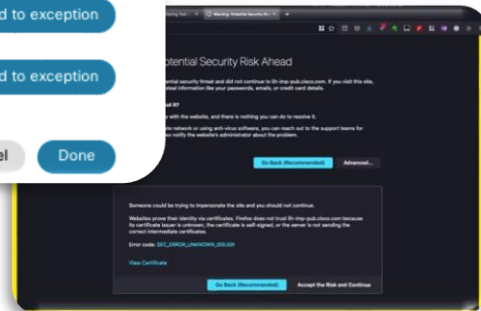
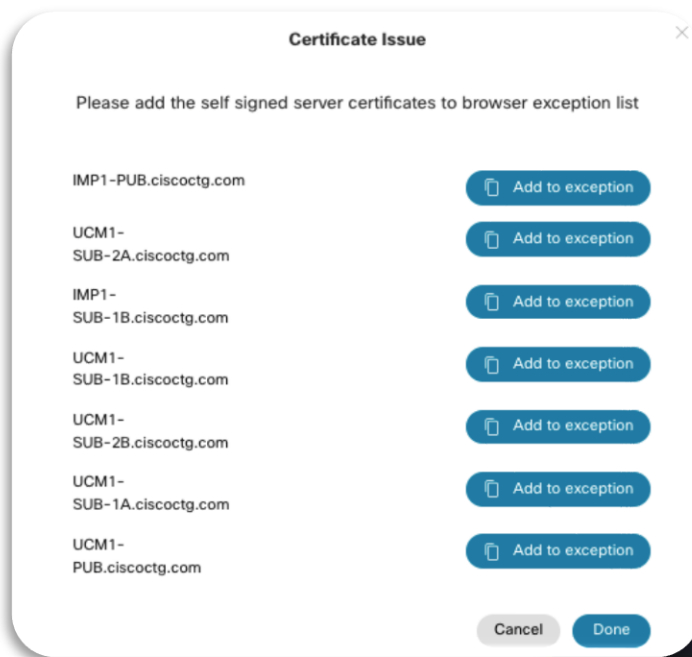
The screenshot shows the 'Real Time Monitoring Tool' interface for Cisco Unified Communications Solutions. The navigation path is highlighted with dashed purple boxes: 'Trace/Log' in the top menu, 'System Logs' in the sub-menu, and 'Audit Logs' in the further sub-menu. The 'Audit Logs' section contains the following options:

- Action:**
 - ☒ Download Audit Logs
 - ☐ Schedule Download of Audit Logs
- Select Server Option:**
 - ☒ ALL
 - ☐ lii-imp-pub.cisco.com
 - ☐ lii-pub.cisco.com
- Download Log Options:**
 - Collection Time:**
 - ☒ Absolute Range
 - ☐ Relative Range
 - Select Reference Server Time Zone:**
 - lii-imp-pub.cisco.com :: (GMT-8:0)Pacific Dayli...
 - Date / Time:**
 - From:** 31 Aug 2021 12:10 PM
 - To:** 31 Aug 2021 12:20 PM
 - Files Generated in last:**
 - 1 Hours
- Download File Options:**
 - Select Partition:**
 - Active
 - ☐ Delete Collected Logs from Server

A 'Download' button is located at the bottom right of the interface.

Trace and Log

Prior to download of logs from Web RTMT, add the self-signed certificates to the browser exception list if it's not already trusted.



04

Features – Device Search

Device Search

Displays an intuitive interface to search and filter devices in your environment from the same UI, no more wizards.

The screenshot shows the 'Device Search' interface of the 'Real Time Monitoring Tool For Cisco Unified Communications Solutions'. The interface includes a top navigation bar with tabs for Overview, Alert Central, Performance, Trace/Log, and Device Search. A dropdown menu for 'Phone' is open, showing options like Gateway Devices, H323 Devices, CTI Devices, Voice Mail Devices, Media Resources, and SIP Trunk. The 'Device Search' section has a 'Search By' dropdown set to 'Device Name' and a text input field containing 'SEPxxx'. A 'Search' button is located to the right. Below the search bar, there are three filter sections: 'Device Status' (with radio buttons for Registered, Unregistered, Partial Registered, Rejected, Any Status, and Device only Configured in Database), 'Call Manager' (with radio buttons for Any and various UCM1-SUB and UCM1-PUB entries), and 'Download Status' (with radio buttons for Any, Successful, Downloading, and Failed). A 'Reset filters' link is also present. At the bottom, there is a 'Specific Device Model' dropdown set to 'Cisco 8841'. To the right of the filters, there is a section titled 'Monitor following attributes:' with two columns of checkboxes. The first column includes Name, Node, IpAddress, Ipv6Address, LoginUserId, TimeStamp, ActiveLoadId, RequestedLoadId, and DownloadFailureReason. The second column includes Status, DirectoryNumber, Description, Model, StatusReason, Protocol, InactiveLoadId, DownloadStatus, and DownloadServer. Annotations with arrows point to various elements: 'Available filters' points to the filter sections; a dashed purple box highlights the 'Search By' dropdown and the 'Specific Device Model' dropdown; a dashed purple box highlights the 'Phone' dropdown menu; and a dashed purple box highlights the 'Search' button. A text box on the right states 'Flexible/Customizable reports can include all of some of these attributes' with an arrow pointing to the 'Monitor following attributes:' section. A search results list on the left shows a list of Cisco devices with their model numbers (e.g., Cisco 8811, Cisco 8821, Cisco 8831, Cisco 8832, Cisco 8832NR, Cisco 8841, Cisco 8845, Cisco 8851).

Available filters

Search By

Device Name

SEPxxx

Search

Additional Filters [Reset filters](#)

Device Status:

- ☒ Registered
- ☐ Unregistered
- ☐ Partial Registered
- ☐ Rejected
- ☐ Any Status
- ☐ Device only Configured in Database

Call Manager:

- ☒ Any
- ☐ UCM1-SUB-1A.ciscoctg.com
- ☐ UCM1-SUB-2A.ciscoctg.com
- ☐ UCM1-PUB.ciscoctg.com
- ☐ UCM1-SUB-1B.ciscoctg.com
- ☐ UCM1-SUB-2B.ciscoctg.com

Download Status:

- ☒ Any
- ☐ Successful
- ☐ Downloading
- ☐ Failed

Monitor following attributes:

☒ Name	☒ Status
☒ Node	☒ DirectoryNumber
☒ IpAddress	☒ Description
☐ Ipv6Address	☒ Model
☒ LoginUserId	☒ StatusReason
☐ TimeStamp	☒ Protocol
☐ ActiveLoadId	☐ InactiveLoadId
☐ RequestedLoadId	☐ DownloadStatus
☐ DownloadFailureReason	☐ DownloadServer

Flexible/Customizable reports can include all of some of these attributes

Search results list:

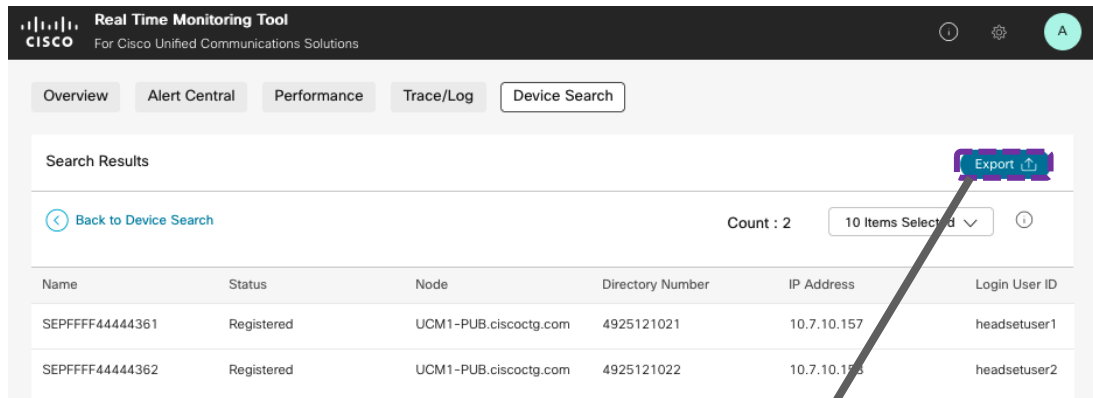
- Cisco 8811
- Cisco 8821
- Cisco 8831
- Cisco 8832
- Cisco 8832NR
- Cisco 8841
- Cisco 8845
- Cisco 8851

Device Search

- If there's any result matching the search, Web RTMT will display the results along with the selected attributes
- Export option also available, device inventory can be exported now to CSV or PDF formats
- Create reports with the attributes selected in the search:

Monitor following attributes:

☒ Name	☒ Status
☒ Node	☒ DirectoryNumber
☒ IpAddress	☒ Description
☐ Ipv6Address	☒ Model
☐ LoginUserId	☐ StatusReason
☐ TimeStamp	☐ Protocol
☐ ActiveLoadId	☐ InactiveLoadId
☐ RequestedLoadId	☐ DownloadStatus
☒ DownloadFailureReason	☐ DownloadServer



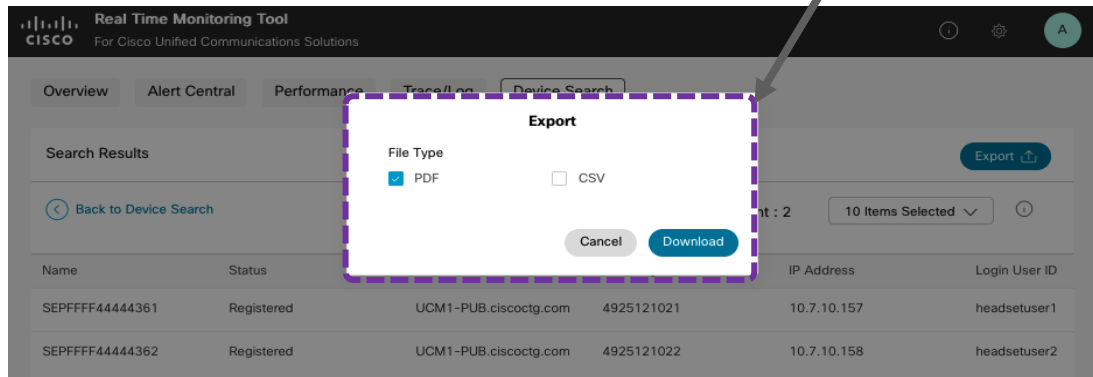
Real Time Monitoring Tool
For Cisco Unified Communications Solutions

Overview Alert Central Performance Trace/Log Device Search

Search Results

[Back to Device Search](#) Count : 2 10 Items Selected

Name	Status	Node	Directory Number	IP Address	Login User ID
SEPFFFF44444361	Registered	UCM1-PUB.ciscoctg.com	4925121021	10.7.10.157	headsetuser1
SEPFFFF44444362	Registered	UCM1-PUB.ciscoctg.com	4925121022	10.7.10.158	headsetuser2



Real Time Monitoring Tool
For Cisco Unified Communications Solutions

Overview Alert Central Performance Trace/Log Device Search

Search Results

[Back to Device Search](#) Count : 2 10 Items Selected

Export

File Type

☒ PDF ☐ CSV

Cancel Download

Name	Status	Node	Directory Number	IP Address	Login User ID
SEPFFFF44444361	Registered	UCM1-PUB.ciscoctg.com	4925121021	10.7.10.157	headsetuser1
SEPFFFF44444362	Registered	UCM1-PUB.ciscoctg.com	4925121022	10.7.10.158	headsetuser2

Troubleshooting

Troubleshooting Contents

01 Overview

General Summary

02 Prerequisites

Necessary steps to get started with Troubleshooting

03 Troubleshooting in CCUC

04 Analysis Components

05 Serviceability Connector Deployment

01 Overview

CCUC troubleshooting goals



Simplify data collection

Support log collection based on symptoms



Drill down into calls with failures and/or quality issues



Admin productivity

Focus on improving MTTR, Single pane of glass



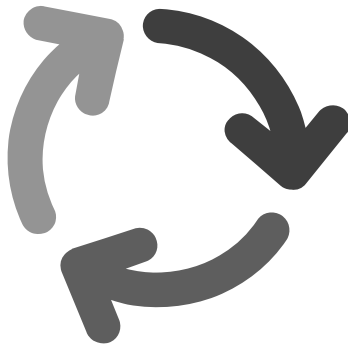
Analysis results

organized in tabs to provide visuals and consolidated data

CCUC and CX CSA integration

CCUC Integrates with Collaboration Solutions Analyzer (CSA) to collect and analyze SDL logs and provide results directly in the Control Hub UI

Metrics and Events from
Webex Cloud-Connected UC



Log collection using Webex Serviceability
Connector

Log analysis using Cisco Collaboration
Solutions Analyzer (CSA)



Collaboration Solutions Analyzer

The suite for your collaboration solution

Tools What is new About CSA



We recommend Google Chrome for a better experience while using CSA.

Log analysis

Upload logs from your collaboration devices to automatically detect, troubleshoot and resolve issues

CollabEdge validator

Speed up your Mobile and Remote Access feature deployment or troubleshooting by doing a step-by-step validation

SRV checker

Check your public domain for DNS service records and connectivity for various collaboration services

B2B call tester

Test inbound and outbound calls to and from your deployment

Cisco Collaboration Solution Analyzer and Serviceability Connector

What is CSA?

A suite of tools for troubleshooting and validating your Cisco Collaboration deployments.

<https://cway.cisco.com/csa/>

What is Webex Serviceability Connector?

Serviceability Connector is a virtual machine that resides in your network ('connector host'). It securely collects data/logs from your on-premises equipment based on the request, for analysis by CSA. cisco.com/go/serviceability

CCUC and CX CSA integration

Webex Control Hub
Onboarding
Trigger Data
Collection

Troubleshooting

Filter

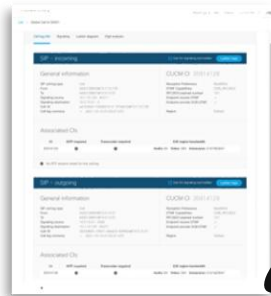
Jul 18, 2021 7:48 PM No Jul 24, 2021 8:48 PM (GMT -05:00) America/Mexico_City Showing 1 - 200 of 1680 2 3

Calling number	Log window	Called number	Final called number	Call status	Reason for failure	Call quality	Start time
4905120024	Collect logs	4905120022	4905120022	Success	N/A	Good	Jul 24, 2021 06:09 pm
4905120024	Collect logs	4905120023	4905120023	Success	N/A	Good	Jul 24, 2021 06:07 pm
4905120024	Collect logs	4905120027	4905120027	Success	N/A	Good	Jul 24, 2021 06:04 pm
4905121017	Collect logs	4905121013	4905121013	Success	N/A	Good	Jul 24, 2021 06:04 pm
4905121012	Collect logs	4905121015	4905121015	Success	N/A	Good	Jul 24, 2021 06:03 pm
4905120024	Collect logs	4905121016	4905121016	Success	N/A	Good	Jul 24, 2021 06:00 pm
4905121017	Collect logs	4905121018	4905121018	Success	N/A	Good	Jul 24, 2021 05:59 pm

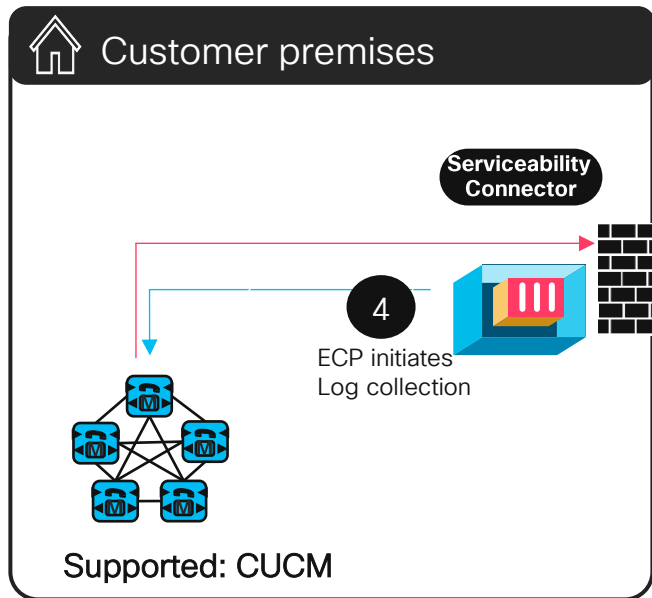


2

Admin triggers log collection



CH renders analysis results

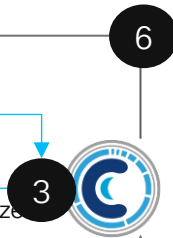


Analytics
+ Ops meta

1



Collaboration Solutions Analyzer
Trigger Data Collection



6 CSA analyze logs and send them to CH

5

Logs



Cisco Storage Connect

API Data Collection request

Webex Cloud-Connected UC metrics/events

API Data Collected

02 Prerequisites

Requirements

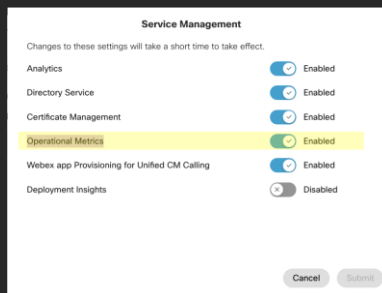
Platform

Minimum supported versions:

- UCM 11.5(x), 12.0(x), 12.5(x), 14(x)
- UCM cluster onboarded
- Install Serviceability connector using the ECP container (See ECP section for details)
- Add Call Processing nodes to ECP for log collection
- Trace Level set up to detail

Control Hub

- Administrator has to enable Operational Metrics (PII Opt-in)



Connectivity

- For a larger cluster, ensure that you have at least 2 MBPS of bandwidth.
- Cloud-Connected UC cloud port 443 must be reachable from customer environment.

*.ucmgmt.cisco.com

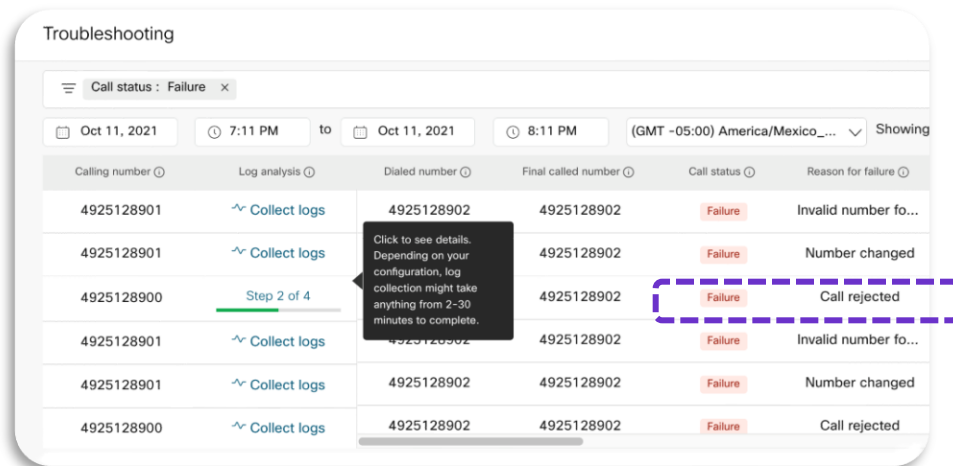
*.webex.com

03 Troubleshooting in CCUC

CCUC troubleshooting-> call issue diagnostics

- ✓ Drill down of call failure and call quality issues by user-id*, calling/called party, failure cause, quality status and more..
- ✓ Correlate CDRs and CMRs for a given end to end call.
- ✓ Admin can trigger collecting relevant logs based on call issue from various nodes
- ✓ Provide log analysis of various call legs including a visual ladder diagram of the call

Cross launch available from operations Dashboards*



Troubleshooting

Call status: Failure x

Oct 11, 2021 7:11 PM to Oct 11, 2021 8:11 PM (GMT -05:00) America/Mexico... Showing

Calling number	Log analysis	Dialed number	Final called number	Call status	Reason for failure
4925128901	Collect logs	4925128902	4925128902	Failure	Invalid number fo...
4925128901	Collect logs	4925128902	4925128902	Failure	Number changed
4925128900	Step 2 of 4	4925128902	4925128902	Failure	Call rejected
4925128901	Collect logs	4925128902	4925128902	Failure	Invalid number fo...
4925128901	Collect logs	4925128902	4925128902	Failure	Number changed
4925128900	Collect logs	4925128902	4925128902	Failure	Call rejected

Click to see details. Depending on your configuration, log collection might take anything from 2-30 minutes to complete.

Troubleshooting in CCUC

CH > Monitoring > Troubleshooting > Connected UC

webex Control Hub

All troubleshooting alerts are now in the [notification center](#).

Troubleshooting

Meetings & Calls Status: **Connected UC**

Filter

Oct 11, 2021 7:09 PM to Oct 11, 2021 8:09 PM (GMT -05:00) America/Mexico... Showing 1 - 12 of 12

Calling number	Log analysis	Dialed number	Final called number	Call status	Reason for failure	Call quality	Start time
4925128902	Collect logs	4925128900	4925128900	Success	N/A	Poor	Oct 11, 2021 08:04 pm
4925128900	Collect logs	4925128901	4925128901	Success	N/A	Poor	Oct 11, 2021 08:03 pm
4925128901	Collect logs	4925128902	4925128902	Failure	Invalid number fo...	Ungraded	Oct 11, 2021 08:03 pm
4925128901	Collect logs	4925128902	4925128902	Failure	Number changed	Ungraded	Oct 11, 2021 08:02 pm
4925128900	Collect logs	4925128902	4925128902	Failure	Call rejected	Ungraded	Oct 11, 2021 08:02 pm
4925128902	Collect logs	4925128900	4925128900	Success	N/A	Poor	Oct 11, 2021 07:37 pm
4925128900	Collect logs	4925128901	4925128901	Success	N/A	Poor	Oct 11, 2021 07:36 pm
4925128902	Collect logs	4925128900	4925128900	Success	N/A	Poor	Oct 11, 2021 07:34 pm
4925128900	Collect logs	4925128901	4925128901	Success	N/A	Poor	Oct 11, 2021 07:33 pm
4925128901	Collect logs	4925128902	4925128902	Failure	Invalid number fo...	Ungraded	Oct 11, 2021 07:33 pm
4925128901	Collect logs	4925128902	4925128902	Failure	Number changed	Ungraded	Oct 11, 2021 07:32 pm
4925128900	Collect logs	4925128902	4925128902	Failure	Call rejected	Ungraded	Oct 11, 2021 07:32 pm

Troubleshooting dashboard

Filter bar to drill down into specific calls

Log collection relies on the availability of logs in the clusters

Information displayed contains
Calling number, Dialed number,
Final called number, Call status,
Reason for failure, call quality, call
general info, device type, cluster
name, UCM ip address and more

Troubleshooting

Filter

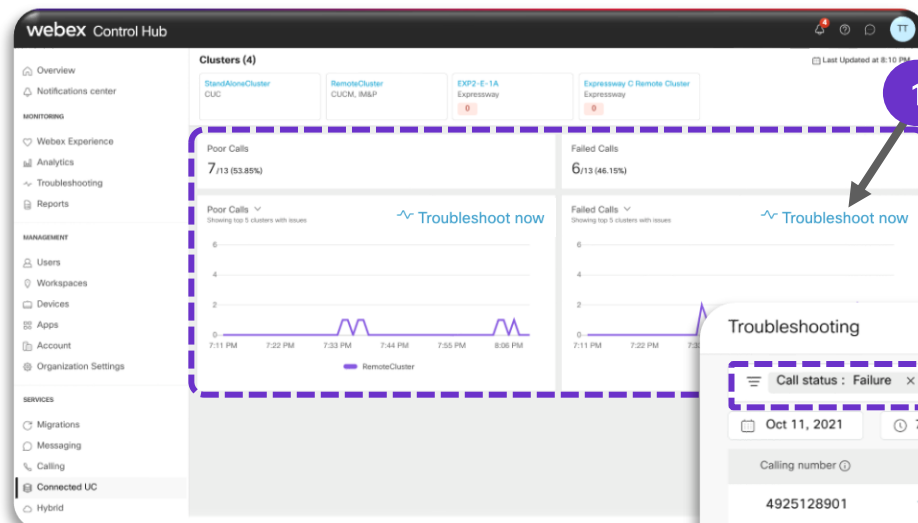
Oct 11, 2021 7:11 PM to Oct 11, 2021 8:11 PM (GMT -05:00) America/Mexico_City Showing 1 - 14 of 14

Calling number	Log analysis	Final called number	Call status	Reason for failure	Call quality	Start time	Duration	End time	Call ID	Source device	Destination device	Cluster name
4925128900	Collect logs	4925128900	Success	N/A	Poor	Oct 11, 2021 08:07 pm	31 sec	Oct 11, 2021 08:07 pm	58993	SEPDA4905129902	SEPDA4905129903	RemoteCluster
4925128900	Collect logs	4925128900	Success	N/A	Poor	Oct 11, 2021 08:07 pm	31 sec	Oct 11, 2021 08:07 pm	58992	SEPDA4905129904	SEPDA4905129902	RemoteCluster
4925128900	Show analysis	4925128901	Success	N/A	Poor	Oct 11, 2021 08:07 pm	31 sec	Oct 11, 2021 08:07 pm	58991	SEPDA4905129902	SEPDA4905129903	RemoteCluster
4925128901	Collect logs	4925128902	Failure	Invalid number fo...	Ungraded	Oct 11, 2021 08:03 pm	0 sec	Oct 11, 2021 08:03 pm	58990	SEPDA4905129903	-	RemoteCluster
4925128901	Collect logs	4925128902	Failure	Number changed	Ungraded	Oct 11, 2021 08:02 pm	0 sec	Oct 11, 2021 08:02 pm	58989	SEPDA4905129903	-	RemoteCluster
4925128900	Show analysis	4925128902	Failure	Call rejected	Ungraded	Oct 11, 2021 08:02 pm	0 sec	Oct 11, 2021 08:02 pm	58988	SEPDA4905129902	-	RemoteCluster
4925128902	Collect logs	4925128900	Success	N/A	Poor	Oct 11, 2021 07:37 pm	31 sec	Oct 11, 2021 07:37 pm	58987	SEPDA4905129904	SEPDA4905129902	RemoteCluster
4925128900	Collect logs	4925128901	Success	N/A	Poor	Oct 11, 2021 07:36 pm	31 sec	Oct 11, 2021 07:37 pm	58986	SEPDA4905129902	SEPDA4905129903	RemoteCluster
4925128902	Collect logs	4925128900	Success	N/A	Poor	Oct 11, 2021 07:34 pm	31 sec	Oct 11, 2021 07:34 pm	58985	SEPDA4905129904	SEPDA4905129902	RemoteCluster
4925128900	Collect logs	4925128901	Success	N/A	Poor	Oct 11, 2021 07:33 pm	31 sec	Oct 11, 2021 07:34 pm	58984	SEPDA4905129902	SEPDA4905129903	RemoteCluster
4925128901	Collect logs	4925128902	Failure	Invalid number fo...	Ungraded	Oct 11, 2021 07:33 pm	0 sec	Oct 11, 2021 07:33 pm	58983	SEPDA4905129903	-	RemoteCluster
4925128901	Show analysis	4925128902	Failure	Number changed	Ungraded	Oct 11, 2021 07:32 pm	0 sec	Oct 11, 2021 07:32 pm	58982	SEPDA4905129903	-	RemoteCluster
4925128900	Collect logs	4925128902	Failure	Call rejected	Ungraded	Oct 11, 2021 07:32 pm	0 sec	Oct 11, 2021 07:32 pm	58981	SEPDA4905129902	-	RemoteCluster

web

Cross launch troubleshooting from CCUC dashboard

Services > Connected UC > Dashboard



The dashboard shows by default “Poor Calls” and “Failed calls” Click on Troubleshoot now to to cross launch troubleshooting feature.

A new tab is open, and the calls listed are accordingly filtered

Click on “Collect Logs” to initiate the log collection of the call in question

3

The screenshot shows the 'Troubleshooting' tab. At the top, there's a filter bar with 'Call status : Failure' selected. Below this is a date and time range selector (Oct 11, 2021, 7:11 PM to Oct 11, 2021, 8:11 PM) and a dropdown for time zone (GMT-07:00 America/M...). The main table lists calls with columns: Calling number, Log analysis, Dialed number, Final called number, Call status, Reason for failure, and Call quality. A dashed purple box highlights the 'Log analysis' column, with a callout '2' pointing to the 'Collect logs' link for the first row. Another dashed purple box highlights the 'Call status' and 'Reason for failure' columns, with a callout '3' pointing to the 'Collect logs' link for the first row.

Calling number	Log analysis	Dialed number	Final called number	Call status	Reason for failure	Call quality
4925128901	Collect logs	4925128902	4925128902	Failure	Invalid number fo...	Ungraded
4925128901	Collect logs	4925128902	4925128902	Failure	Number changed	Ungraded
4925128900	Collect logs	4925128902	4925128902	Failure	Call rejected	Ungraded
4925128901	Collect logs	4925128902	4925128902	Failure	Invalid number fo...	Ungraded
4925128901	Collect logs	4925128902	4925128902	Failure	Number changed	Ungraded
4925128900	Collect logs	4925128902	4925128902	Failure	Call rejected	Ungraded

Show Analysis (1)

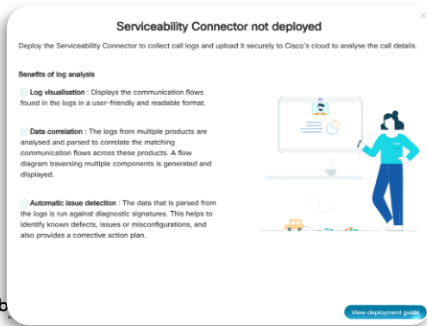
Note: If logs are overwritten ECP won't collect anything from the UCM clusters

Note: Currently only 2 log collections can be run simultaneously

Note: If Serviceability Connector is not deployed the administrator get a pop up and gets redirected to the deployment guide

After starting log collection in point #3 the progress of the task is shown, once it's ready the administrator can click on Show Analysis to launch the results.

4



Troubleshooting

Call status : Failure x

Oct 11, 2021 7:11 PM to Oct 11, 2021 8:17 PM (GMT -05:00) America/Mexico... Showing 1 - 6 of 6

Calling number	Log analysis	Dialed number	Final called number	Call status	Reason for failure	Call quality
4925128901	Collect logs		3902	Failure	Invalid number fo...	Ungraded
4925128901	Collect logs		3902	Failure	Number changed	Ungraded
4925128900	Collect logs		3902	Failure	Call rejected	Ungraded
4925128901	Collect logs		3902	Failure	Invalid number fo...	Ungraded
4925128901	Collect logs	4925128902	4925128902	Failure	Number changed	Ungraded
4925128901	Collect logs	4925128902	4925128902	Failure	Call rejected	Ungraded

Step 2 of 4

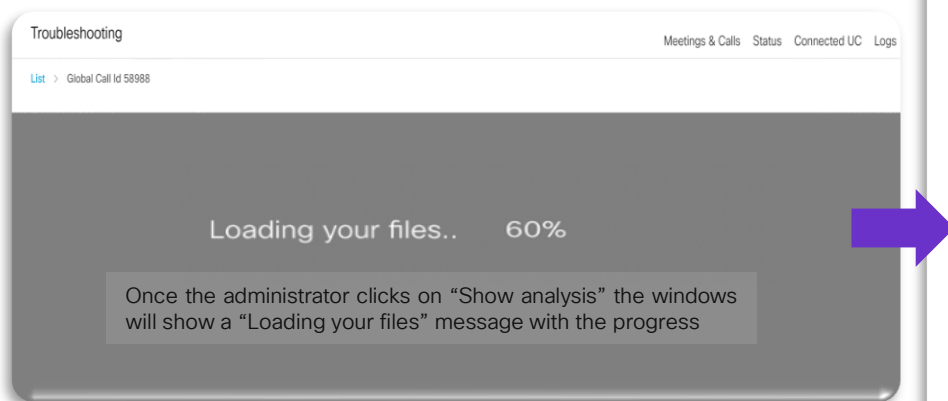
Click to see details. Depending on your configuration, log collection might take anything from 2-30 minutes to complete.

Show analysis

Depends on a number of factors such as call duration, number of clusters, communication to the cloud etc.

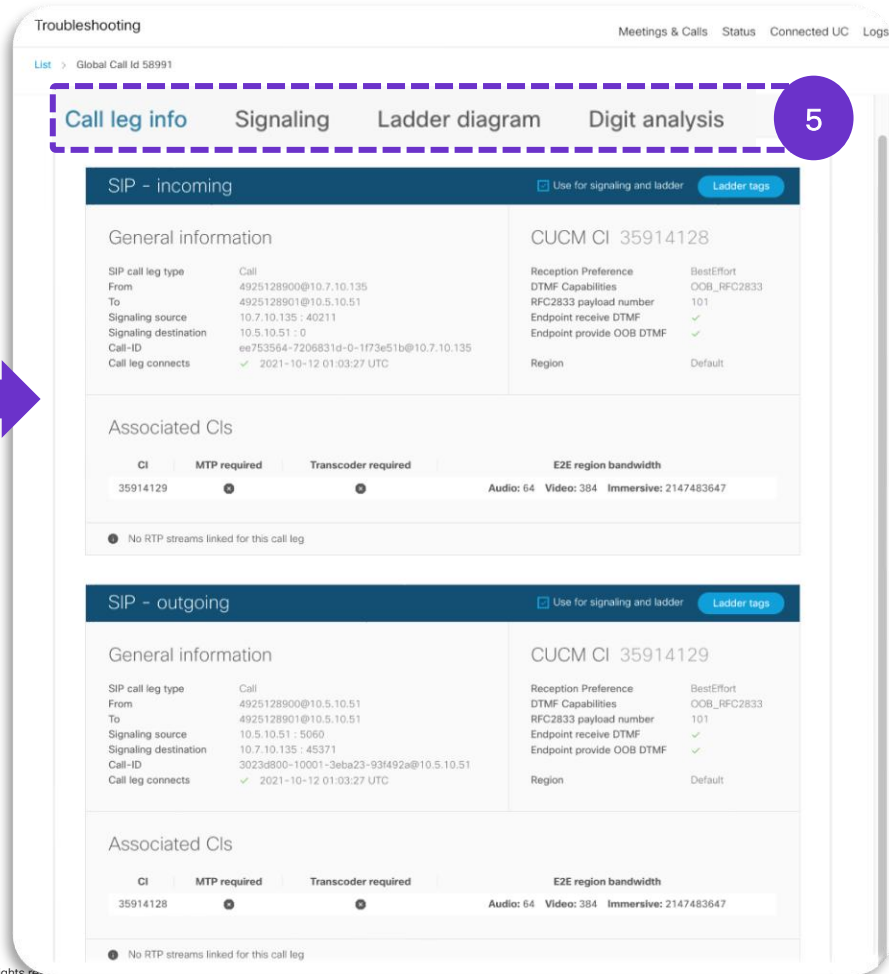
© 2022 Cisco and/or its affiliates. All rights reserved. Cisco Confidential

Show Analysis (2)



The results will be displayed organized in tabs:

1. Call Leg info
2. Signaling
3. Ladder Diagram
4. Digit Analysis



Show Analysis (2) - Warnings

Troubleshooting

Meetings

Call status : Failure x						
Oct 18, 2021	8:40 AM	to	Oct 18, 2021	9:40 AM	(GMT -05:00) America/Mexico_City	Showing 1 - 6 of 6
Calling number	Log analysis	Dialed number	Final called number	Call status	Reason for failure	Call quality
4925128901	Show analysis	4925128902	4925128902	Failure	Invalid number fo...	Ungraded
4925128901	Show analysis	4925128902	4925128902	Failure	Number changed	Ungraded
4925128900	Collect logs	4925128902	4925128902	Failure	Call rejected	Ungraded
4925128901	Collect logs	4925128902	4925128902	Failure	Invalid number fo...	Ungraded
4925128901	Collect logs	4925128902	4925128902	Failure	Number changed	Ungraded
4925128900	Collect logs	4925128902	4925128902	Failure	Call rejected	Ungraded

Log collection status for call ID 61194

Depending on your configuration, the log collection might take anything from 2-30 minutes to complete.

Step 4 of 4

Triggered

Collecting

Uploading

Completed

Log collection completed with errors. This could lead to partial analysis.

Transaction Id: 383TEPHWHI29MFEWM5SX

To restart the log collection for this call, click **Retry**.

Hide errors (1)

Node: 10.6.10.51

Error: The connection to the remote server could not be established. Please verify the logs for more information.

Retry

Close

04 Analysis components

Call leg info

Call Leg Info tab displays
the call information per
SIP leg that was
established.

Troubleshooting

Meetings & Calls Status Connected UC Logs

List > Global Call Id 58991

Call leg info Signaling Ladder diagram Digit analysis

SIP - incoming

☒ Use for signaling and ladder [Ladder tags](#)

General information	
SIP call leg type	Call
From	4925128900@10.7.10.135
To	4925128901@10.5.10.51
Signaling source	10.7.10.135 : 40211
Signaling destination	10.5.10.51 : 0
Call-ID	ee753564-7206831d-0-1f73e51b@10.7.10.135
Call leg connects	✓ 2021-10-12 01:03:27 UTC

CUCM CI 35914128	
Reception Preference	BestEffort
DTMF Capabilities	OOB_RFC2833
RFC2833 payload number	101
Endpoint receive DTMF	✓
Endpoint provide OOB DTMF	✓
Region	Default

Associated CIs

CI	MTP required	Transcoder required	E2E region bandwidth
35914129	✗	✗	Audio: 64 Video: 384 Immersive: 2147483647

● No RTP streams linked for this call leg

SIP - outgoing

☒ Use for signaling and ladder [Ladder tags](#)

General information	
SIP call leg type	Call
From	4925128900@10.5.10.51
To	4925128901@10.5.10.51
Signaling source	10.5.10.51 : 5060
Signaling destination	10.7.10.135 : 45371
Call-ID	3023d800-10001-3eba23-93f492a@10.5.10.51
Call leg connects	✓ 2021-10-12 01:03:27 UTC

CUCM CI 35914129	
Reception Preference	BestEffort
DTMF Capabilities	OOB_RFC2833
RFC2833 payload number	101
Endpoint receive DTMF	✓
Endpoint provide OOB DTMF	✓
Region	Default

Associated CIs

CI	MTP required	Transcoder required	E2E region bandwidth
35914128	✗	✗	Audio: 64 Video: 384 Immersive: 2147483647

● No RTP streams linked for this call leg

Each leg section
contains General
Information, call-id,
Call Identifier and
negotiation
capabilities along
with other
associated Call
Identifiers

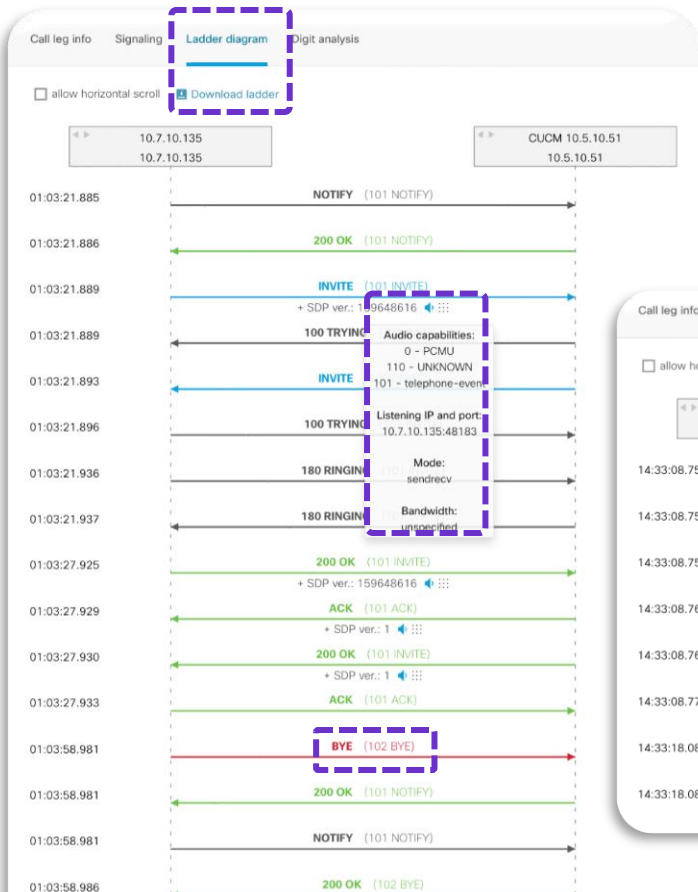
Signaling tab

Displays the sip messages indicating the directions, media type, ip address and ports involved.

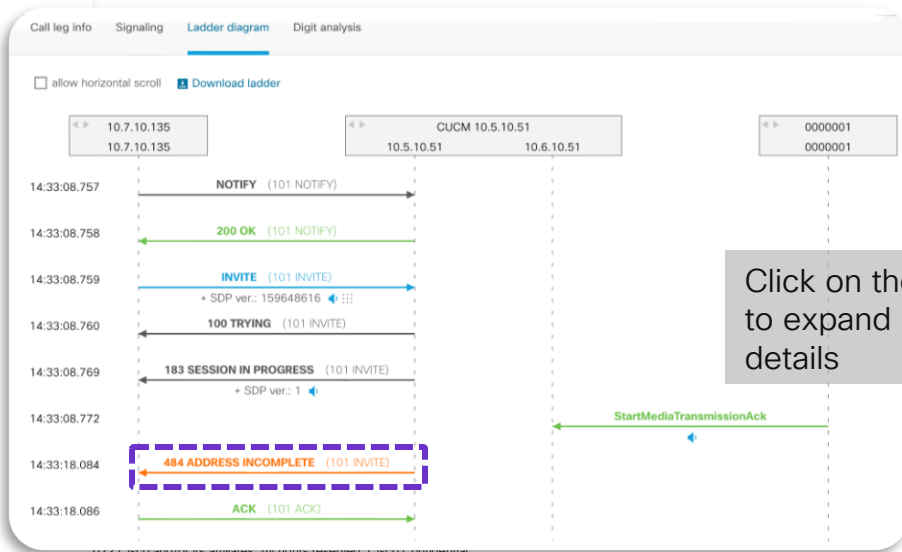
Media type and negotiated capabilities easily visible

Click on the SIP message of your interest to expand and review headers and SDP lines.

Ladder diagram tab



- ✓ Creates a ladder diagram of the call in question
- ✓ Hover over the SDP message to display the media capabilities negotiated
- ✓ Download and store the diagram



Click on the SIP message to expand and review the details

Digit Analysis tab

Show the list of all digit analysis involved in the call set up of the analyzed call

Troubleshooting

Call leg info

Signaling

Ladder diagram

Digit analysis

List of DAs

1634000601.891, CI 35914128

Digit analysis details

CI	35914128
Timestamp	2021-10-12 01:03:21 +0000 UTC
Complexity	0
Line number	50174904.000
Calling Number	
Dialed digits	4925128901

Digit translations

05 Serviceability Connector Deployment

What is Webex Serviceability Connector?

Serviceability Connector is a virtual machine that resides in your network ('connector host'). It securely collects data/logs from your on-premises equipment based on the request, for analysis by CSA.

cisco.com/go/serviceability

For Cloud-Connected UC the recommended Serviceability Connector is ECP.

Prerequisites - Connectivity

Ensure that firewall allows the below communication outbound communication:

Purpose	Direction	Destination	Protocol	Dst. Port
Data upload path	Outbound	cx.d.cisco.com	HTTPS	443
Download upgrades	Outbound	binaries.webex.com	HTTPS	443
Authorize to obtain token for cloud registration	Outbound	idbroker.webex.com	HTTPS	443
Registration with cloud	Outbound	*.wbx2.com	HTTPS	443
Register with cloud to receive commands for data acquisition				
Feature toggles				
Metrics				

Prerequisites – Platform

We recommend the larger option with 4 CPU, 8GB RAM, and 20GB HDD.
If you have limited resources, you can choose the smaller option.

VM Property	Value
Select virtual disk format	Thick provision lazy zeroed
VM storage policy	Datastore default

Serviceability Connector

Cisco Webex
Control Hub

Overview

MONITORING

Organization Health

Analytics

Troubleshooting

MANAGEMENT

Users

Workspaces

Devices

Apps

Account

Organization Settings

SERVICES

Updates

Messaging

Calling

Connected UC

Hybrid

Hybrid

Hybrid Calendar

Exchange

Use @webex or @meet in a meeting location to insert join details, show upcoming meetings in the Webex app, and provide One Button to Push (OBTP) to join.

View Prerequisites

Set Up

Hybrid Calendar

Office 365

Use @webex or @meet in a meeting location to insert join details, show upcoming meetings in the Webex app, and provide One Button to Push (OBTP) to join.

View Prerequisites

Set Up

Serviceability Service

Enables TAC to collect on-demand diagnostic data. This speeds up case resolution, reduces impact on your technical staff, and increases your on-premises infrastructure uptime.

View Prerequisites

Set Up

Video Integration

Microsoft Teams

Join Microsoft Teams meetings from Webex devices. Enable Hybrid Calendar Service to join meetings with One Button To Push (OBTP).

View Prerequisites

Learn More

Webex Monitoring Service

The Webex Monitoring Service collects diagnostics data from your on-premises video devices and sends the information to your Cisco Webex Control Hub account for troubleshooting purposes.

Set Up

1

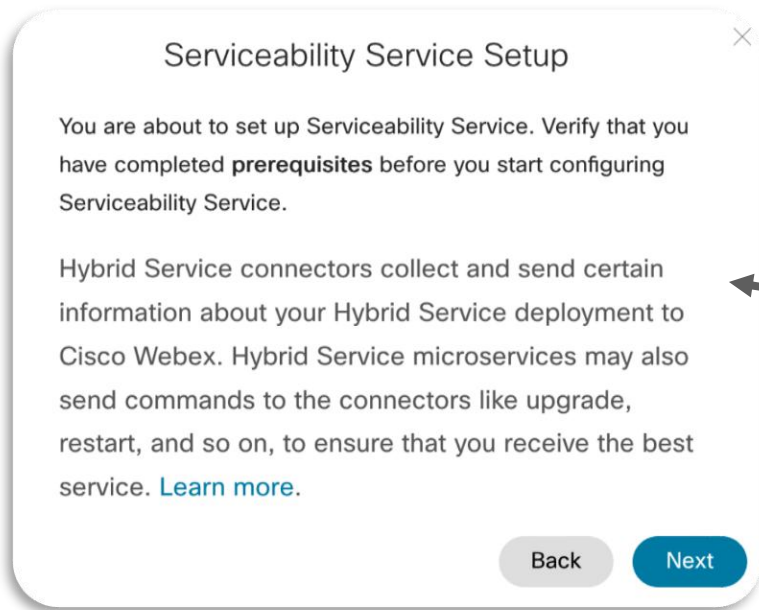
2

CCUC PM_TME demos

webex

CCUC PM_TME demos

Serviceability Service Setup in Control Hub



A wizard will guide the administrator through the steps

Serviceability Service Setup in Control Hub

Serviceability Service Prerequisites

Before you set up the Serviceability Service, deploy the on premises connector host by following the instructions in the "Prepare Your Environment" section of the documentation. Note that the documentation includes specific sections depending on whether you are deploying an ECP Connector host or an Expressway Connector host.

The "Deploy Cisco Webex Serviceability Service" section of the document will then guide you through the process of deploying Serviceability Service using your chosen host.

[Open Documentation](#)

- In the comments find the Serviceability connector documentation
- Or visit Serviceability Connector section in this document

Serviceability Service Setup in Control Hub

Add Resource ✕

What platform do you want to deploy on?

☒ Enterprise Compute Platform ←

☐ Expressway

Cancel Next

We recommend use of ECP. Our future development will focus on this platform. Some new features won't be available if you install the Serviceability Connector on an Expressway™.

Serviceability Service Setup in Control Hub

Set Up Webex Monitoring Service Software

Download Software

The Webex Monitoring Agent collects diagnostics data from your network connection and on-premises video devices, and then sends the information to your Cisco Webex Control Hub account. Install an agent on each computer or virtual machine that you want to use for the Webex Monitoring Service.

The minimum requirements for the Webex monitoring Service is an operating system running 64-bit Windows Server 2016 or later version, an audio driver enabled on the physical or virtual machine, a 4-core processor or higher, 8-GB of RAM, and 20-GB free storage. You can have up to 250 devices per agent.

Back

Download

Download the ovf

Serviceability Service Setup in Control Hub

Create a new
cluster

Register Serviceability Service on ECP Node

Before you can register your Serviceability Node you must install and configure the software on your server
Please click [here](#) to download the software
Once the software is installed and configured, continue with the Cluster and Node information:

Cluster info

Create a new or select an existing ECP Serviceability Cluster where you want to add the Serviceability Service Node.

☐ Create a new cluster
☐ Select an existing cluster

Serviceability Node

Enter the Internal FQDN or IP address of the Serviceability Node that you want to register with Webex.

Cancel

Next

Serviceability Service Setup in Control Hub

Fill with the FQDN assigned to the ECP

This FQDN or IP address should be used to launch the ECP registration in former steps

Register Serviceability Service on ECP Node

Before you can register your Serviceability Node you must install and configure the software on your server
Please click [here](#) to download the software
Once the software is installed and configured, continue with the Cluster and Node information below:

Cluster info

Create a new or select an existing ECP Serviceability Cluster where you want to add the Serviceability Service Node.

☒ Create a new cluster

PA-ECP-Serviceability

☐ Select an existing cluster

Serviceability Node

Enter the Internal FQDN or IP address of the Serviceability Node that you want to register with Webex.

serv-connt1.cisco.com

Cancel Next

Opening serab_ecp.ova

You have chosen to open:

serab_ecp.ova
which is: OVA file (2.2 GB)
from: <https://binaries.webex.com>

What should Firefox do with this file?

☐ Open with VMware Fusion (default)

☒ Save File

☐ Do this automatically for files like this from now on.

Cancel OK

Serviceability Connector – VMware Client

Review first prerequisites platform to ensure that hardware requirements are met.

Deploy OVF Template

- ✓ 1 Select an OVF template
- ✓ 2 Select a name and folder
- ✓ 3 Select a compute resource
- ✓ 4 Review details
- ✓ 5 Configuration
- ✓ 6 Select storage
- ✓ 7 Select networks
- 8 Customize template**
- 9 Ready to complete

Networking Properties. [Note: These network settings are optional. If preferred, you can configure them using the node console or web interface after the node is deployed. Note that DHCP is not supported. See the documentation at www.cisco.com/go/video-mesh for guidance.] [Warning: The system performs minimum checks on these network values, so be careful as you enter these values.] 6 settings

Hostname	Enter the hostname or FQDN (optional).
IP Address	Enter the IP address. Leave blank if DHCP is desired.
Subnet Mask	Enter the subnet mask address in the format 0.0.0.0. Leave blank if DHCP is desired.
Gateway	Enter the gateway address. Leave blank if DHCP is desired.
DNS Servers	Enter Domain Name Servers (comma separated and not more than 4 entries). Leave blank if DHCP is desired.
NTP Servers	Enter NTP servers (comma separated and not more than 6 entries).

CANCEL BACK NEXT

Serviceability Connector – Setup on ECP

Only if Proxy is required

Cisco Webex Serviceability Node

Trust Store & Proxy

Trust Store Management and Proxy Settings

☐ No Proxy
☐ Transparent Non-Inspecting Proxy
☐ Transparent Inspecting Proxy
☒ Explicit Proxy

Proxy IP/FQDN:
Proxy Port:
Proxy Protocol: ☒ Http ☐ Https
Authentication Type: ☒ None ☐ Basic ☐ Digest

Check Proxy Connection This proxy setting was verified to function correctly.

To complete proxy setup, reboot this node

Reboot

To complete proxy setup, enable routing through the explicit proxy

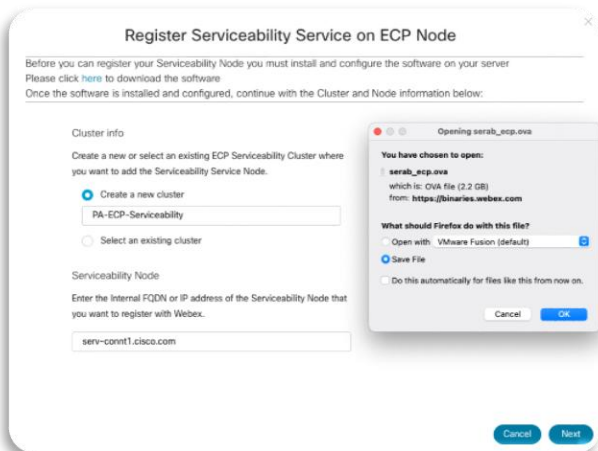
☐ Route all port 443/444 https requests from this node through the explicit proxy (requires 15 seconds to complete).

Upload a Root Certificate or End Entity Certificate (.crt or .pem file)

Install All Certificates into the Trust Store This node will reboot while installing the certificates into the trust store.

- Access Setup page:
<https://<IP or FQDN>:443/setup>
- Configure Proxy information if required and test out the connection
- Check Proxy Connection, if connectivity test succeeds reboot the node and complete the proxy setup
- Return to Control Hub to finalize ECP setup

Serviceability Connector – Registration in Control Hub



Register Serviceability Service on ECP Node

Before you can register your Serviceability Node you must install and configure the software on your server. Please click [here](#) to download the software. Once the software is installed and configured, continue with the Cluster and Node information below:

Cluster info

Create a new or select an existing ECP Serviceability Cluster where you want to add the Serviceability Service Node.

☒ Create a new cluster
PA-ECP-Serviceability

☐ Select an existing cluster

Serviceability Node

Enter the Internal FQDN or IP address of the Serviceability Node that you want to register with Webex.

serv-cont1.cisco.com

Opening serab_ecp.ova

You have chosen to open:
serab_ecp.ova
which is: OVA file (2.2 GB)
from: <https://binaries.webex.com>

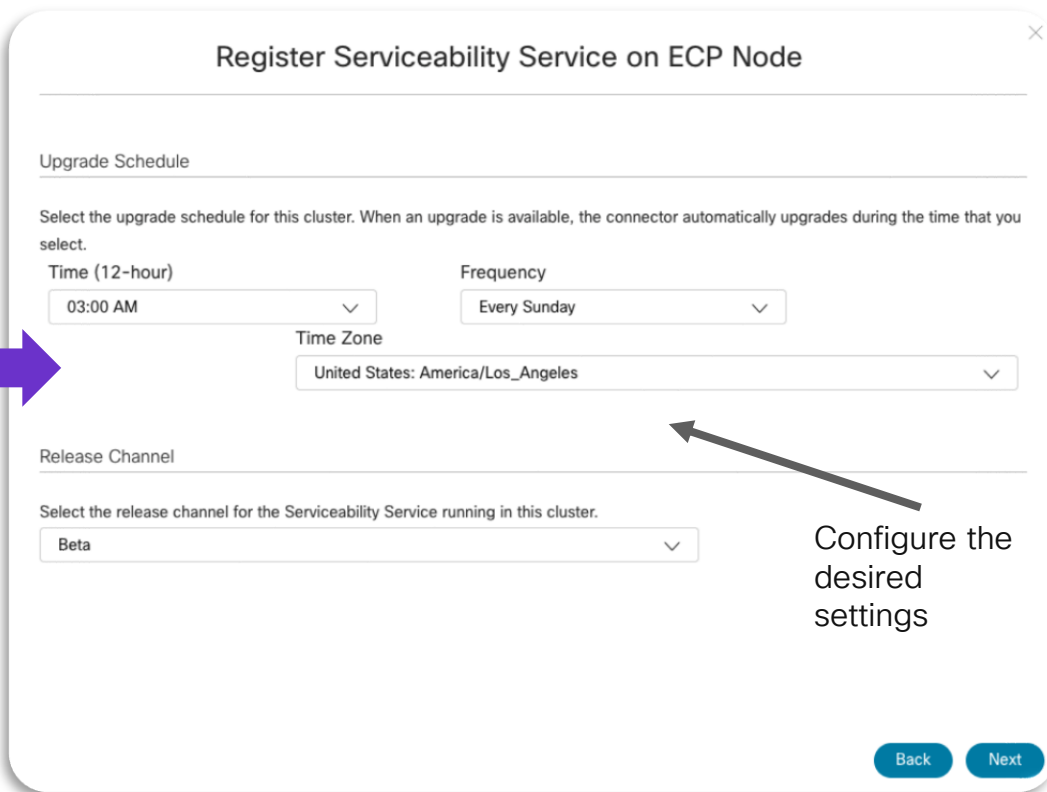
What should Firefox do with this file?

☐ Open with: VMware Fusion (default)

☒ Save File

☐ Do this automatically for files like this from now on.

Cancel OK



Register Serviceability Service on ECP Node

Upgrade Schedule

Select the upgrade schedule for this cluster. When an upgrade is available, the connector automatically upgrades during the time that you select.

Time (12-hour): 03:00 AM

Frequency: Every Sunday

Time Zone: United States: America/Los_Angeles

Release Channel

Select the release channel for the Serviceability Service running in this cluster.

Beta

Back Next

Configure the desired settings

Serviceability Connector – Registration in CH (2)

Register Serviceability Service on ECP Node

Next, click Go to Node to continue the registration of the Serviceability Service node to the Webex cloud.

After you complete this step, the node will be visible in Control Hub and ready to use.

Node Internal IP Address/FQDN: **serv-connt1.cisco.com**

Cluster: **PA-ECP-Serviceability**

[Back](#)[Go to Node](#)

The wizard will launch a local connection to the ECP in a new tab, make sure you have vpn or local access to the ECP

Serviceability Connector – Registration in CH (3)



Cisco Webex Hybrid Services



Laurent TME
CCUC PM_TME dem...

Hybrid Services

Allow Access to Hybrid Services

Permissions are required to allow your Cisco Webex organization to create, read, update, and delete user accounts, as well as read and update information about your organization.

Organization

CCUC PM_TME demos

FQDN or IP Address

serv-connt1.cisco.com

☒ Allow Access to the Hybrid Services Node
Only allow access to hosts you know and trust

Continue



By using Cisco Webex Services you accept the
[Terms of Service](#) and [Privacy Statement](#).

Allow Webex
services to access
the ECP

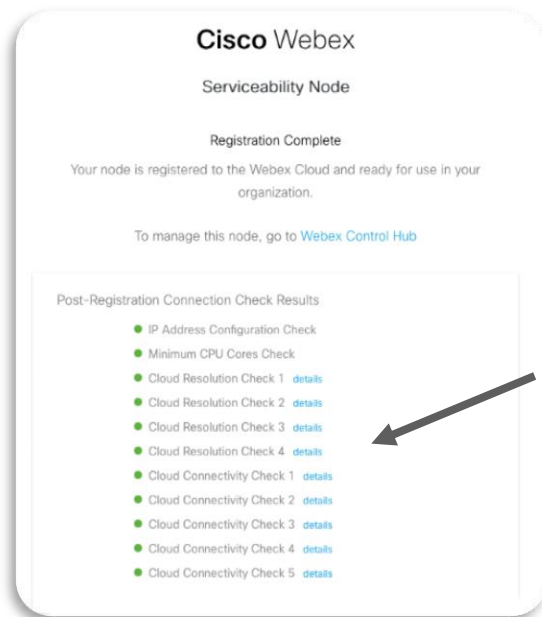
Wait for the
Registration
confirmation

Hybrid Services

Registration Confirmed

You will now be redirected to [serv-connt2.cisco.com](#).

Serviceability Connector – ECP



All post registration
Connection checks
should succeed

Serviceability Connector – Resources page in CH

Cisco Webex Control Hub

< Serviceability Service

Resources Settings

1 notification

Add Resource

Enterprise Compute Platform Clusters	Service Status
PA-ECP-Serviceability	No Nodes Registered

Now the ECP is registered

Next step is to add UCM clusters to the ECP

Remember:

- Add call processing nodes to ECP for log collection and analysis
- Trace level of logs in the cluster should be set up as detailed

Overview

MONITORING

- Organization Health
- Analytics
- Troubleshooting

MANAGEMENT

- Users
- Workspaces
- Devices
- Apps
- Account
- Organization Settings

SERVICES

- Updates
- Messaging
- Calling
- Connected UC
- Hybrid

Serviceability Connector – Resources page in CH

Cisco Webex
Control Hub

Overview

MONITORING

Organization Health

Analytics

Troubleshooting

MANAGEMENT

Users

Workspaces

Devices

Apps

Account

Organization Settings

SERVICES

Updates

Messaging

Calling

Connected UC

Hybrid

CCUC PM_TME demos

< On-Premises Resources

All 2Expressway 0ECP Serviceability 2

Expressway

No clusters found.

ECP Serviceability

PA-ECP-Serviceability

1 node

Upgrade Schedule

03:00 AM Every Sunday

America/Los_Angeles

Release Channel

Beta

Serviceability

Settings

06

Serviceability Connector Add Call Processing Nodes

Add Call Processing Nodes

Navigate to ECP <https://x.x.x.xcom:8443/home> > Managed Devices > New Device

The screenshot shows a 'Device details' form with three main sections: 'Managed Device', 'Application User', and 'OS Admin (SSH)'. The 'Managed Device' section has fields for Type (Unified CM), Device name (cucm-lrh), Address (x.x.x.x), Role (Publisher), and TLS verify mode (off). The 'Application User' section has fields for Username (ccucadmin) and Password (masked). The 'OS Admin (SSH)' section has a checkbox for 'Do SSH Credentials differ from those of Application User?' (No). At the bottom are buttons for Save, Verify, Delete, and Cancel. Annotations include an arrow pointing to the 'Unified CM' type, another pointing to the 'Save' button, and a third pointing to the 'Verify' button.

Device details

Managed Device

Type:

Device name:

Address:

Role:

TLS verify mode:

Application User

Username:

Password:

OS Admin (SSH)

Do SSH Credentials differ from those of Application User:

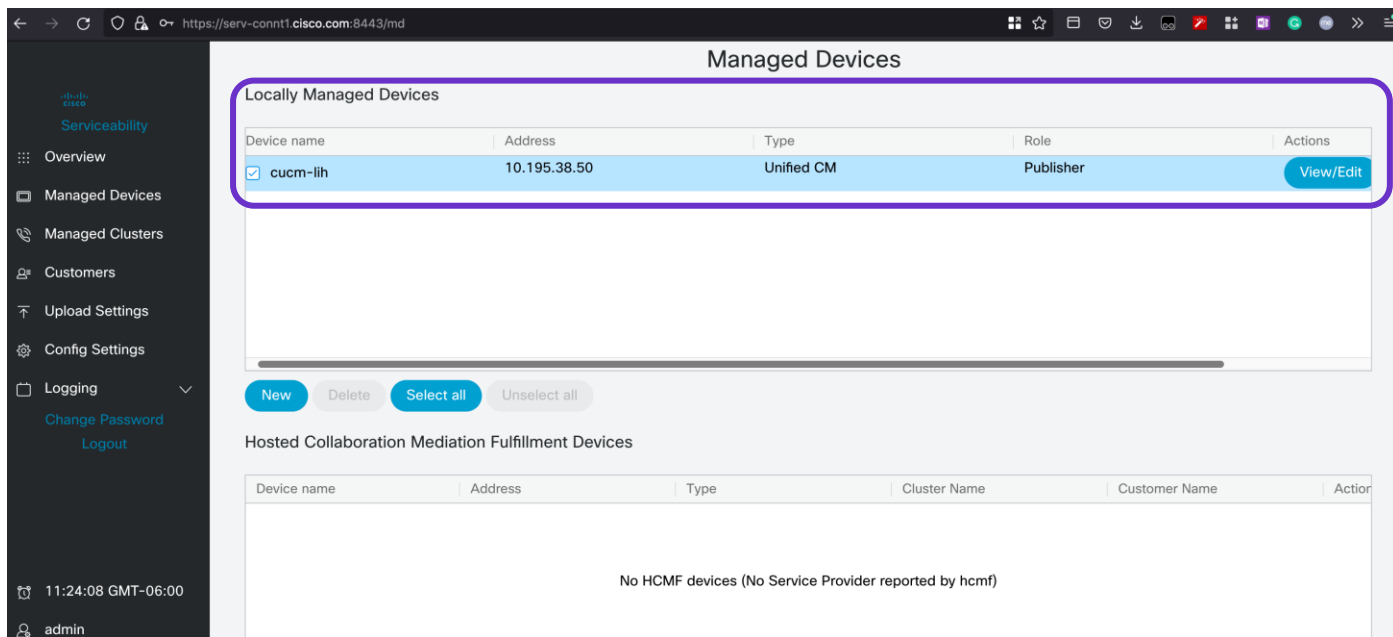
Select Unified CM type

If application user does not have permission to access the node via CLI, provide the credentials

Save and Verify

Add Call Processing Nodes

Added devices will show up in the list:



The screenshot shows a web browser window with the URL `https://serv-conn1.cisco.com:8443/md`. The page title is "Managed Devices". On the left is a dark sidebar with navigation links: "Serviceability", "Overview", "Managed Devices", "Managed Clusters", "Customers", "Upload Settings", "Config Settings", "Logging", "Change Password", and "Logout". The main content area is titled "Managed Devices" and contains a section for "Locally Managed Devices". This section features a table with the following data:

Device name	Address	Type	Role	Actions
☒ cucm-lih	10.195.38.50	Unified CM	Publisher	View/Edit

Below the table are buttons for "New", "Delete", "Select all", and "Unselect all". Below this is a section for "Hosted Collaboration Mediation Fulfillment Devices" with a table that is currently empty. At the bottom of this section, it says "No HCMF devices (No Service Provider reported by hcmf)". The bottom of the sidebar shows a clock for "11:24:08 GMT-06:00" and the user "admin".

Verify Nodes

Click on Verify button

Device details

Managed Device

Type:

Device name:

Address: * 10.195.38.50 ⓘ

Role: * Publisher ⓘ

TLS verify mode: * off ⓘ

Application User

Username: * ccucadmin ⓘ

Password: * ⓘ

Verifying Configuration. This may take up to 30 seconds.

Cancel OK

Wait for completion

Device details

Managed Device

Type:

Device name: * cucm-lrh ⓘ

Address: * 10.195.38.50 ⓘ

Role: * Publisher ⓘ

TLS verify mode: * off ⓘ

Application User

Username: * ccucadmin ⓘ

Password: * ⓘ

Success: Configuration Verified

OK

Troubleshooting - recap

Call processing Nodes

Add all nodes to ECP for log collection and analysis.

Trace Level

In order to allow CSA to execute the analysis, ensure that in all call processing nodes the Debug Trace is Detailed and in the Trace Filter Settings section, check the Enable SIP Call Processing Trace checkbox for Cisco Call Manager Service

Meet Network requirements

ECP & CCUC network prerequisites

ECP is the recommended platform

Roles

Login to Control Hub with Full administrator, read-only administrator, or a support administrator role.



Collaboration
Technical Marketing

Touchless Webex App Provisioning

Touchless Webex App Provisioning Contents

- 01 Overview
General Summary
- 02 Prerequisites
- 03 Touchless Webex App
Provisioning flow
- 04 Configuration Steps

01 Overview

Touchless Webex App provisioning

Cloud-based service that simplifies the process of device creation, provisioning and usage of soft client devices in UCM based on self-provisioning of devices by the end user by just logging into Webex App



Admin productivity

No admin intervention required to provision device in UCM



Avoid over provisioning for Webex App



Set up Control Hub policies to clean up devices that are inactive



Minimize cluster impact in scale and licensing

Webex App auto-provisioning

Device creation

Devices:

- Android device (BOT)
- Chromebook/iPad devices (TAB)
- Windows/MAC devices (CSF)
- iPhone device (TCT)

Control Hub policy for device creation and deletion of auto-provisioned devices

Unified CM Settings

Allow Unified CM registration without trusted certificate

When enabled, the security of the Unified CM connection is not authenticated. We recommend that you only use this setting in a lab environment.



Allow auto-provisioning for Webex

Turn on for automatic device provisioning when a Unified Communications Manager user signs in with their registered email address. You can also change this through individual user settings.



Allowed device types

Allow auto-provisioning Webex app for desktop, mobile, or both.



Desktop and mobile



Desktop only



Mobile only

Delete inactive auto-provisioned devices



Don't delete inactive devices



Delete after 30 days

02 Prerequisites

Requirements

Platform

Minimum supported versions:

- UCM 11.5(x), 12.0(x), 12.5(x), 14(x)
- Webex App version 41.12 and above
- Expressway x14.0.2
- User ID or email ID of Unified CM users matches with the User ID in Webex Identity Service.

Control Hub

- Webex App Provisioning for UCM Calling under Service Management

Connectivity

- For a larger cluster, ensure that you have at least 2 MBPS of bandwidth.
- Cloud-Connected UC cloud port 443 must be reachable from customer environment.

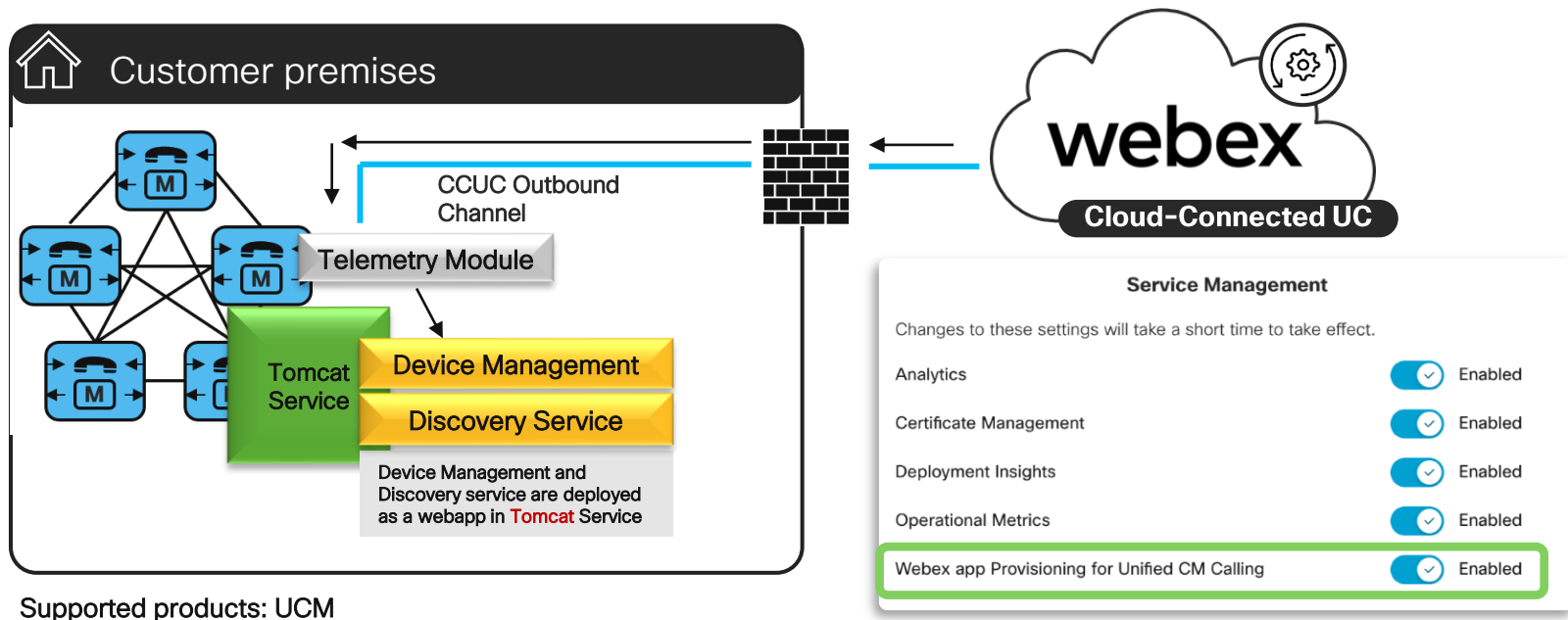
*.ucmgt.cisco.com

*.webex.com

03 Touchless Webex App Provisioning flow

Webex App auto-provisioning

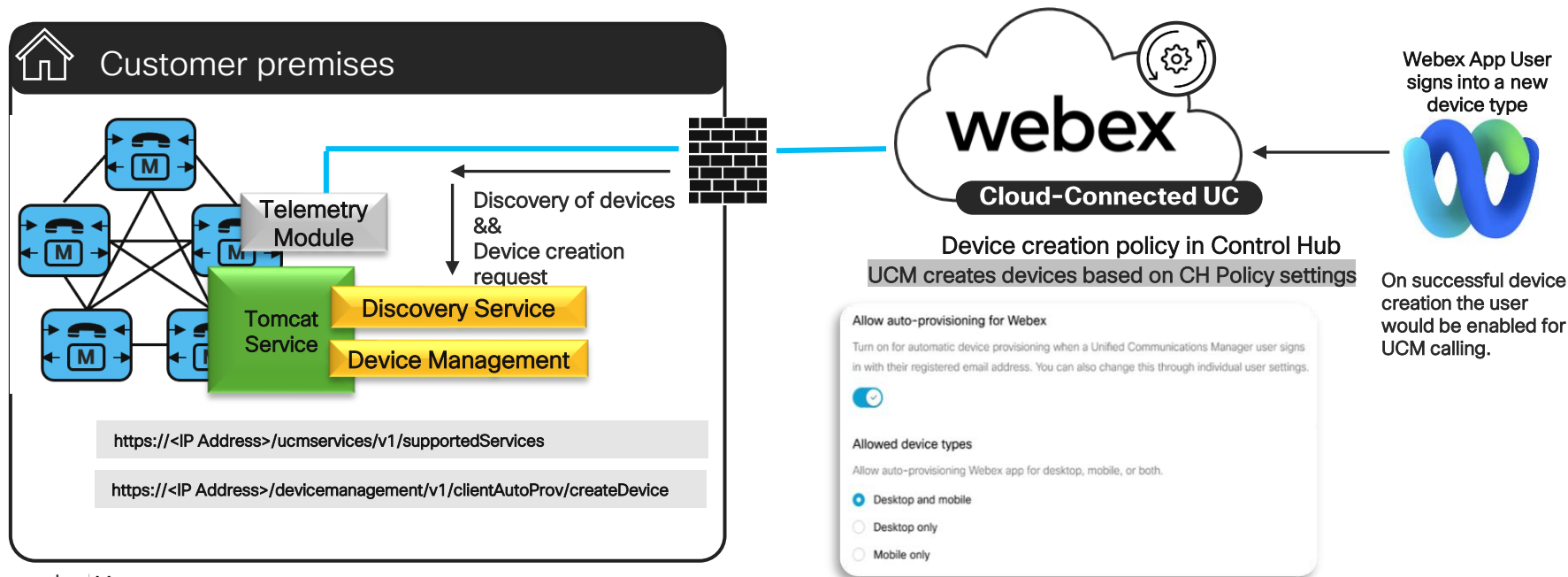
Flow after service is enabled



Supported products: UCM

Webex App auto-provisioning

Flow when the user logs in to Webex App on a new device



Considerations

If the device does not exist in the Unified CM:

Device Settings—User profile device template is used in the following order of preference.

1. Mobile and desktop Universal Device Template
2. Cisco IP Phone Universal Device Template
3. Auto-registration template (system defined) of the cluster is used, if UDT isn't available.

Phone Button Template—Standard Model Specific Template

NOTE: Device creation fails if none of the above 3 options are available.

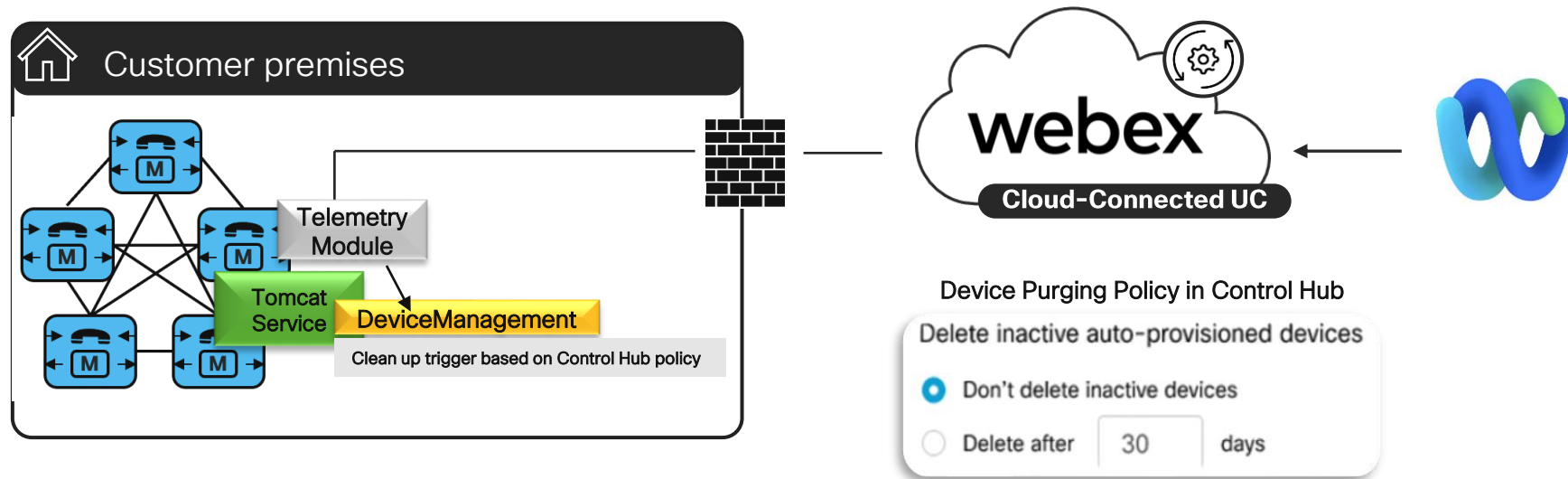
If the device exist in Unified CM already at the time of the user signing into the Webex App:

Device Settings—Users' existing device settings is used for the new device in the following order of preference: Mobile, Desktop, or Cisco IP Phone.

Phone Button Template— Standard Model Specific Phone Button Template

Webex App auto-provisioning

Flow auto – clean up

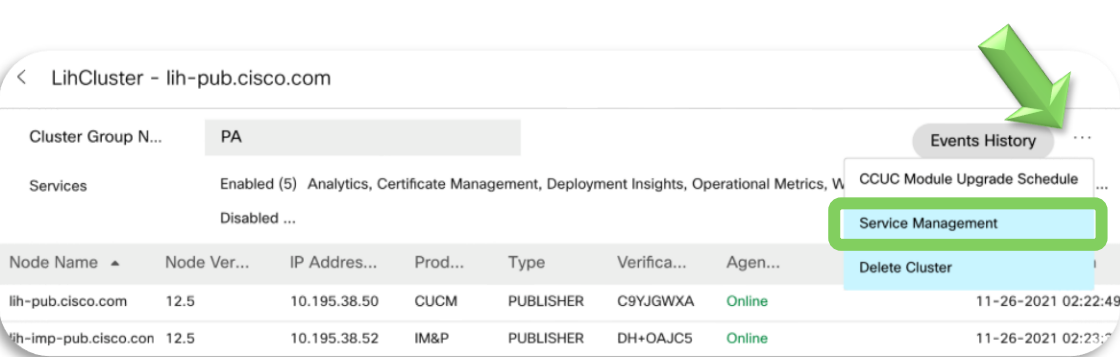


04 Configuration Steps

Configuration

Go to the **Connected UC > UC Management card > Inventory > Cluster Group details > Cluster details>**

Click the Ellipsis and choose Service Management and select the service

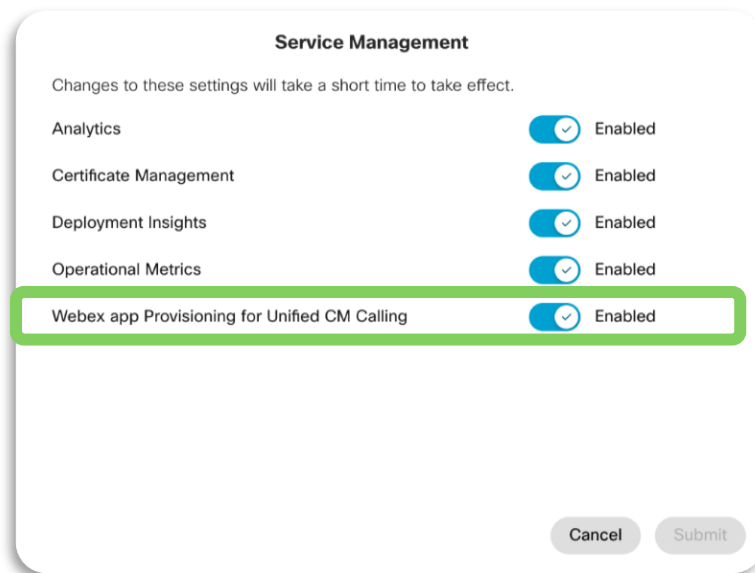


Cluster Group N... PA

Services Enabled (5) Analytics, Certificate Management, Deployment Insights, Operational Metrics, W CCUC Module Upgrade Schedule ...

Disabled ...

Node Name ▲	Node Ver...	IP Address...	Prod...	Type	Verifica...	Agen...	
lih-pub.cisco.com	12.5	10.195.38.50	CUCM	PUBLISHER	C9YJGWXA	Online	11-26-2021 02:22:49
lih-imp-pub.cisco.com	12.5	10.195.38.52	IM&P	PUBLISHER	DH+OAJC5	Online	11-26-2021 02:23:0



Service Management

Changes to these settings will take a short time to take effect.

- Analytics ☒ Enabled
- Certificate Management ☒ Enabled
- Deployment Insights ☒ Enabled
- Operational Metrics ☒ Enabled
- Webex app Provisioning for Unified CM Calling ☒ Enabled

Cancel Submit

Configuration

Calling > Unified CM Settings > Allow auto-provisioning for Webex

Unified CM Settings

Allow Unified CM registration without trusted certificate

When enabled, the security of the Unified CM connection is not authenticated. We recommend that you only use this setting in a lab environment.



Allow auto-provisioning for Webex

Turn on for automatic device provisioning when a Unified Communications Manager user signs in with their registered email address. You can also change this through individual user settings.



By default, this option is disabled even if the Cisco Webex app provisioning for Unified CM calling service is turned on for provisioning

Allowed device types

Allow auto-provisioning Webex app for desktop, mobile, or both.

- ☒ Desktop and mobile
☐ Desktop only
☐ Mobile only

To define the policy to determine which devices are allowed for Auto-provisioning

Delete inactive auto-provisioned devices

- ☒ Don't delete inactive devices
☐ Delete after

0 days

Set the period to delete the inactive auto-provisioned devices in your Unified CM clusters.

Expiration timer for OAuth refresh token

Set how many days until a user needs a new OAuth refresh token. Updating the number of days takes effect immediately. Newly issued refresh tokens use the new expiry timer. Previously issued refresh tokens aren't valid and will force all users to re-authenticate within one hour. ⓘ

60 days

You can configure the new OAuth refresh token expiry timer

Confidential

Configuration

Enable or disable auto-provisioning per individual user

- Go to **Management > Users**, and Select the user to modify
- Click **Calling > Unified Communications Manager settings**.
Use the toggle button to decide whether you want to enable or disable the Allow auto-provisioning for Webex feature for this user.

Unified Communications Manager settings

Allow auto-provisioning for Webex ⓘ



Allowed device types

Desktop only



Desktop only or Mobile only or both



Collaboration
Technical Marketing

Directory Service

Directory Service Contents

01 Overview

General Summary

02 Prerequisites

Necessary steps to get started with Directory Service

03 Flow

04 Configuration

01 Overview

Webex Cloud-Connected UC Directory Service



Seamless Sync
from Webex CI
into UCM and
Unity Connection



Enable
Migrations
options
incrementally
from On-prem
AD to fully cloud



Ubiquitous Identity
and Access
Management while
moving directory to
Cloud



Dry Run for
previewing Sync
before is
committed

Cloud Connected UC Directory Service

Synchronize users from cloud-based directories into on-premises deployments.



Currently Common Identity Service syncs users from Cloud AD via **Cross-domain Identity Management** (SCIM) API

Webex will synchronize existing users within the Common Identity Service container into **UCM** and **Unity Connection** via Cloud-Connected UC web socket and Directory Service feature

02 Prerequisites

Requirements

Platform

- Minimum supported versions:

UCM & Unity Connection
11.5(x), 12.5(x) and 14(x)

- UCM cluster onboarded

[How to onboard clusters
HERE](#)

Control Hub

- Administrator has to enable Directory Service under Service Management (Disabled by default)

Connectivity

- For a larger cluster, ensure that you have at least 2 MBPS of bandwidth.
- Cloud-Connected UC cloud port 443 must be reachable from customer environment.

*.ucmgt.cisco.com

*.webex.com

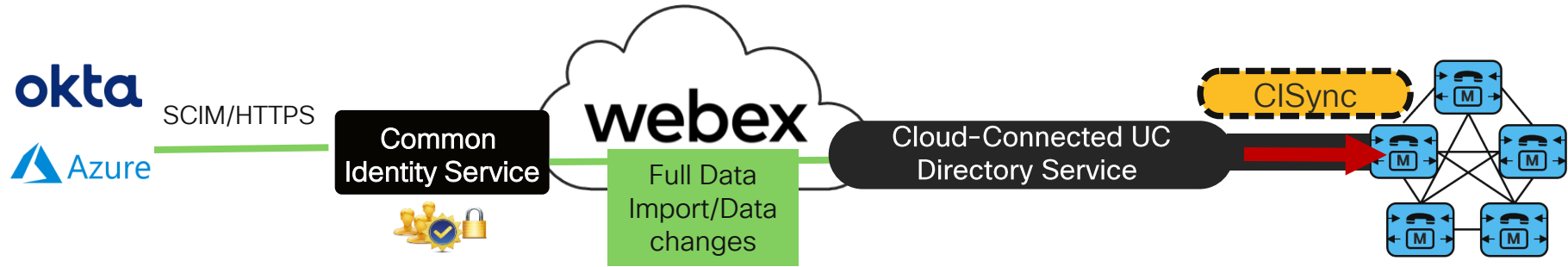
Considerations

- Only SSO authentication Node Based supported, no ldap based
- No ldap group configuration
- Sync occurs every 24 hours
- Existing LDAP agreements are deleted after sync is accepted
- During Sync process new logins of inactive users will not work, only if end users are already logged in will work. (applies for self-care portal, Extension Mobility)
- During Sync process call across Endpoints are not affected
- During Sync Process users cannot login to Jabber/Webex App

Note: This Could be leveraged for a fully Cloud migrated deployment (UCMC / Dedicated Instance) as well.

03 Flow

Cloud Connected UC Directory Service



- 1 Admin enables CCUC Directory Service and Directory Service card becomes available under **CH>Services>CCUC**
- 2 Admin starts Directory Service Wizard for a cluster
- 3 CCUC creates a Ldap agreement in the on-prem server to Sync with CI (**CISync**)
- 4 Admin reviews the **preview** of the user list about to be synchronized from CI
- 5 Admin once ready accepts and runs the synchronization (**18 hour window**)
- 6 CCUC initiates CI sync and UCM stores user data in **local LDAP** into it's DB

04 Configuration

Directory Service Configuration

Enable Directory Service under

CH> Services > CCUC > Inventory > Select the Cluster Group details> Click details > Click “...” next to Events History

The screenshot shows a 'Service Management' dialog box overlaid on a web interface. The dialog box has a title bar 'Service Management' and a subtitle 'Changes to these settings will take a short time to take effect.' It contains a list of services with their status and a toggle switch. The 'Directory Service' row is highlighted with a red rectangle. The background interface shows a table of cluster groups and an 'Events History' button.

Cluster Group Name	CCUC_ClusterGroup	
StandAloneCluster - ccuc-cm-pub.tmedemo.com		
Services	Enabled (5)	
	Disabled (1)	
Node Name	Node Version	IP Address
cm-pub.tmedemo.com	12.5	10.50.55.64

Events History

Agent Status: Online

Events: Warning, Info

Last Connection: 05-04-2022 20:34:45

Service Management

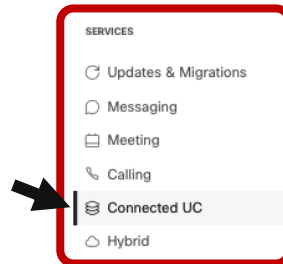
Changes to these settings will take a short time to take effect.

- Analytics: Enabled
- Directory Service: Enabled
- Certificate Management: Enabled
- Operational Metrics: Enabled
- Webex app Provisioning for Unified CM Calling: Disabled
- Deployment Insights: Enabled

Cancel Submit

Directory Service Configuration

- Navigate to Directory Service card under *Services > Connected UC services*
- Click on “*View Clusters*” to list UCM and Unity clusters and launch the Directory Service Wizard
- Click “*Learn More*” to launch the documentation



Connected UC

UC Management

[Inventory](#)
[Agent Install Files](#)
[Telemetry Module Inventory](#)

Operations

[Dashboard](#)

Certificate Management

[Alerts](#)
[Clusters](#)
[Profiles](#)
[Settings](#)

Directory Service

Synchronizes users from cloud-based directories into on-premises UC infrastructure. For example, Azure Active Directory, Okta, and other similar cloud-based directories.

[Learn more](#)
[View clusters](#)

Directory Service Configuration

Select target Cluster on Control Hub

Onboarded UCM and Unity Connection clusters will be listed along with filter bar and details:

- Status of provisioning
- Last Synchronized state
- Product type

< Cloud Connected UC

Directory Service

Search Filter by status (GMT -05:00) America/Mexi... 3 Cluster(s)

Cluster	Status	Last synced ↓	Product
EMEAcluster - emea-cm-pub.ent-pa.com	Not provisioned	N/A	CUCM
StandAloneCluster - ucba-aricent-vm33.cisco.com	Not provisioned	N/A	CUC
LihCluster - lih-pub.cisco.com	Provisioned	Mar 31, 2022 08:20 AM	CUCM

Directory Service Configuration

Click “Start Provisioning” to launch the Wizard

Note: Launching the Wizard
does not cause any changes

Cloud Connected UC > Dashboard

StandAloneCluster - ucbu-aricent-vm33.cisco.com



Provision this cluster now

Provision this cluster to enable the synchronization.



Directory Service Configuration

FIELD MAPPING: Select the appropriate UCM User ID field mapping

Note:

- New User account will be created if the mapping cannot be done successfully for an existing user in UCM/UC.
- Email ID of the user will be used as the unique identifier for the newly created user account
- Ensure that the mapping chosen uniquely identifies the user within the cluster after you start provisioning.

The screenshot shows a three-step progress bar at the top: 'Field mapping' (active, blue circle), 'Agreement selection' (inactive, grey circle), and 'Enable synchronization' (inactive, grey circle). A large black arrow points down to the 'Field mapping' step. Below the progress bar, the section is titled 'Field mapping' with the instruction: 'Ensure that the mapping chosen for Unity Connection User ID field uniquely identifies the user within the cluster.' The main instruction reads: 'Choose the Unity Connection User ID field mapping for synchronizing the user from Webex.' There are three radio button options: 1. 'User ID field in Unity Connection maps to email ID of the user in Webex.' (selected, blue circle with a black arrow pointing to it). 2. 'Mail ID field in Unity Connection maps to email ID of the user in Webex.' (inactive, grey circle). 3. 'User ID field in Unity Connection maps to email ID without domain part of the user in Webex.' (inactive, grey circle).

Directory Service Configuration

FIELD MAPPING: Select Ldap Agreement

WebexCommonIdentitySync is the default LDAP agreement that CCUC will create

Administrator can opt to select any local existing LDAP agreements to get overrited by CCUC for the CISync about to be launched

The screenshot shows a three-step progress bar at the top: 'Field mapping' (completed), 'Agreement selection' (current step), and 'Enable synchronization' (pending). Below the progress bar, the section is titled 'Agreement selection' with the instruction 'Select an agreement for creating a new synchronization agreement.' A box labeled 'Select agreement' contains the instruction 'Select an agreement to copy the settings for creating a new synchronization agreement. You can make edits to the new synchronization agreement in Unity Connection, once it is created.' Below this is a dropdown menu with a red border. The dropdown is open, showing the text 'Select' at the top and 'WebexCommonIdentitySync (Default)' as the selected option. A mouse cursor is pointing at the dropdown arrow. To the right of the dropdown, partially visible text reads '...onting to the on-premises directory will be deleted once the new agreement is'.

Directory Service Configuration

FIELD MAPPING: Ldap agreement preview becomes available after selection

Dry run option to preview the changes that would be applied

The screenshot shows the 'Agreement selection' step in a three-step process: Field mapping, Agreement selection, and Enable synchronization. The 'Select agreement' section has a dropdown menu set to 'WebexCommonIdentitySync (Default)'. Below it, an 'Important' note states that existing synchronization agreements pointing to the on-premises directory will be deleted. The 'AGREEMENT PREVIEW' section is highlighted with a red box and contains four expandable items: 'LDAP directory information', 'LDAP directory synchronization schedule', 'Standard user fields to be synchronized', and 'LDAP server information'. An arrow points from the 'Dry run option' text to the 'AGREEMENT PREVIEW' section.

Field mapping Agreement selection Enable synchronization

Agreement selection
Select an agreement for creating a new synchronization agreement.

Select agreement
Select an agreement to copy the settings for creating a new synchronization agreement. You can make edits to the new synchronization agreement in Unity Connection, once it is created.

WebexCommonIdentitySync (Default) ▼

Important: All the existing synchronization agreement(s) pointing to the on-premises directory will be deleted once the new agreement is created.

AGREEMENT PREVIEW

- ▼ LDAP directory information
- ▼ LDAP directory synchronization schedule
- ▼ Standard user fields to be synchronized
- ▼ LDAP server information

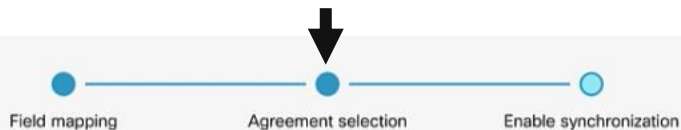
© 2022 Cisco and/or its affiliates. All rights reserved. Cisco Confidential

Agreement configurations can be clicked on and expanded for revision

LDAP directory information	
Configuration name	WebexCommonIdentitySync (Default)
Manager distinguished name	uid=admin,ou=system
User search base	ou=users,ou=ci
Custom filter for users	None

Directory Service Configuration

AGREEMENT SELECTION IS A THREE STAGE PROCESS



Enable synchronization

You can enable the synchronization once the system copies the user data into temporary storage and a new synchronization agreement is created automatically. The processing might take some time, you may close the window and come back later.

1

In progress

Step 1: Copy data into temporary storage and synchronize dry run

User data is copied from Webex into a temporary storage in on-premise infrastructure and a dry run of synchronization is performed. A dry run allows you to see what users will be added, modified, or deleted before you run a full synchronization and commit the changes to the on-premise infrastructure.

2

Pending

Step 2: New agreement creation

A new synchronization agreement is created for syncing the users.

3

Pending

Step 3: Synchronization agreement verification

You can click on the preview button to log in to your on-premises infrastructure and make changes to newly created synchronization agreement. VPN access is required.

[Preview in Unity Connection](#)

Directory Service Configuration

Step 1 allows admin to review the list of users about to get synchronized

Synchronization must be enabled within the next 18 hours.

After CI sync LDAP agreement is created the existing users in UCM that are LDAP sync will become inactive and removed after 24 hours of inactivity.

Once ready, admin can click on **Continue** to proceed with Step2 or **Abandon** to quit process



Enable synchronization

You can enable the synchronization once the system copies the user data into temporary storage and a new synchronization agreement is created automatically. The processing might take some time, you may close the window and come back later.

In progress

Step 1: Copy data into temporary storage and synchronize dry run

User data is copied from Webex into a temporary storage in on-premise infrastructure and a dry run of synchronization is performed. A dry run allows you to see what users will be added, modified, or deleted before you run a full synchronization and commit the changes to the on-premise infrastructure.

Dry run results :

Users Matched : 0

New Users : 56

Mismatched Users : 0

[Download report](#)

Mismatched users will be deleted in order to enable automatic synchronization, do you want to continue?

Important: You must enable synchronization **within 18 hours** from the time the new agreement is created. LDAP synchronized users will become inactive as soon as the new agreement is generated. These users are removed after 24 hour of inactivity. Inactive users will not be able to login and use services of the Call Manager.

Abandon

Continue

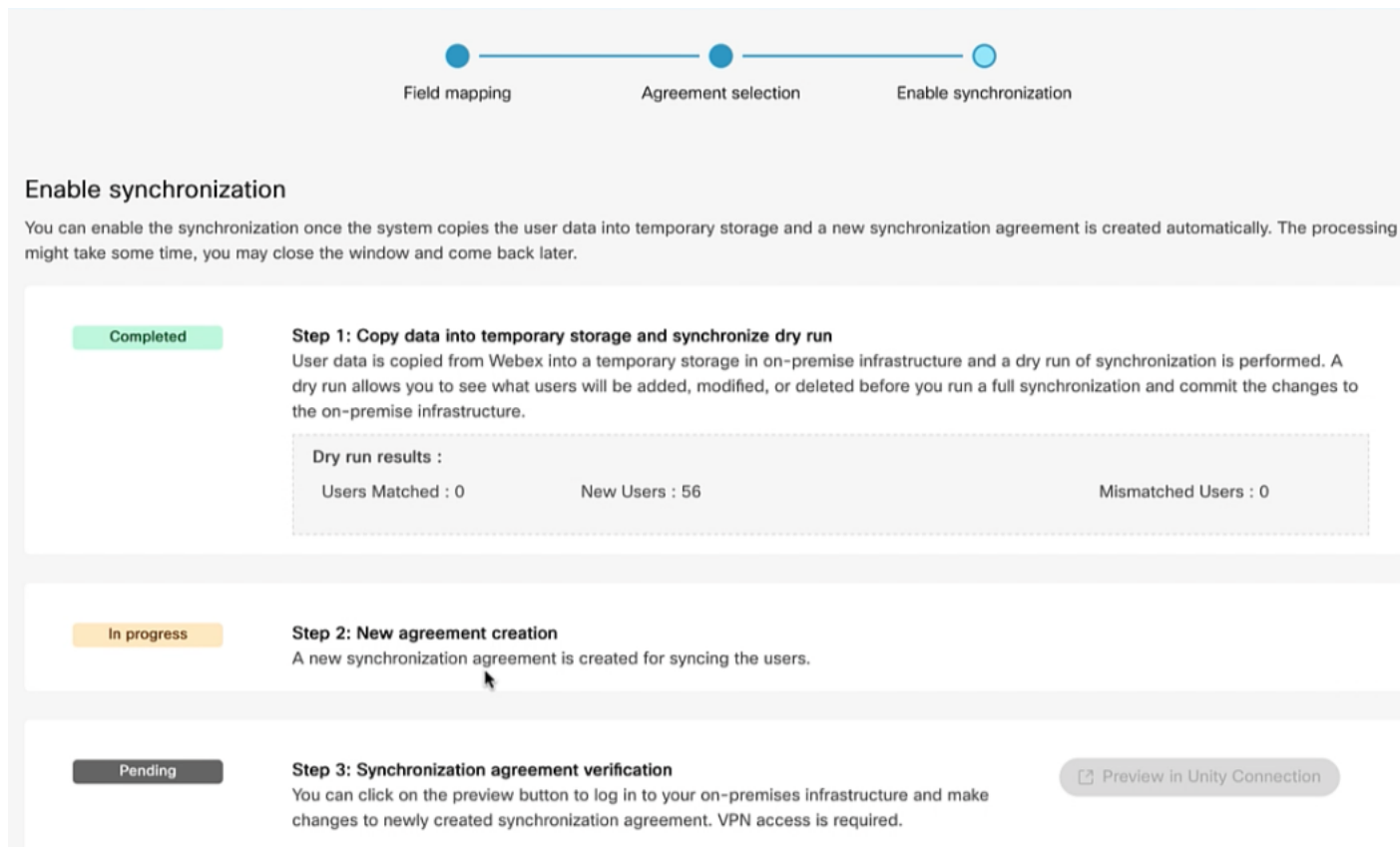
Pending

Step 2: New agreement creation

A new synchronization agreement is created for syncing the users.

Directory Service Configuration

- In **Step 2** CCUC will start the creation of the LDAP agreement in UCM or Unity Connection
- Existing Ldap synchronization agreement(s) in UCM/CUC pointing to the on-premises directory are deleted



Directory Service Configuration

- **Step 3** Once the sync agreement is created on the cluster, admin can preview and/or edit on UCM (cross launch from ControlHub UI)

Field mapping Agreement selection Enable synchronization

Enable synchronization

You can enable the synchronization once the system copies the user data into temporary storage and a new synchronization agreement is created automatically. The processing might take some time, you may close the window and come back later.

Completed

Step 1: Copy data into temporary storage and synchronize dry run

User data is copied from Webex into a temporary storage in on-premise infrastructure and a dry run of synchronization is performed. A dry run allows you to see what users will be added, modified, or deleted before you run a full synchronization and commit the changes to the on-premise infrastructure.

Dry run results :

Users Matched : 0	New Users : 56	Mismatched Users : 0
-------------------	----------------	----------------------

Completed

Step 2: New agreement creation

A new synchronization agreement is created for syncing the users.

In progress

Step 3: Synchronization agreement verification

You can click on the preview button to log in to your on-premises infrastructure and make changes to newly created synchronization agreement. VPN access is required.

You should verify and acknowledge the following check box:

☐ I have reviewed the new synchronization agreement and agree to proceed.

[Preview in Unity Connection](#)

Directory Service Configuration

Previewing LDAP agreement will launch a new tab to display the configured item.

Settings can be modified at this step or later after synchronization

Local Access or VPN connection to UCM or CUC is required

The screenshot shows the 'LDAP Directory Configuration' page in a web browser. The page has a top navigation bar with 'LDAP Directory Configuration', 'Refresh', and 'Help'. Below this is a 'Save' button. The main content area is divided into several sections:

- Status:** Shows 'Status: Ready' with an information icon.
- LDAP Directory Information:** Contains fields for 'LDAP Configuration Name*' (WebexCommonIdentitySync), 'LDAP Manager Distinguished Name*' (uid=admin,ou=system), 'LDAP Password*', 'Confirm Password*', 'LDAP User Search Base*' (ou=users,ou=ci), and 'LDAP Custom Filter for Users' (set to '< None >').
- LDAP Directory Synchronization Schedule:** Includes a checkbox for 'Perform Sync Just Once' (checked), a field for 'Perform a Re-sync Every*' (7) with a 'DAY' dropdown, and a 'Next Re-sync Time (YYYY-MM-DD hh:mm)*' (2022-04-29 22:24).
- Standard User Fields To Be Synchronized:** A table mapping Cisco Unified Communications Manager User Fields to LDAP Attributes.
- Custom User Fields To Be Synchronized:** A section with a note that custom user field names must be the same across all synchronization agreements. It includes input fields for 'Custom User Field Name' and 'LDAP Attribute', along with add and delete icons.
- LDAP Server Information:** Contains fields for 'Host Name or IP Address for Server*' (localhost), 'LDAP Port*' (10389), and a 'Use TLS' checkbox. It also has an 'Add Another Redundant LDAP Server' button.

At the bottom, there are buttons for 'Save', 'Delete', 'Copy', 'Perform Full Sync Now', and 'Add New'. A footer note states: 'Fields marked with an asterisk (*) are required.'

Cisco Unified Communications Manager User Fields	LDAP Attribute
User ID	uid
Middle Name	initials
Manager ID	manager
Phone Number	telephonenumber
Title	title
Mobile Number	mobile
Directory URI	mail

Directory Service Configuration

After preview and accepting the sync agreement, admin triggers “enable sync” within 18 hours for the users to get provisioned on the UCM Cluster

Field mapping Agreement selection Enable synchronization

Enable synchronization

You can enable the synchronization once the system copies the user data into temporary storage and a new synchronization agreement is created automatically. The processing might take some time, you may close the window and come back later.

Completed

Step 1: Copy data into temporary storage and synchronize dry run
User data is copied from Webex into a temporary storage in on-premise infrastructure and a dry run of synchronization is performed. A dry run allows you to see what users will be added, modified, or deleted before you run a full synchronization and commit the changes to the on-premise infrastructure.

Dry run results :		
Users Matched : 0	New Users : 56	Mismatched Users : 0

Completed

Step 2: New agreement creation
A new synchronization agreement is created for syncing the users.

You must enable synchronization within 24 hours from the time the new agreement is created. LDAP synchronized users have become inactive. These users are removed after 24 hour of inactivity. Inactive users will not be able to login and use services of the Call Manager. **Time left : 23h 59m 31s**

Completed

Step 3: Synchronization agreement verification
You can click on the preview button to log in to your on-premises infrastructure and make changes to newly created synchronization agreement. VPN access is required.

You should verify and acknowledge the following check box:

☒ I have reviewed the new synchronization agreement and agree to proceed.

[Preview in Unity Connection](#)

Enable synchronization

Directory Service Configuration

Directory Service will require the Admin to enable the Synchronization again before progressing the process

< Cloud Connected UC

Directory Service

Can't find a cluster?

You should take action for the following cluster(s): [StandAloneCluster - ucba-aricent-vm33.cisco.com](#)

Search

Filter by status

(GMT -05:00) America/Mexi...

3 Cluster(s)

Cluster

Status

Last synced

Product

EMEAcluster - emea-cm-pub.ent-pa.com

Not provisioned

N/A

CUCM

StandAloneCluster - ucba-aricent-vm33.cisco.com

Action required

N/A

CUC

23h 57m 52s left to enable synchronization

LihCluster - lih-pub.cisco.com

Provisioned

Mar 31, 2022 | 08:20 AM

CUCM

Directory Service Configuration

Status will change to Processing once the CI Sync starts

< Cloud Connected UC

Directory Service

[Can't find a cluster?](#)

Cluster	Status	Last synced ↓	Product
EMEAcluster - emea-cm-pub.ent-pa.com	Not provisioned	N/A	CUCM
StandAloneCluster - ucba-aricent-vm33.cisco.com	Processing	N/A	CUC
LihCluster - lih-pub.cisco.com	Provisioned	Mar 31, 2022 08:20 AM	CUCM

Directory Service Configuration

Provisioned status means that CI users exist now in DB of the on-prem servers

< Cloud Connected UC

Directory Service

[Can't find a cluster?](#)

Q Search Filter by status (GMT -05:00) America/Mexi... 3 Cluster(s)

Cluster	Status	Last synced ↓	Product
EMEAcluster - emea-cm-pub.ent-pa.com	Not provisioned	N/A	CUCM
StandAloneCluster - ucbu-aricent-vm33.cisco.com	Provisioned	Apr 21, 2022 12:05 PM	CUC
LihCluster - lih-pub.cisco.com	Provisioned	Mar 31, 2022 08:20 AM	CUCM

Considerations for Cisco Unity Connection

Up to this point CI Sync succeeded and users exist in CUC DB

Finalize import of users with two methods:

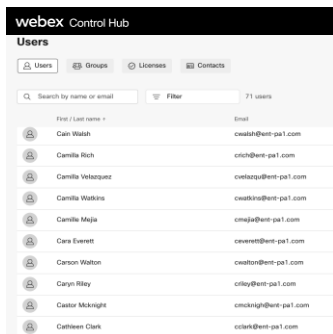
- Import Users tool in Cisco Unity Connection
- Bulk Administration Tool

Directory Service Configuration

Import users in CUC with IMPORT TOOL

Import “Users > Import Users > Select LDAP directory and template”

Select the users you want to import and fill the fields



The screenshot shows the 'Cisco Unity Connection Administration' interface. The left sidebar shows the navigation tree with 'Users' > 'Import Users' selected. The main content area shows the 'Import Users' tool. It includes a 'Status' section indicating 'Found 56 LDAP User(s)'. Below this is a 'Find' section with a dropdown for 'Find End Users In' (set to 'LDAP Directory') and a 'Where' section with 'Alias' and 'Begins With' dropdowns. The 'Import With' section shows 'Based on Template' set to 'Not Selected'. A red box highlights the 'Directory Search Results' table.

Import Selected	Import All	25	Rows Per Page		
	Alias	First Name	Last Name	Phone Number	Extension
☐	caguirre@ent-pa1.com	Chandler	Aguirre		
☐	charrett@ent-pa1.com	Craig	Barrett		
☐	cbenson@ent-pa1.com	Constance	Benson		
☐	cbird@ent-pa1.com	Clementine	Bird		
☐	ccase@ent-pa1.com	Clark	Case		
☐	cclark@ent-pa1.com	Cathleen	Clark		
☐	ccucro@gmail.com	ccucro			
☐	cdickers@ent-pa1.com	Colton	Dickerson		
☐	ceverett@ent-pa1.com	Cara	Everett		
☐	chendric@ent-pa1.com	Coby	Hendricks		

Directory Service Configuration

Under “CUC > Users” the new imports will be listed

Search Users

Search Users

Related Links Bulk Edit By CSV Go

User Refresh Help

Status

Found 8 User(s)

Search Limits

Limit search to All

Users (1 - 8 of 8)Rows per Page 25

Find Users where Alias begins with Find

☐	Alias ^	Extension	First Name	Last Name	Display Name
☐	admin				admin
☐	caguirre@ent-pa1.com	1088	Chandler	Aguirre	Chandler Aguirre
☐	cbarrett@ent-pa1.com	1089	Craig	Barrett	Craig Barrett
☐	cbenson@ent-pa1.com	1090	Constance	Benson	Constance Benson

Directory Service – Remove Configuration

Under Service Management disable Directory Service

If you want to create a new synchronization agreement for the same cluster, you should go to the Service Management page and disable the Directory Service. You can then create a new on-prem ldap agreement for provisioning.

To roll back changes, disable Directory Service Feature and remove CI SYNC ldap Agreement

Borderless CTI

Borderless CTI Contents

01 Overview

General Summary

02 Prerequisites

03 Configuration

01 Overview

Borderless CTI

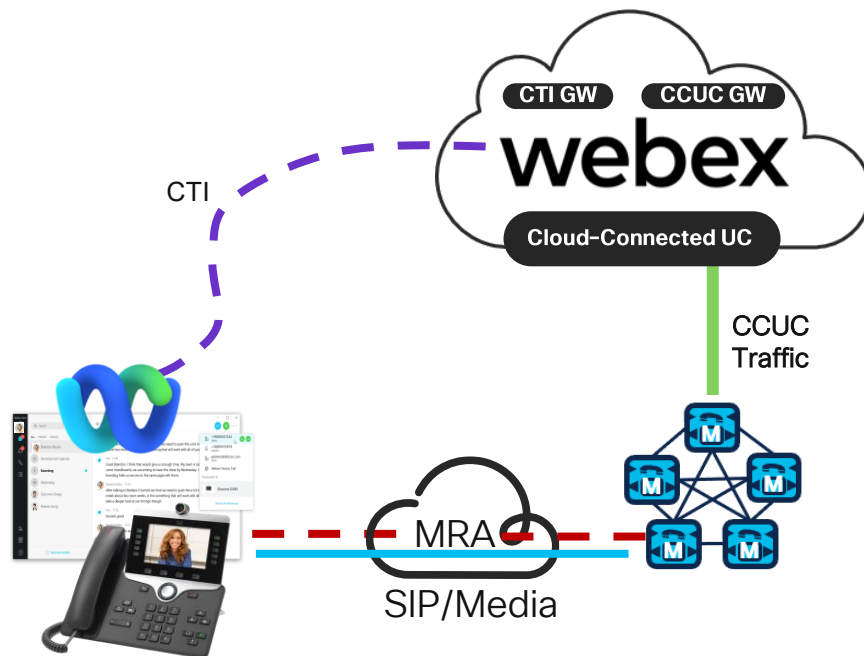
A cloud-based service that uses Cloud Connected UC to execute Webex App CTI application functionalities remotely over Mobile and Remote Access (MRA), without a VPN connection

Key Features

- Desk phone control of Cisco IP Phone 88xx/78xx, CE, and DX from Webex App remotely
- CTI applications include: receive incoming enterprise calls, Make/Disconnect/Hold/Retrieve calls, among other applications

Borderless CTI for on-premises

End user experience



Supported Releases:

- Cisco UCM: 14 onwards
- Webex App Jun 22

02

Prerequisites

Requirements

Platform

Minimum supported versions:

- UCM 14 SU2 onwards
- Webex App version 42.7 onwards
- Expressway™ latest version recommended
- Cisco IP Phone 8800 Series or Cisco IP Phone 7800 Series or Cisco Webex Desk Series (DX)

MRA & CTI configuration in place

Control Hub

- Borderless CTI toggle on under Service Management in CCUC Control Hub

Connectivity

- For a larger cluster, ensure that you have at least 2 MBPS of bandwidth.
- Cloud-Connected UC cloud port 443 must be reachable from customer environment.

*.ucmgmt.cisco.com

*.webex.com

03 Configuration

Borderless CTI

Connected UC > Inventory > Cluster details > Ellipsis icon > Service Management

- Borderless CTI feature gets activated after enabling the toggle
- Requires Mobile and Remote Access (MRA) and CTI configuration in the UCM cluster

Service Management

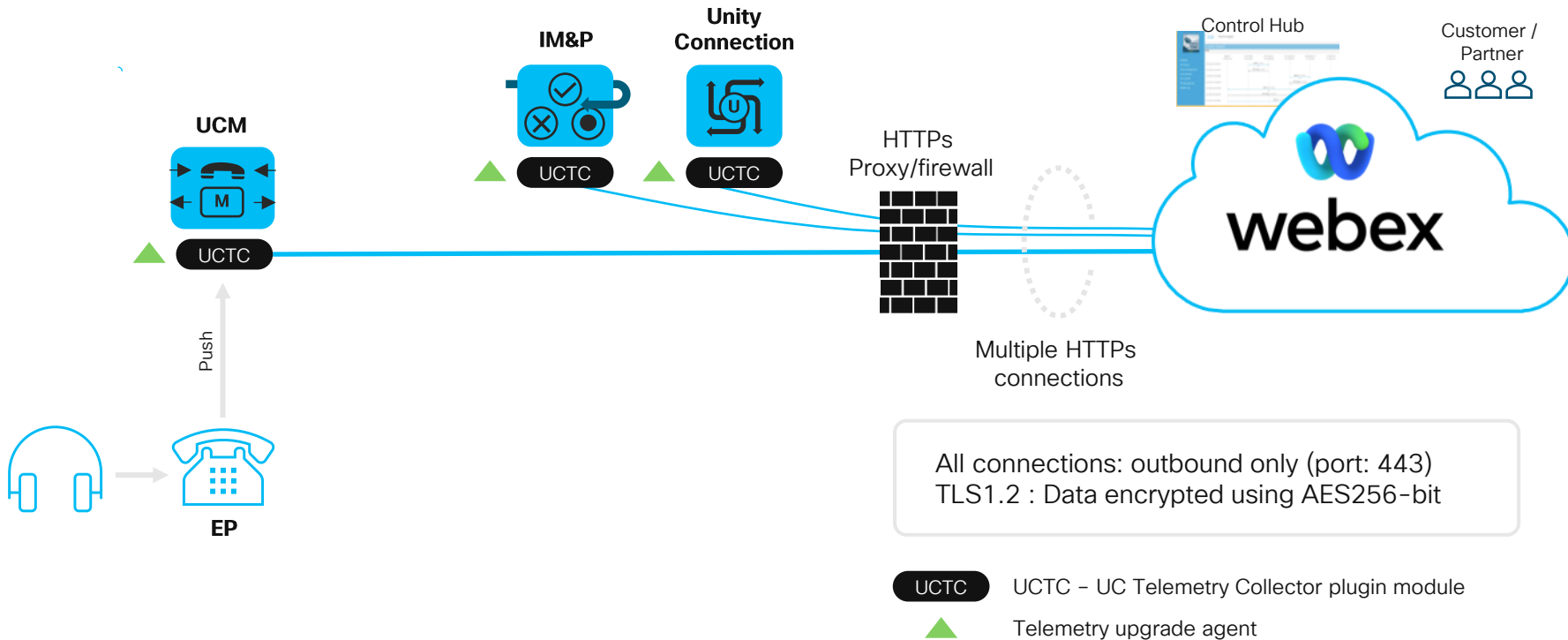
Changes to these settings will take a short time to take effect.

Analytics	☒	Enabled
Borderless CTI *	☒	Enabled
Directory Service	☒	Enabled
Certificate Management	☒	Enabled
Operational Metrics	☒	Enabled
Webex app Provisioning for Unified CM Calling	☒	Enabled
Deployment Insights	☒	Enabled

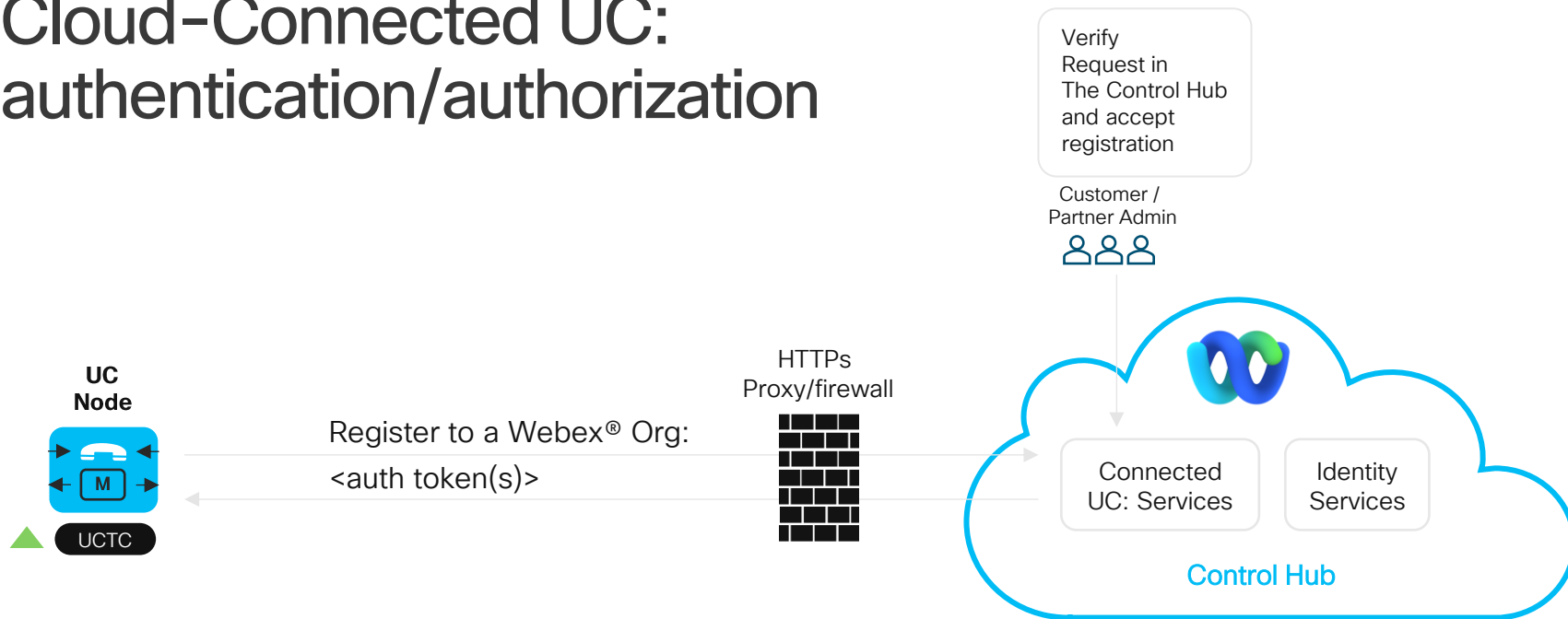
CancelSubmit

Security and Privacy

Cloud-Connected UC: secure connectivity



Cloud-Connected UC: authentication/authorization



UCTC

UCTC – UC Telemetry Collector plugin module



Telemetry upgrade agent

Cloud-Connected UC: proxy access

If a proxy is used, following URLs need to remain accessible

URL	Description
https://tcgw.telemetry.ucmgmt.cisco.com https://controller.ucmgmt.cisco.com	URLs to send telemetry data and manage the agents

Proxy Server configuration: **Support for Unauthenticated Proxy and Authenticated Proxy**

Security and privacy considerations



- All data is encrypted at rest and in transit.
- Webex® Common Identity infrastructure is used to authenticate and authorize cloud connectors to a specific Control Hub ORG.
- All the privacy related data sent by CCUC is outlined in Privacy data sheet maintained at the Cisco® Trust Portal (trustportal.cisco.com).

Privacy data sheet for CCUC

The Privacy data sheet describes the processing of personal data (or personal identifiable information PII).

It includes:

- Personal data processing (types and purpose of processing)
- Cross-border transfers (location of Cisco® and 3rd party data centers)
- Access control (who has access, purpose of access)
- Data deletion & retention
- Personal data security
- Third party service providers

Links:

Trust Portal: <https://trustportal.cisco.com>

CCUC privacy data sheet [direct link](#)

Additional Resources

References

Webex® Cloud-Connected UC in SalesConnect:

<https://salesconnect.cisco.com/#/program/PAGE-16868>

Webex Cloud-Connected UC BDM Deck

<https://salesconnect.cisco.com/open.html?c=83865fa1-cf32-4be2-896c-d28762099c1e>

Webex Cloud-Connected UC Privacy Data Sheet"

<https://trustportal.cisco.com#/1597957746696555>

Voice of the Engineer - Aug 6, 2020

Recording:

<https://salesconnect.cisco.com/open.html?c=83424d15-a7e1-422c-93d8-108f2b815ae7>

Webex Help Center: " CCUC Overview"

<https://help.webex.com/en-us/jv0u1db/Cisco-Webex-Cloud-Connected-UC-Overview>

Blog

<https://blogs.cisco.com/collaboration/webex-cloud-connected-uc-add-the-power-of-cloud-to-your-on-premises-uc-systems>

References – Webex Help Center Articles

Webex CCUC Overview

<https://help.webex.com/en-us/jv0u1db/Cisco-Webex-Cloud-Connected-UC-Overview>

System Requirements for Webex CCUC

<https://help.webex.com/en-us/55mug2/System-Requirements-for-Cisco-Webex-Cloud-Connected-UC>

Analytics for your Webex® CCUC

<https://help.webex.com/en-us/ntm7wvk/Analytics-for-Your-Cisco-Webex-Cloud-Connected-UC>

Network Requirements for Webex CCUC

<https://help.webex.com/en-us/fg3qim/Network-Requirements-for-Cisco-Webex-Cloud-Connected-UC>

Set Up Webex CCUC for On-Premises Devices

<https://help.webex.com/en-us/nzt6c0b/Set-Up-Cisco-Webex-Cloud-Connected-UC-for-On-Premises-Devices>

Resources - feature walkthrough VoD's

The Inside Tour of Webex Cloud-Connected UC

Part 1: Introduction to Webex Cloud-Connected UC -

<https://salesconnect.cisco.com/open.html?c=a29d89cf-31b9-40c0-9541-5c800f3fba99>

Part 2: Introduction to onboarding UC nodes -

<https://salesconnect.cisco.com/open.html?c=41597bce-a2ed-4794-b362-0f8a7e591716>

Part 3: Completing UC node onboarding -

<https://salesconnect.cisco.com/open.html?c=8a816e52-f0b6-4618-b49c-15f7d0390400>

Part 4: Introducing CCUC Analytics -

<https://salesconnect.cisco.com/open.html?c=8a7cf30a-d007-4dfe-9385-5274dcd9dfea>

Part 5: CCUC Analytics Functional Overview -

<https://salesconnect.cisco.com/open.html?c=2b1a4121-e827-43dc-be63-7d67415b070b>

Part 6: Call Status in CCUC Analytics -

<https://salesconnect.cisco.com/open.html?c=b30836ea-e3a8-4d23-9dc0-71db1b8abe2d>

Part 7: Call Quality in CCUC Analytics -

<https://salesconnect.cisco.com/open.html?c=bf359de6-0ea9-4bdd-867f-b867cefd5f54>

Part 8: Asset Usage in CCUC Analytics -

<https://salesconnect.cisco.com/open.html?c=db0c7dfc-b72a-4562-943c-d88e4862c6dd>

Recently uploaded to SalesConnect!

Demo Resources

➞ Dcloud ... //Ask Venky

➞ Instant demo with features already deployed

<https://demotoolbox.cat-dcloud.com/demo/demo-webex-cloud-connected-uc-ccuc>

Additional Resources

- Sales Connect

For partners:
[SalesConnect](#)

For customers:
visit cisco.com

- Deployment Guide

Download





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